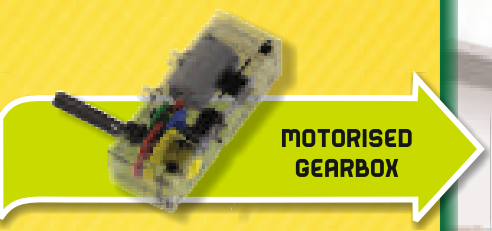
ZEPHYR
More Than Just Play

STEM EDUCATION
SCIENCE, TECHNOLOGY, ENGINEERING, MATH

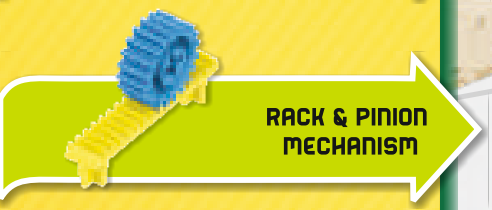
RACK & PINION

Blix



MOTORISED GEARBOX

QUICK CONSTRUCTION



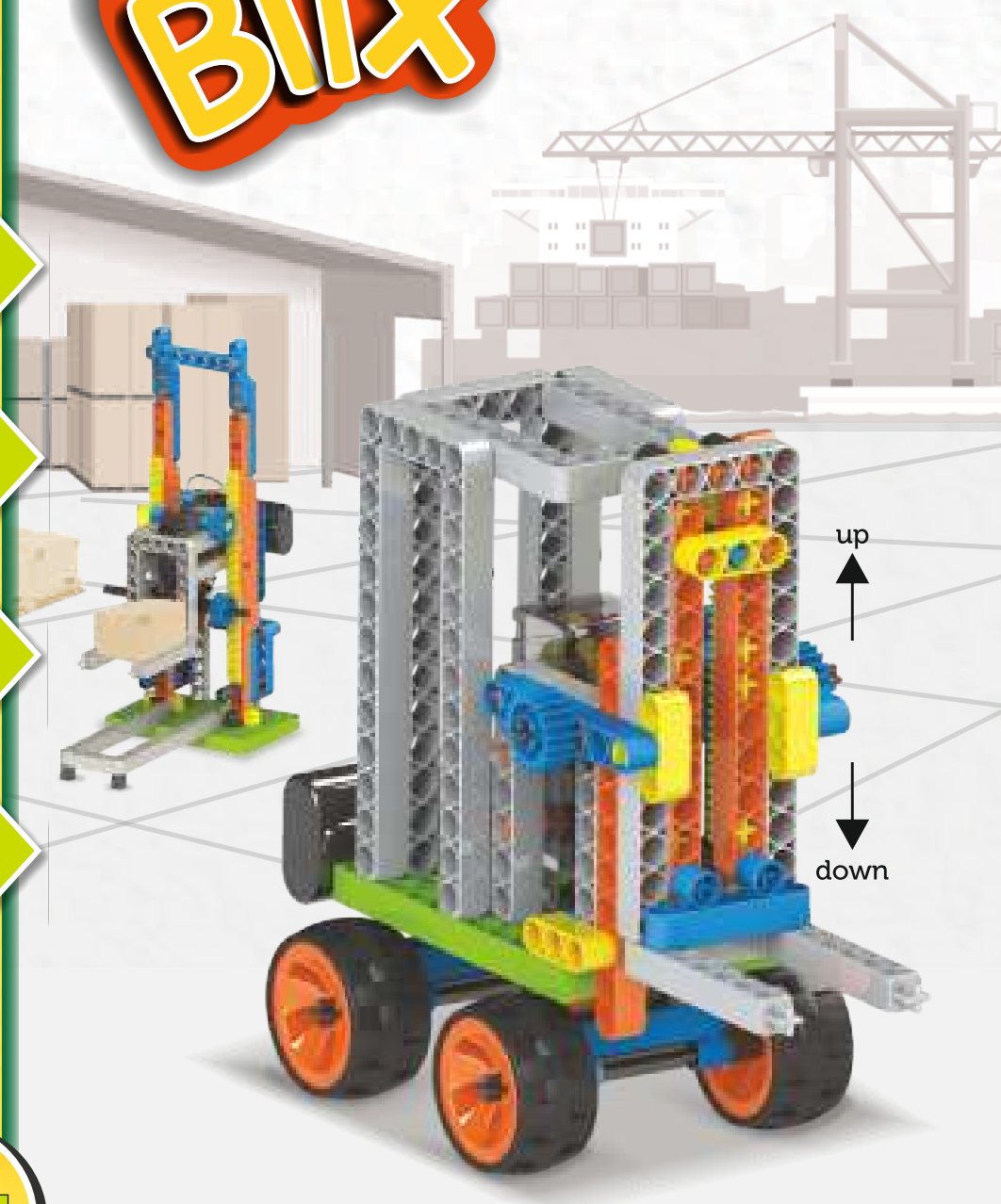
RACK & PINION MECHANISM

2 WAY CONTROL

AGE 8+

MODELS 6

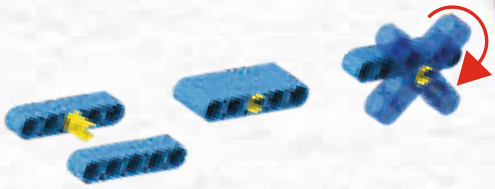
150+ PIECES



Do - it yourself manual

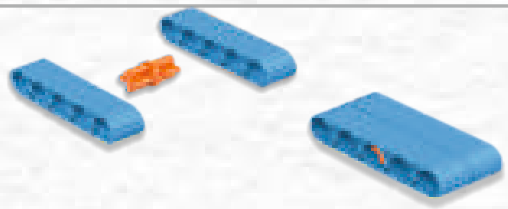
1. Read all instructions carefully before constructing.
2. Place all the pieces used in a step on the side before starting.
3. Always remove batteries when not in use.

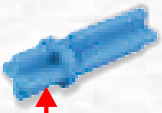
How to construct



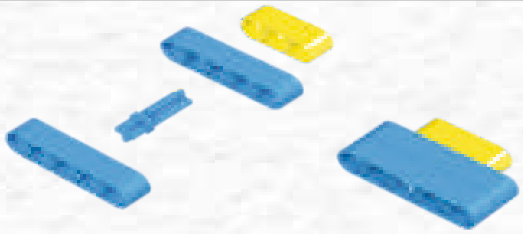
I - CL2 - Use this connector to loosely connect 2 pieces.

II - CT2 - Use this connector to attach 2 pieces.



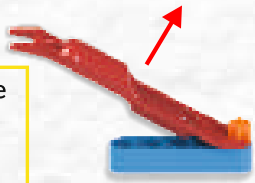


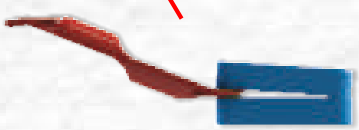
III - CT3 - Use this connector to attach 3 pieces.



During assembly ensure position of collar is same as in the image in manual.

How to dismantle



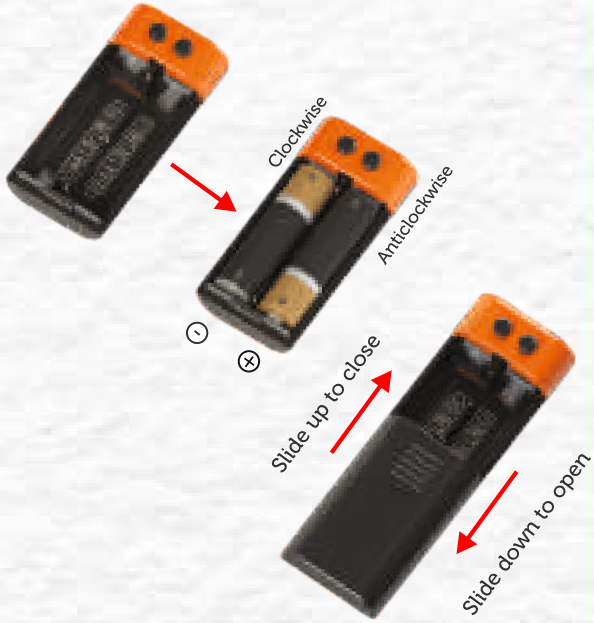


- Narrow edge to remove connectors
- Fit the tool into narrow side of connector collar

- Broad edge to split two pieces

Safety Guidelines

1. Do not mix old and new batteries.
2. Do not mix alkaline, standard (Carbon-Zinc) or rechargeable (Nickel-Cadmium) batteries.
3. Batteries are to be inserted by matching (+) and (-) polarity markings as shown in the image.
4. Non-rechargeable batteries are not to be charged.
5. Rechargeable batteries are to be removed before being charged.
6. Exhausted batteries are to be removed from toy.
7. The supply terminals are not to be short- circuited.
8. The kit is not intended for children under 3 years old.
9. Remove batteries before cleaning any part of the toy.
10. Prevent wire from coming in contact with the wheels and tracks, by using wire clip.



Add 2 x 1.5 V "AA" size batteries

What is Rack and Pinion?

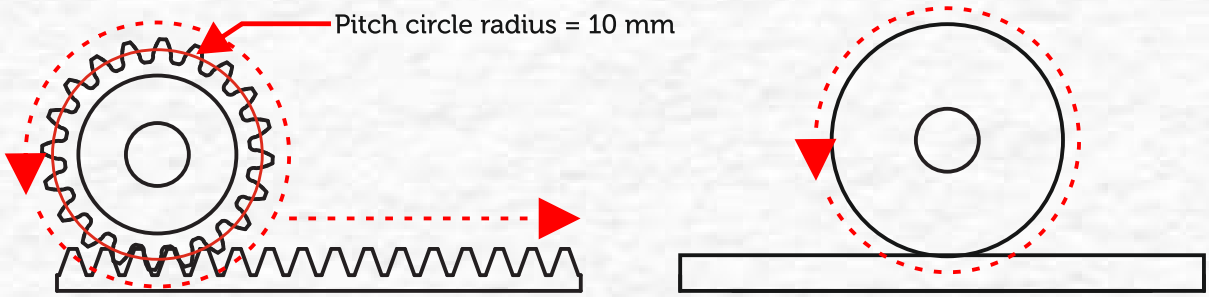
Rack and pinion is a common mechanism used to convert rotational motion into linear motion. This means that this mechanism is suitable if you have an input that creates motion in a circular direction (like and engine or motor), but you want an output where the motion is in a straight line (like a garage door opening up).

How does it work:

This mechanism is not that different from the wheels of a car turning and moving the car forward. Similarly, a pinion is a small gear that turns like wheels and creates a linear motion on the rack.

A rotating wheel can slip, but a rack and pinion's teeth mesh and are capable of transferring much more force.

You might have seen roller coasters using this mechanism to climb slopes. The metal wheels do not have enough friction, and the wheels by themselves would slip. Therefore, they place a rack on the slope and pinion on the car for it to climb.



How can we calculate speed of rack?

Distance travelled by rack in one rotation = Circumference of pinion

= $2\pi r$ ('r' is pitch circle radius of pinion gear which is 10 mm for this application)

= 2π x 10 (π= 3.14)

= 62 .8 mm per rotation of pinion

Distance travelled by rack in one second = 62.8 x R.P.M. of pinion

= 62.8 x 70 R.P.M.

= 4396 mm per minute

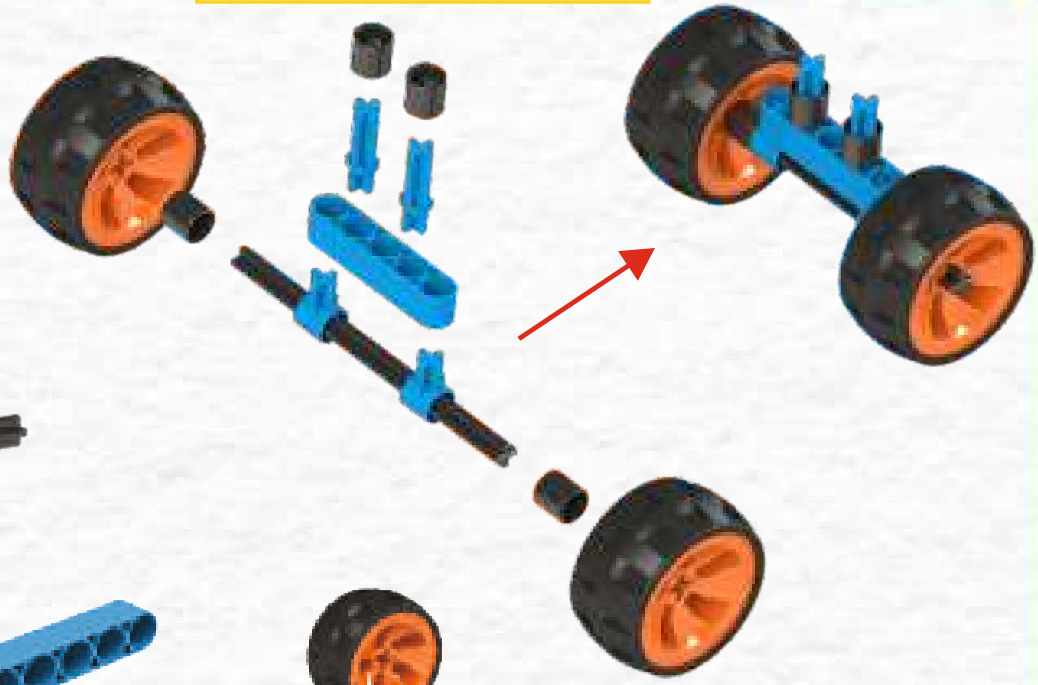
= 4.4 meter per minute (1 meter = 100 centimeter = 1000 millimeter)

= 0.07 meter per second (1 minute = 60 seconds)

MODEL 1 - FORK LIFT

STEP - 1

Make 2 sets of this step.



TW1	8 pcs.
CT3	4 pcs.
SH 100	2 pcs.
CH2	4 pcs.
P5	2 pcs.
Wheels	4 pcs.

STEP - 2

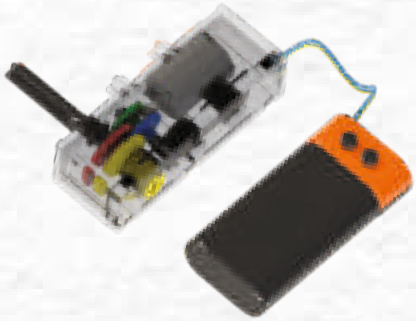
PU5 X 13	1 pc.
P7 X 11	1 pc.
CH2	4 pcs.

STEP - 3

TW1	1 pc.

CT2	2 pcs.

G(20)	1 pc.

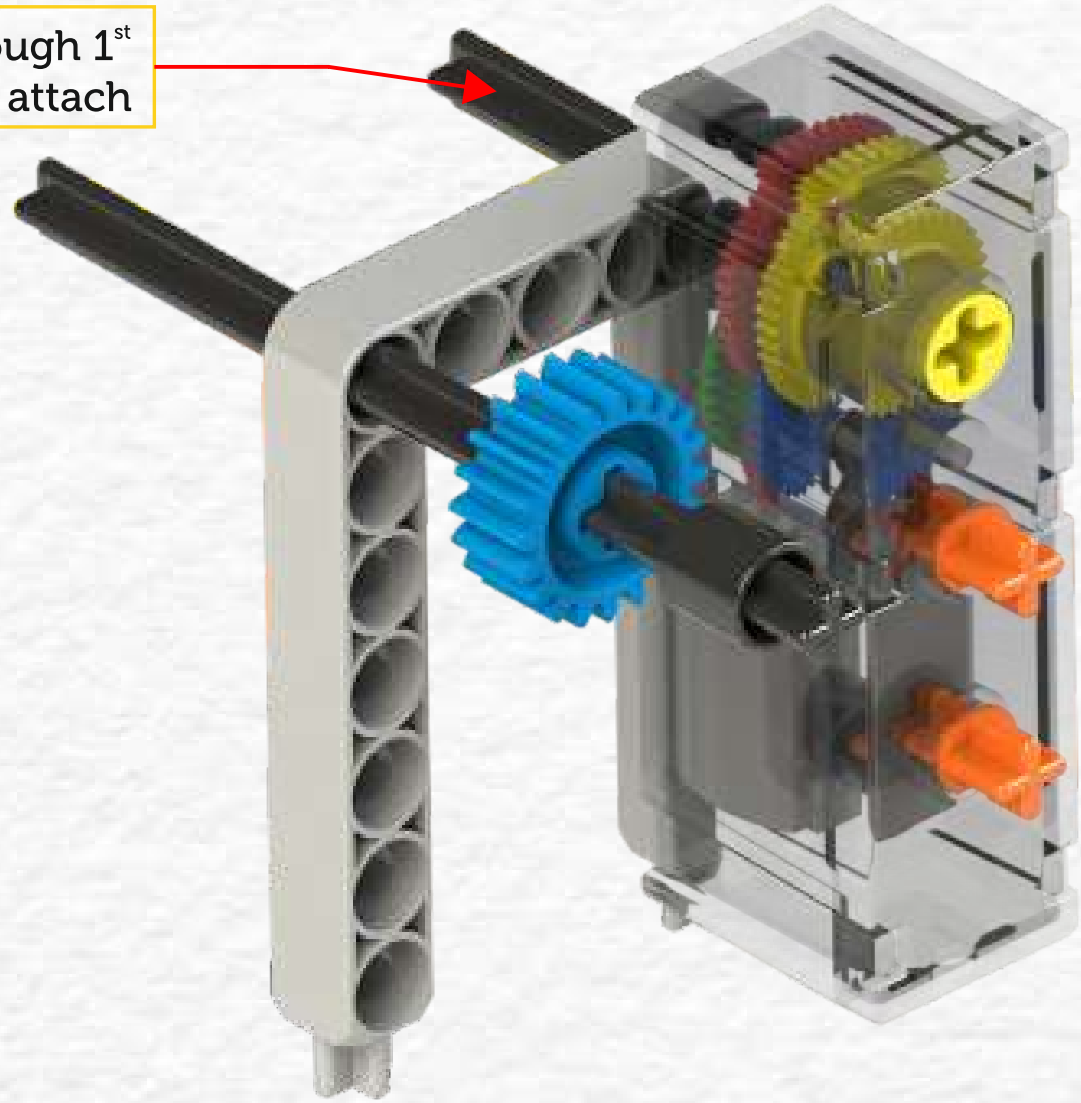


Motor With Battery Box 1 pc.

SH 100	1 pc.

PU5 X 7	1 pc.

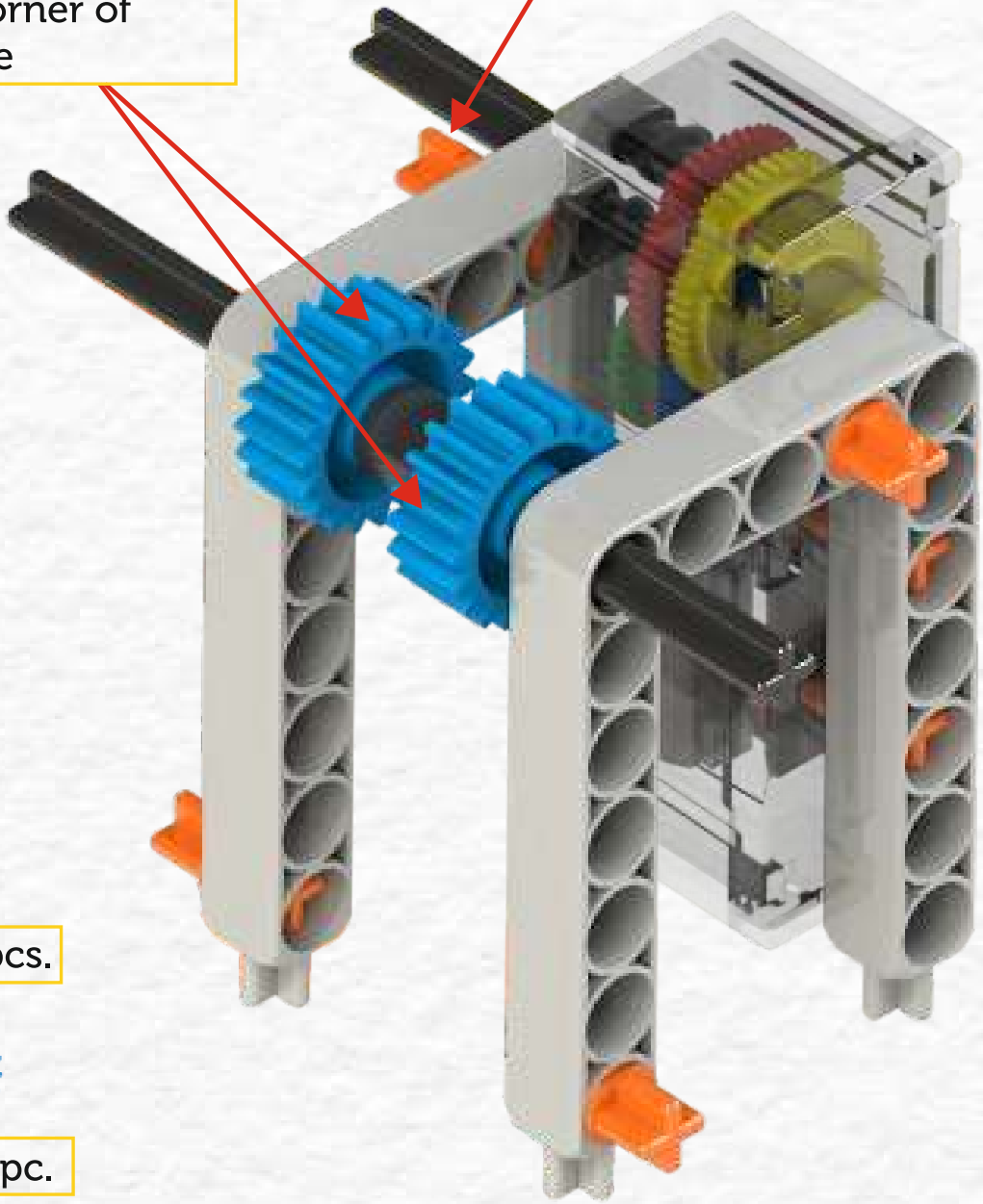
Pass through 1st hole and attach



STEP - 4

Shift the gears G(20) to the corner of both side

Attach CT2 on 2nd Hole



CT2	4 pcs.

G(20)	1 pc.

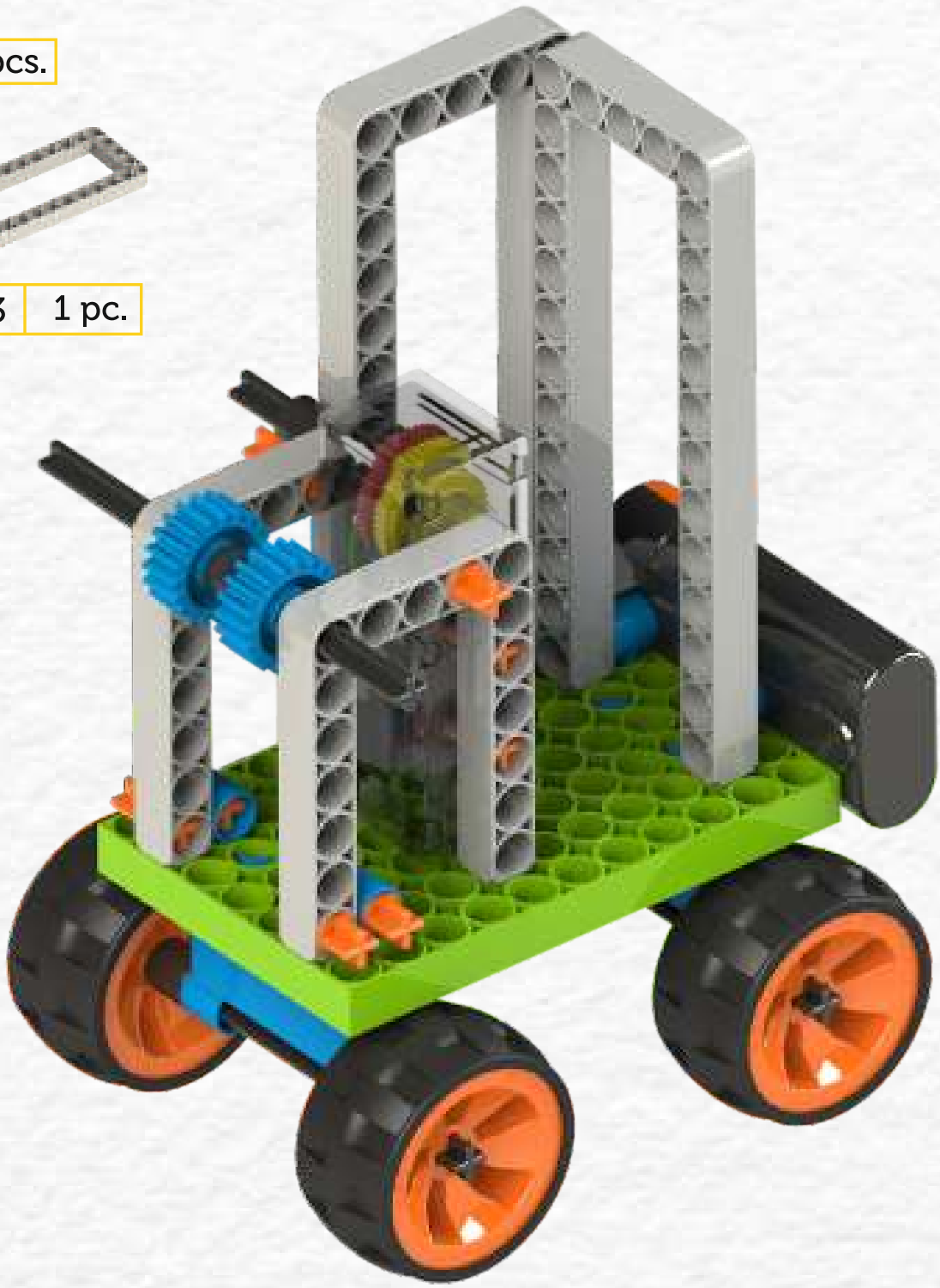
PU5 X 7	1 pc.

STEP - 5

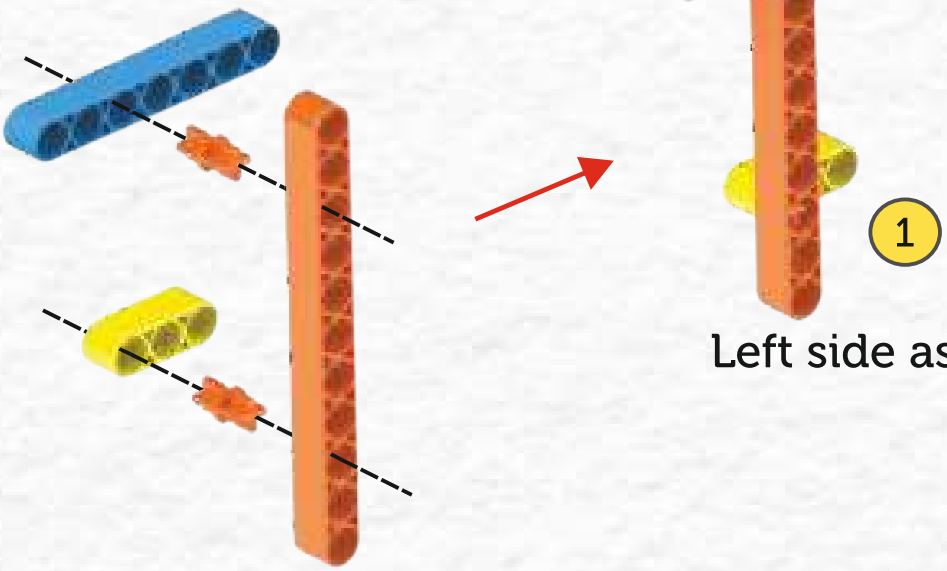
Assembly of step 4 & step 2

CT2	2 pcs.

PU5 X 13	1 pc.

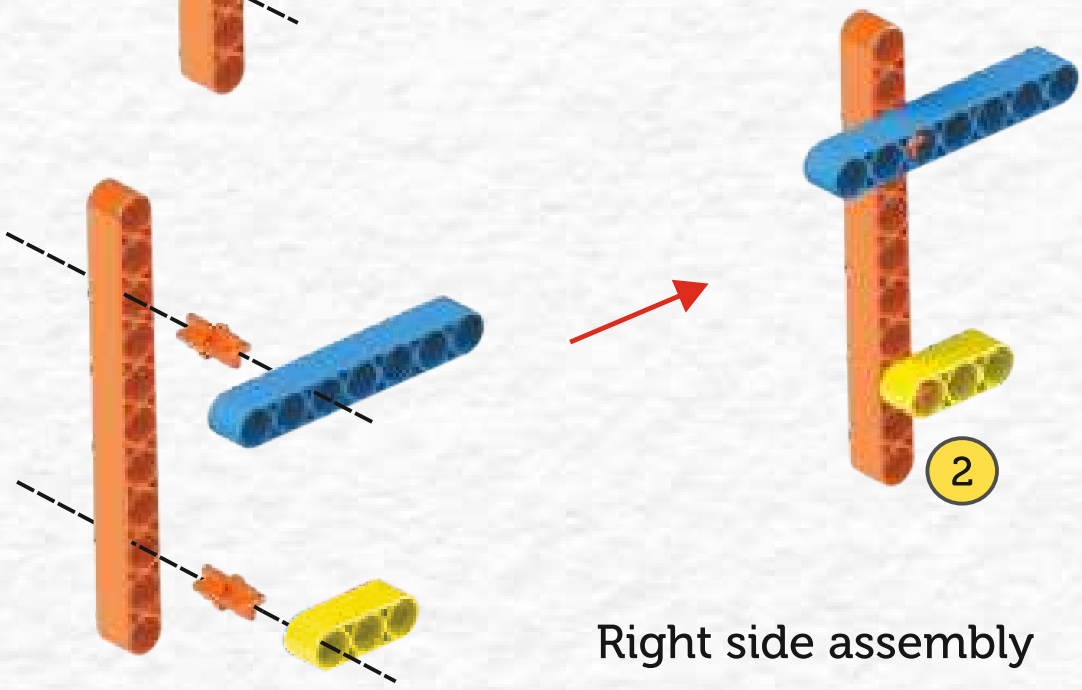


STEP - 6



1

Left side assembly

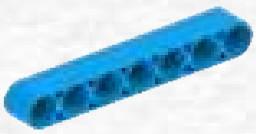


2

Right side assembly

P3	2 pcs.

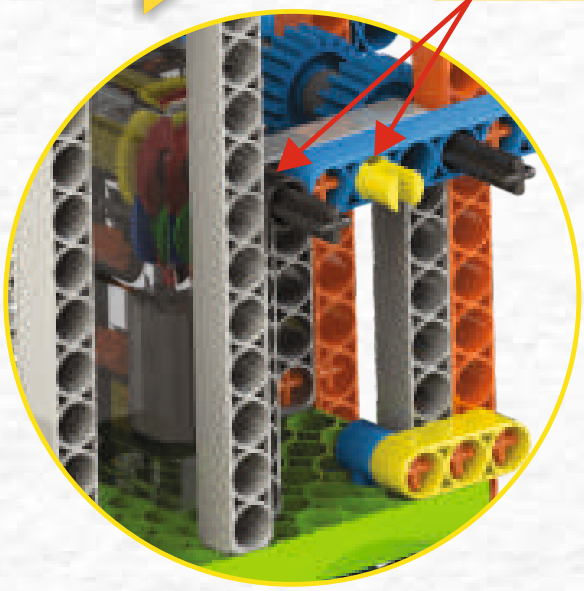
CT2	4 pcs.



P7	2 pcs.

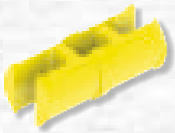
P11	2 pcs.

STEP - 7



Left side view

TW1	1 pc.



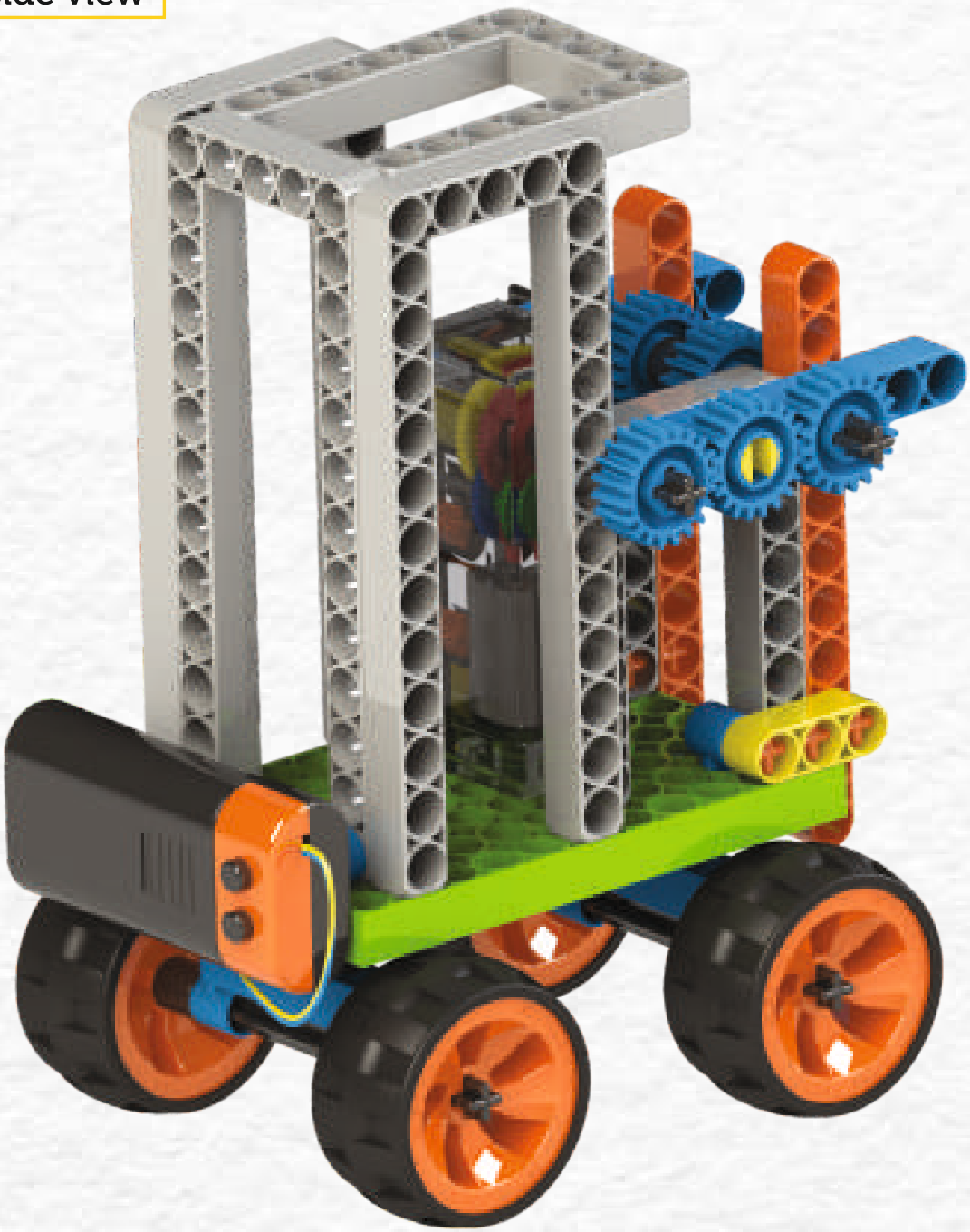
CL2	1 pc.

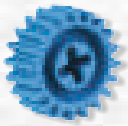
PU5 X 7	1 pc.

PU5 X 13	1 pc.

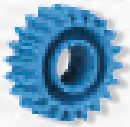
STEP - 8

Left side view



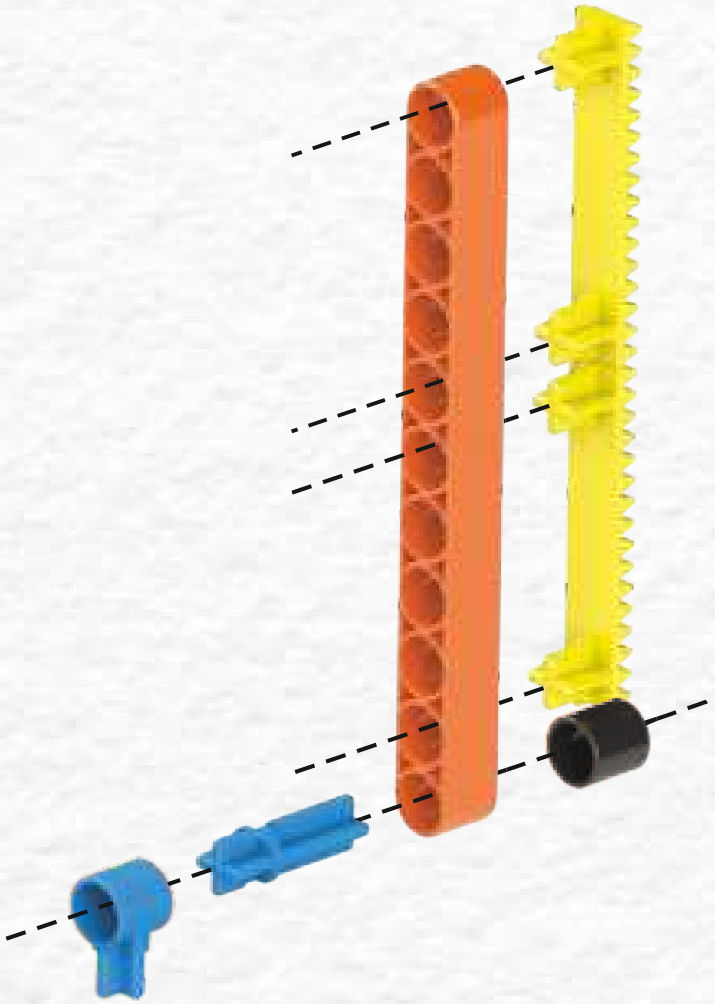


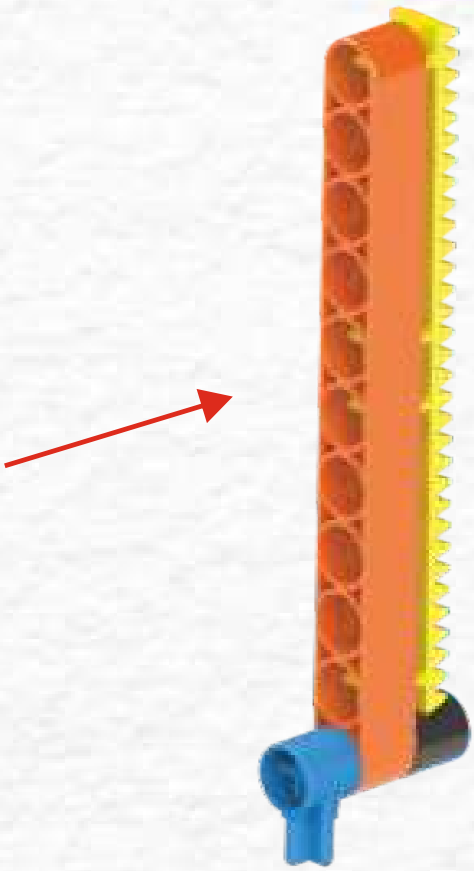
G(20)	2 pcs.



G(20) Idler	1 pc.

STEP - 9



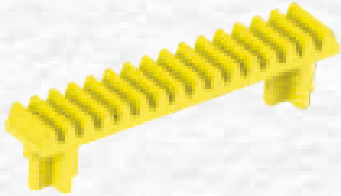


Make 2 sets of this step.

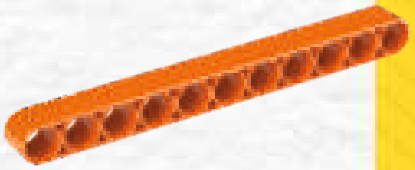
TW1	2 pcs.

CH2	2 pcs.

CT3	2 pcs.



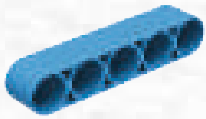
Rack	4 pcs.



P11	2 pcs.

STEP - 10

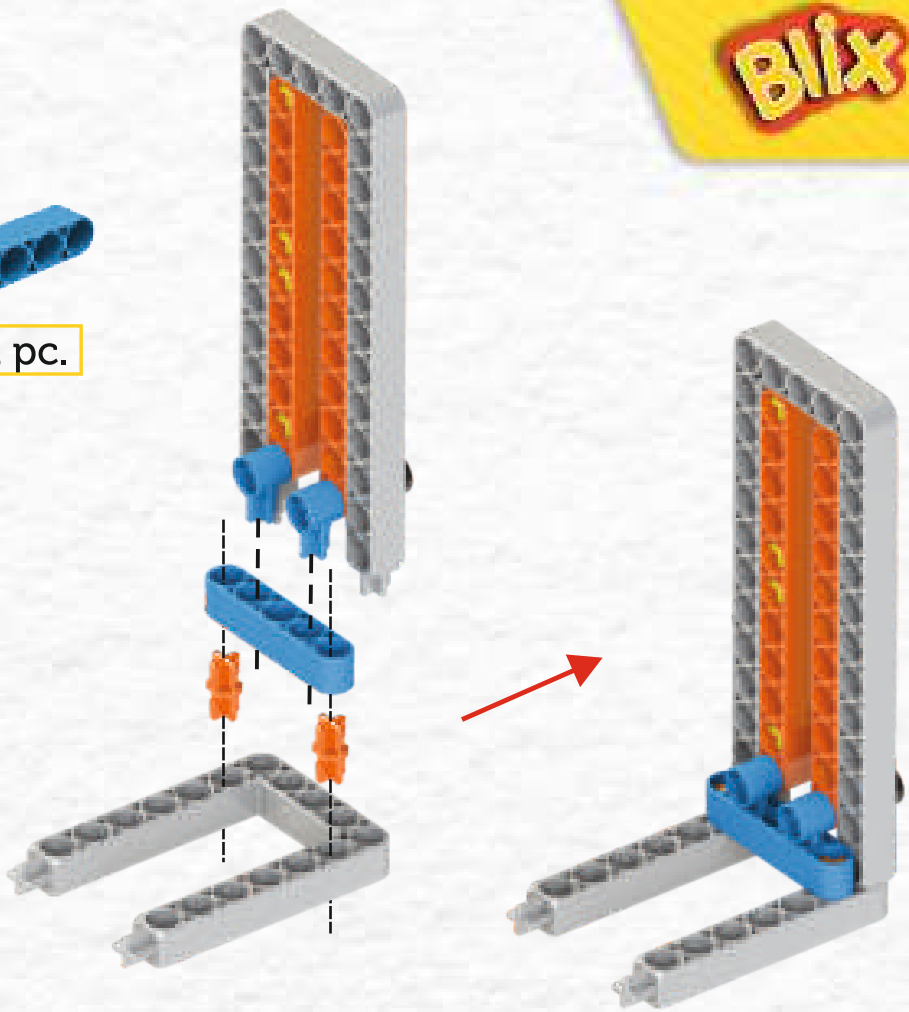
CT2	2 pcs.



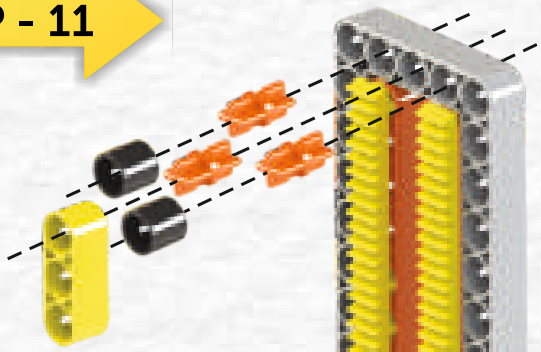
P5	1 pc.

PU5 X 7	1 pc.

PU5 X 13	1 pc.



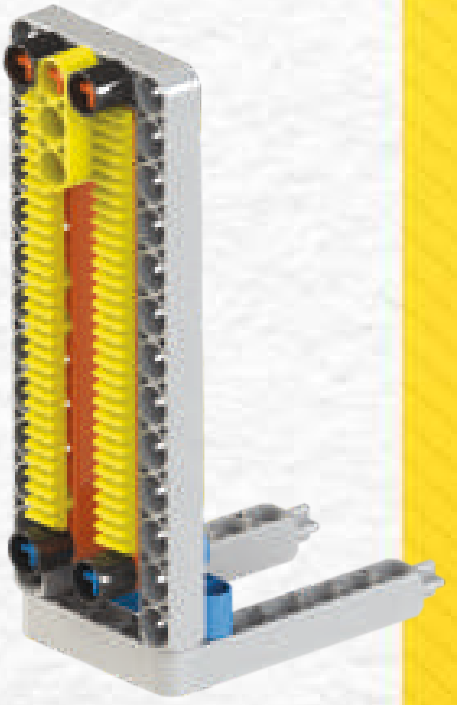
STEP - 11



TW1	2 pcs.

CT2	3 pcs.

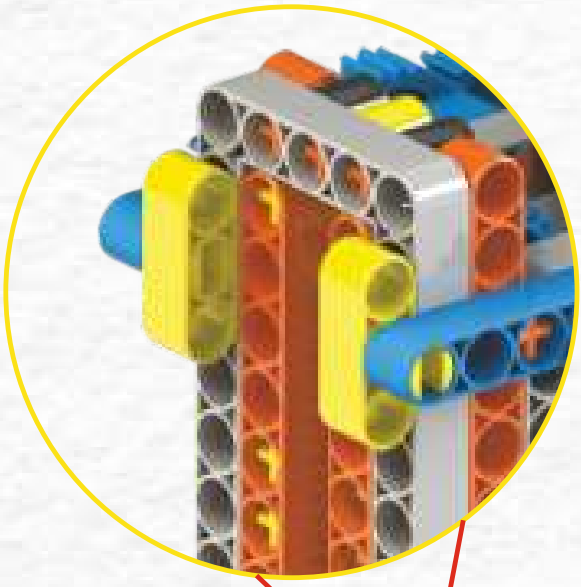
P3	1pc.

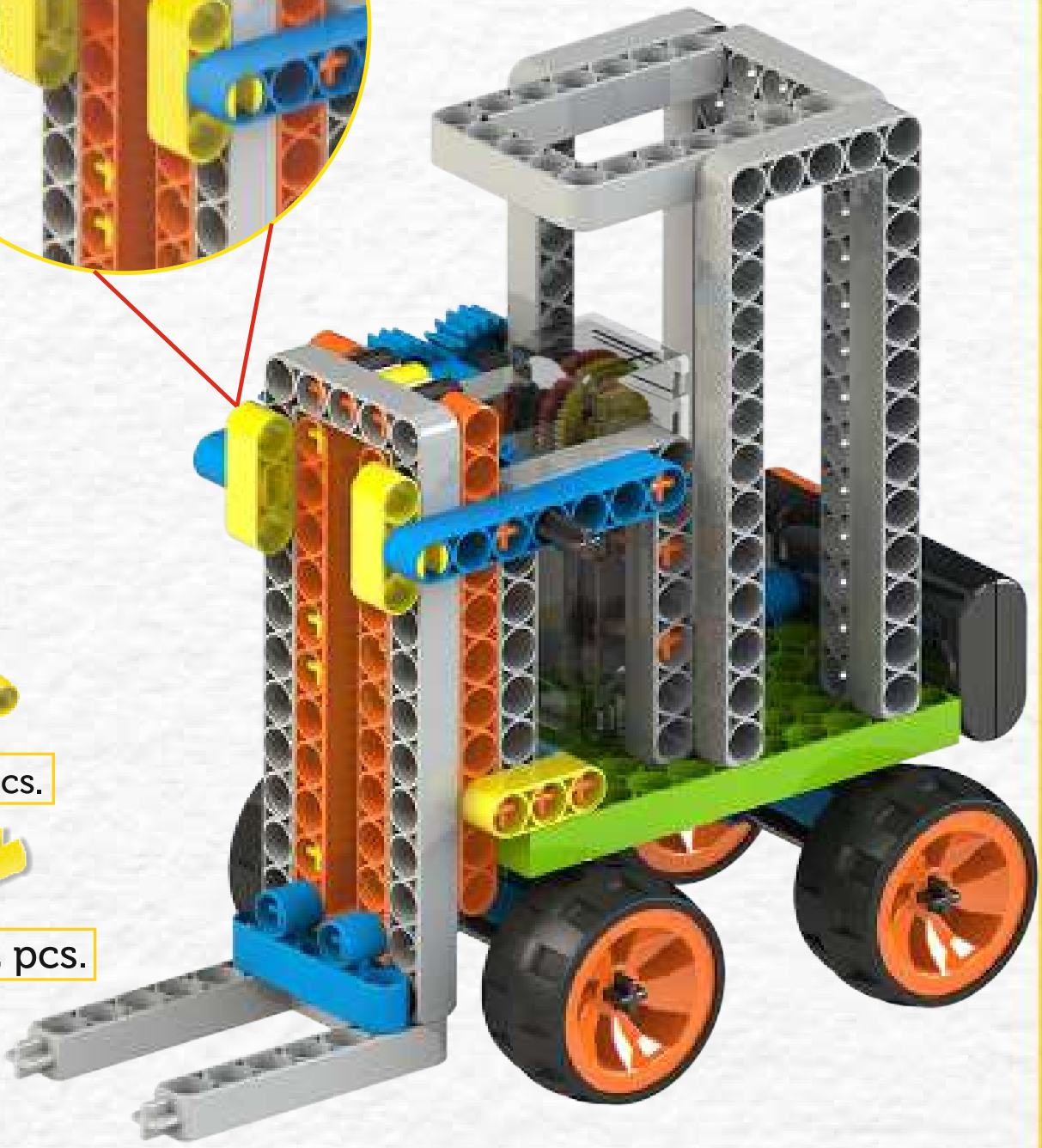


Back side view

STEP - 12

Assembly of step 11 & step 8

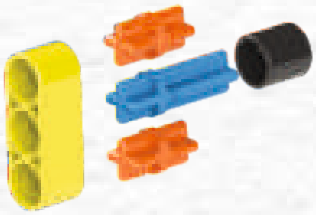


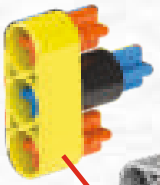


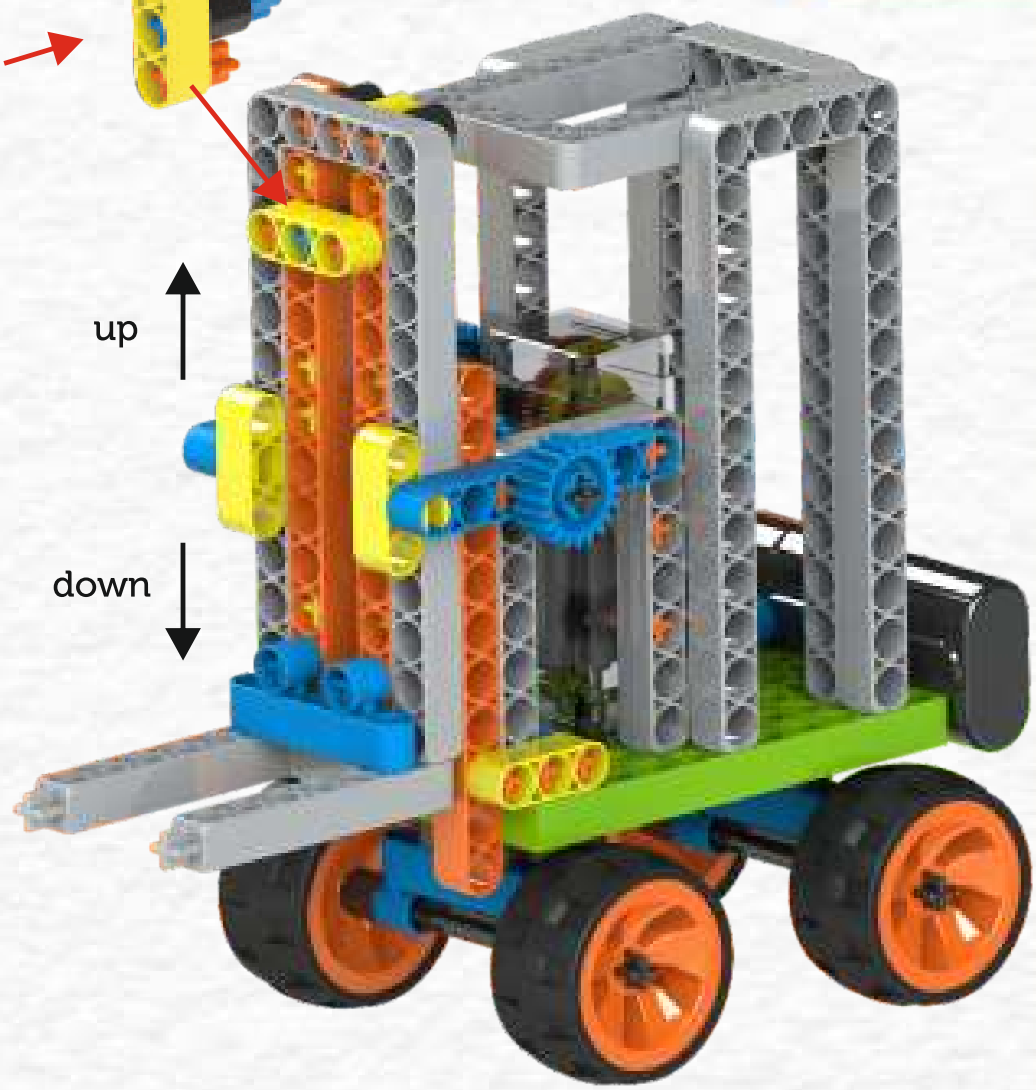
P3	2 pcs.

CL2	2 pcs.

STEP - 13







TW1	1 pc.

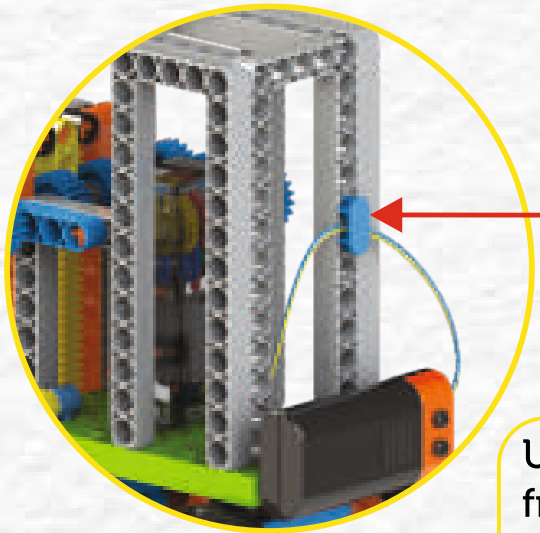
P3	1 pc.

CT2	2 pcs.

CT3	1 pc.

G(20)	1 pc.

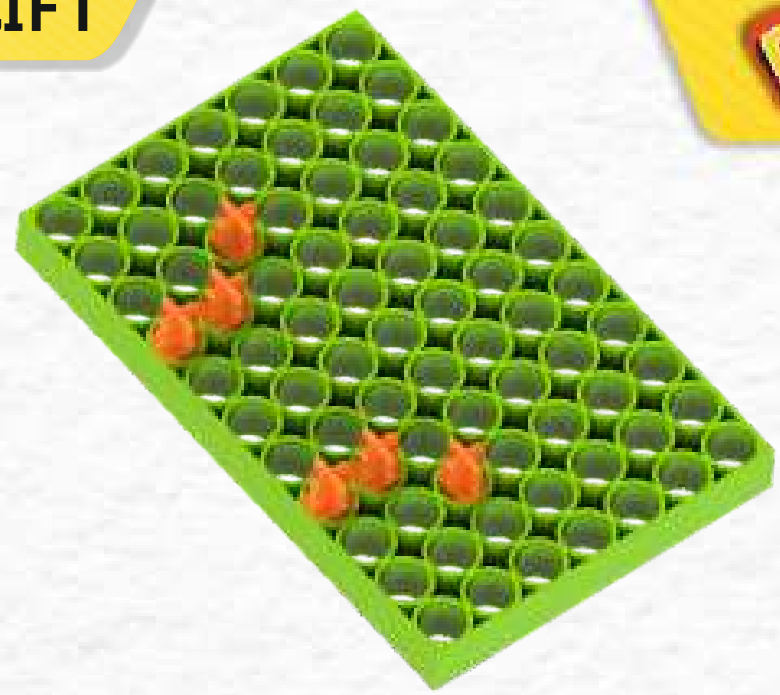
CT1 X 2	1 pc.



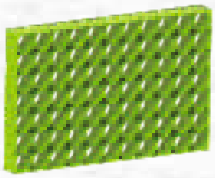
Use CT1 x 2 to prevent wire from coming in contact with moving parts

MODEL 2 - LIFT

STEP - 1



CT2	6 pcs.



P7 X 11	1 pc.

STEP - 2

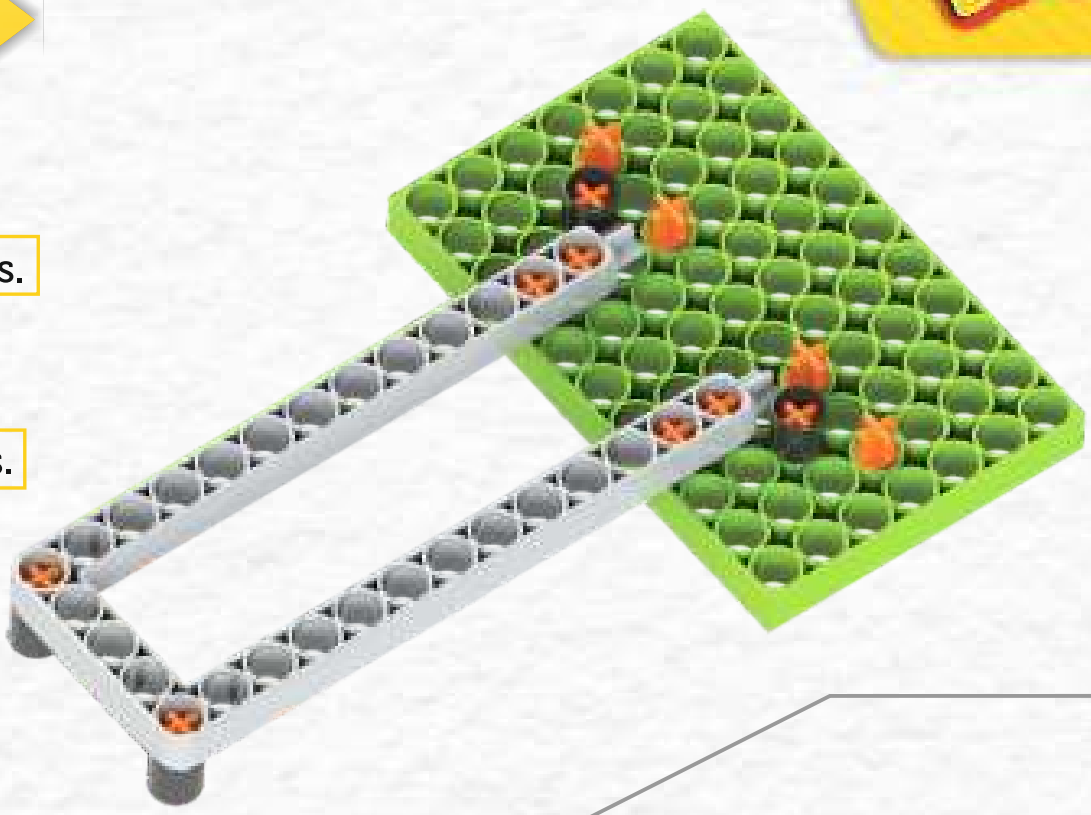
CT2	2 pcs.

PU5 X 13	1 pc.

STEP - 3

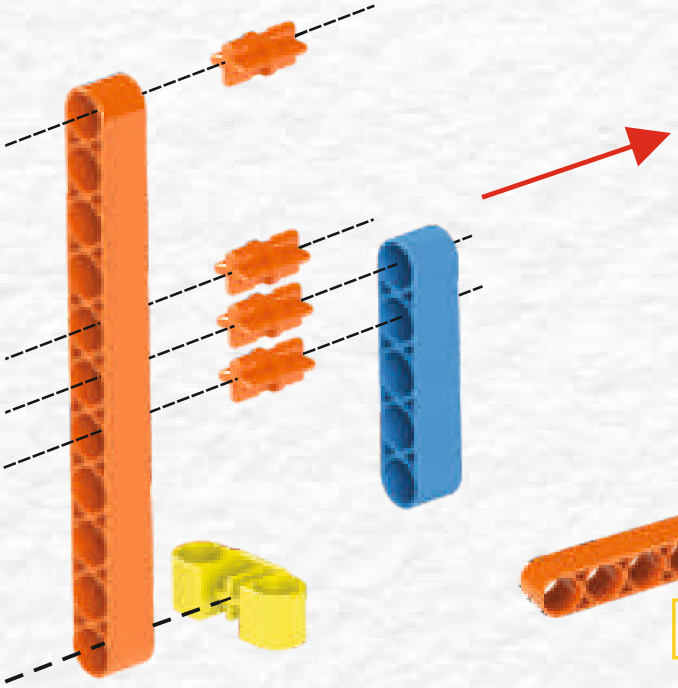
TW1	4 pcs.

CT2	4 pcs.



STEP - 4

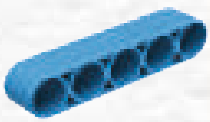
Make 2 Sets of this step



CT2	8 pcs.

PC 3	2 pcs.

P11	2 pcs.



P5	2 pcs.

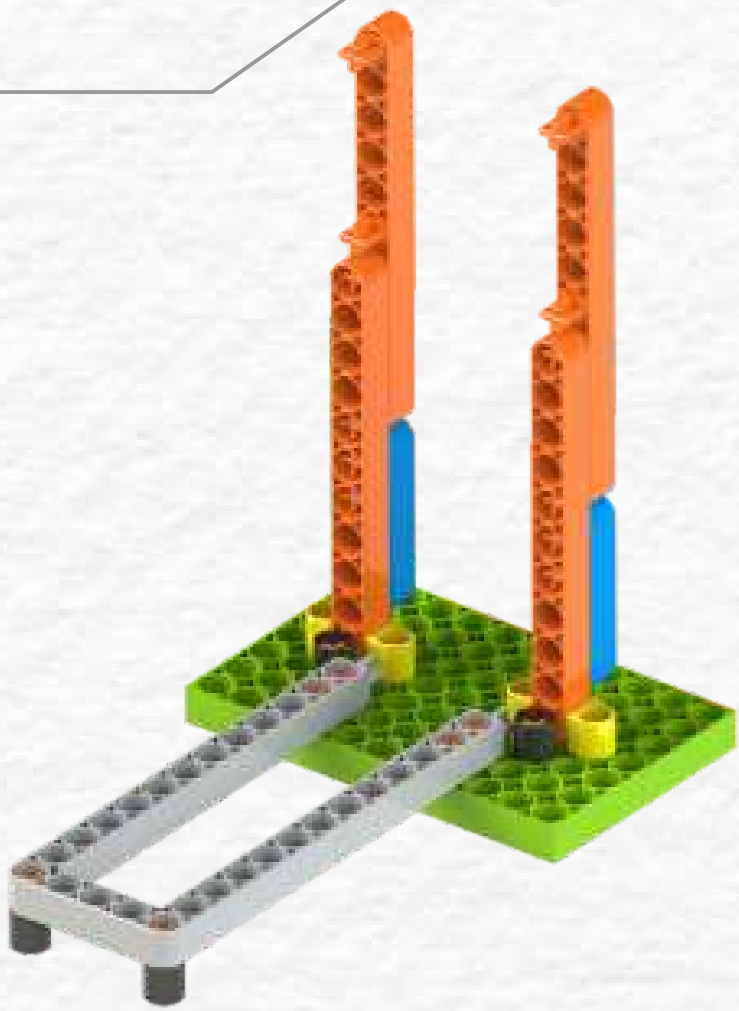
STEP - 5

Assembly of Step 4 & 3

STEP - 6

CT2	4 pcs.

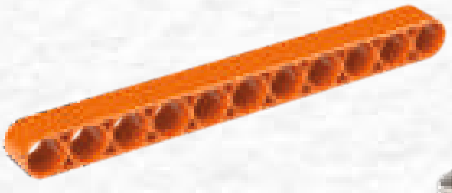
P11	2 pcs.



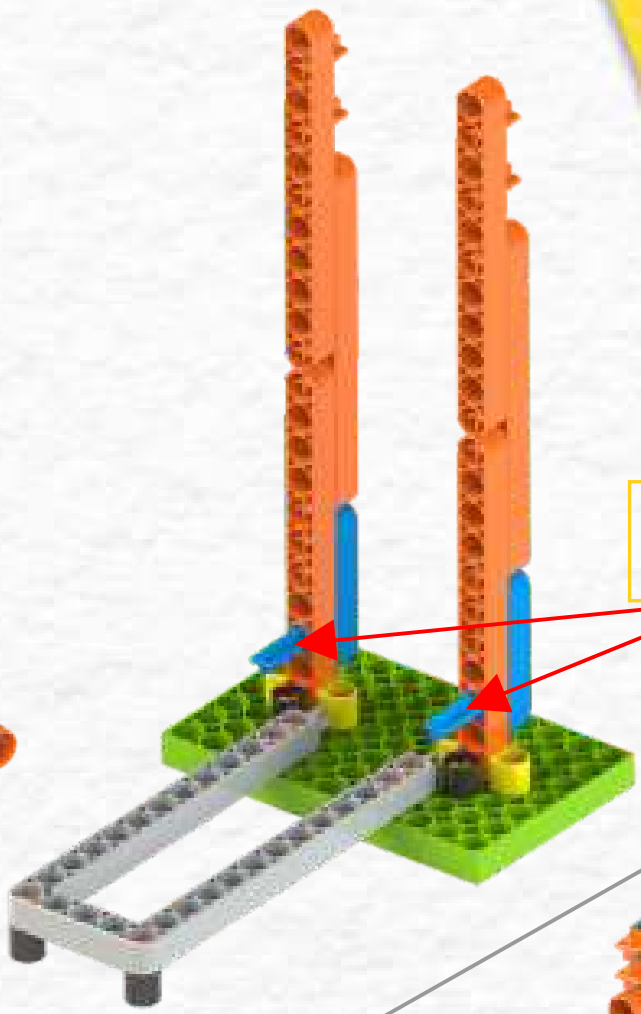
STEP - 7

CT2	4 pcs.

CT3	2 pcs.



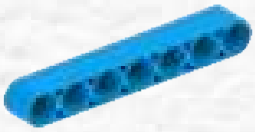
P11	2 pcs.



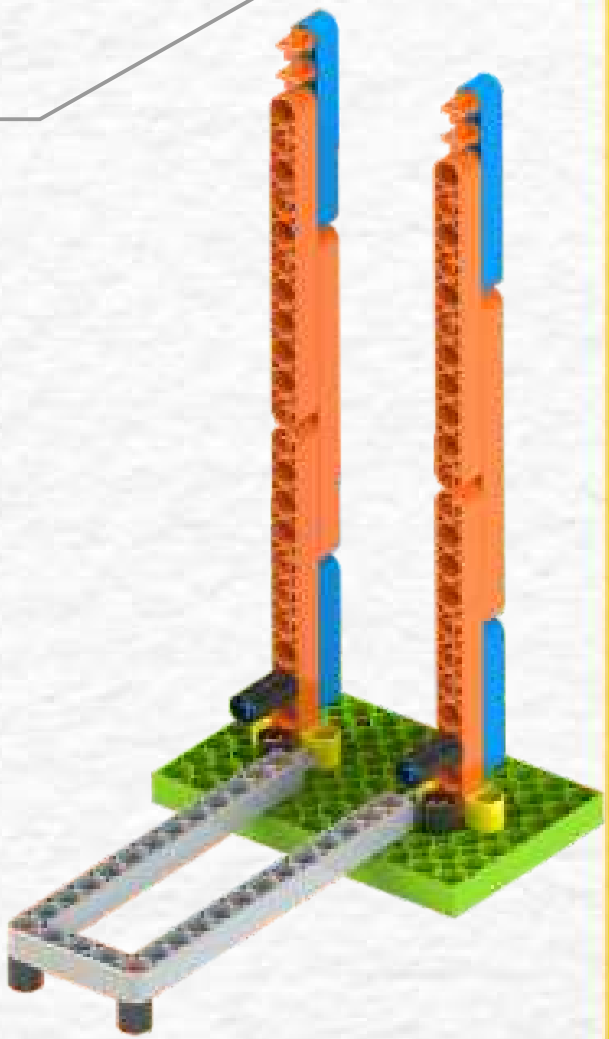
STEP - 8

CT2	4 pcs.

TW1	4 pcs.

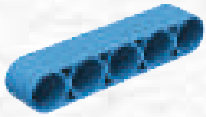


P7	2 pcs.

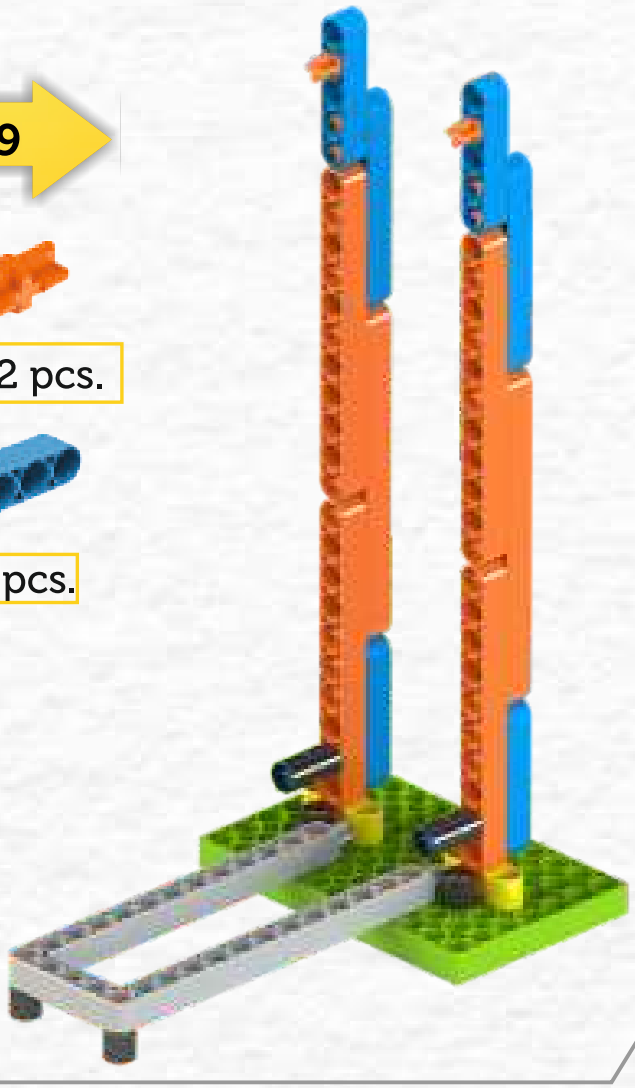


STEP - 9

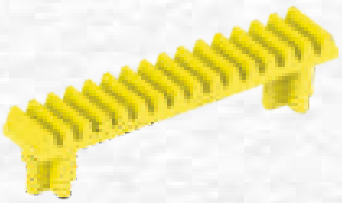
CT2	2 pcs.



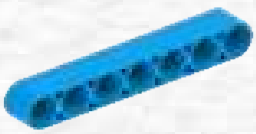
P5	2 pcs.



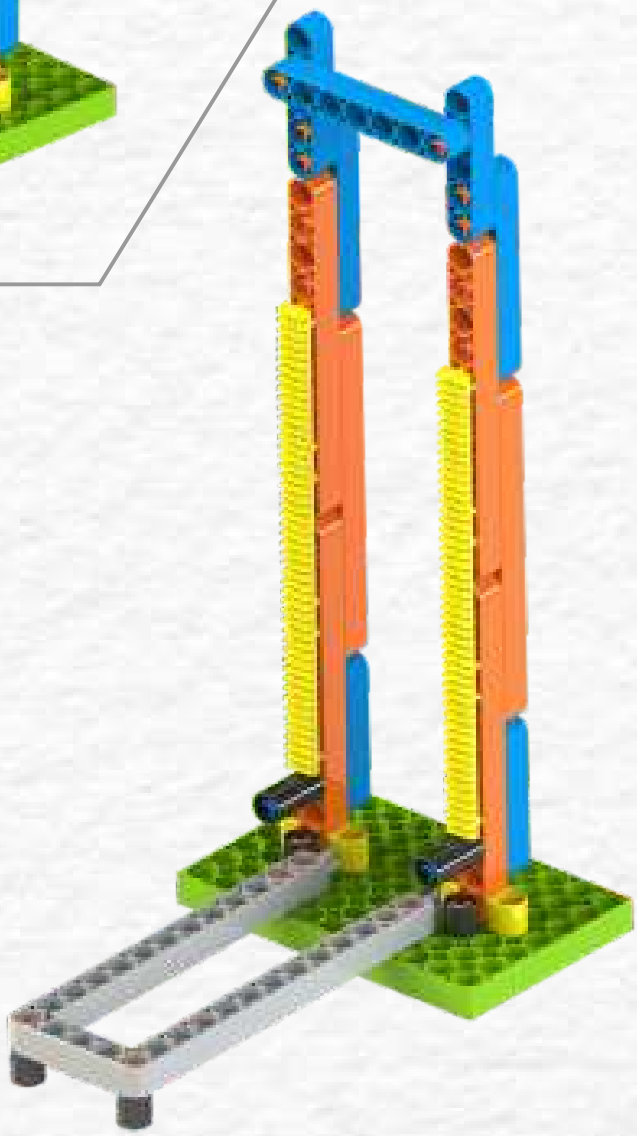
STEP - 10



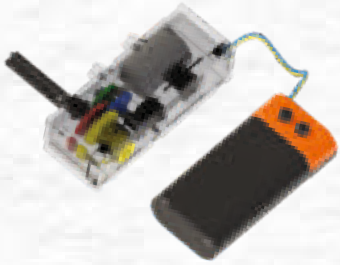
Rack	6 pcs.



P7	1 pc.



STEP - 11

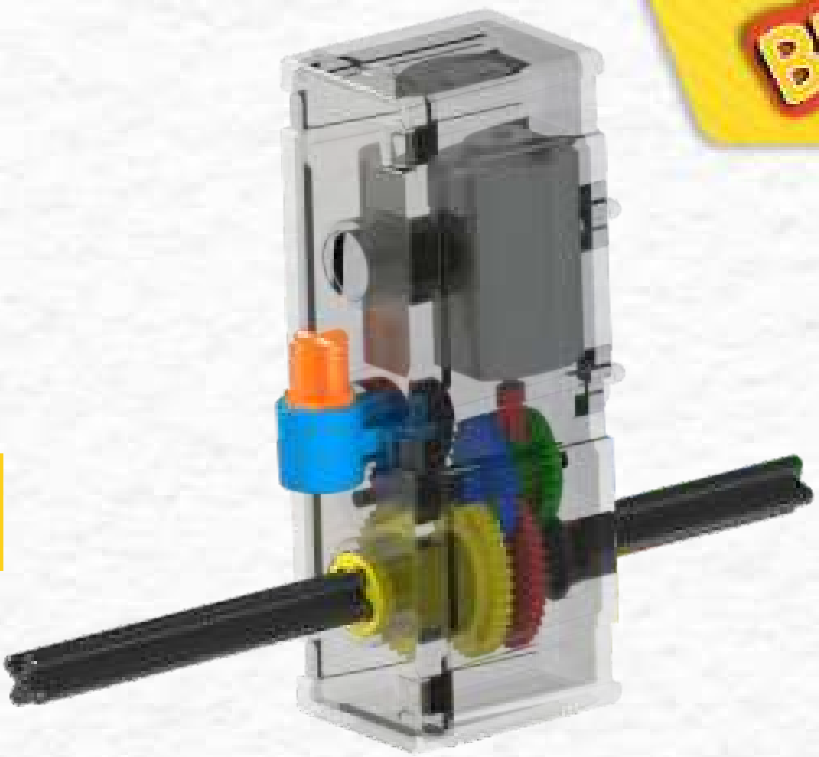


Motor With Battery Box 1 pc.

CT2	1 pc.

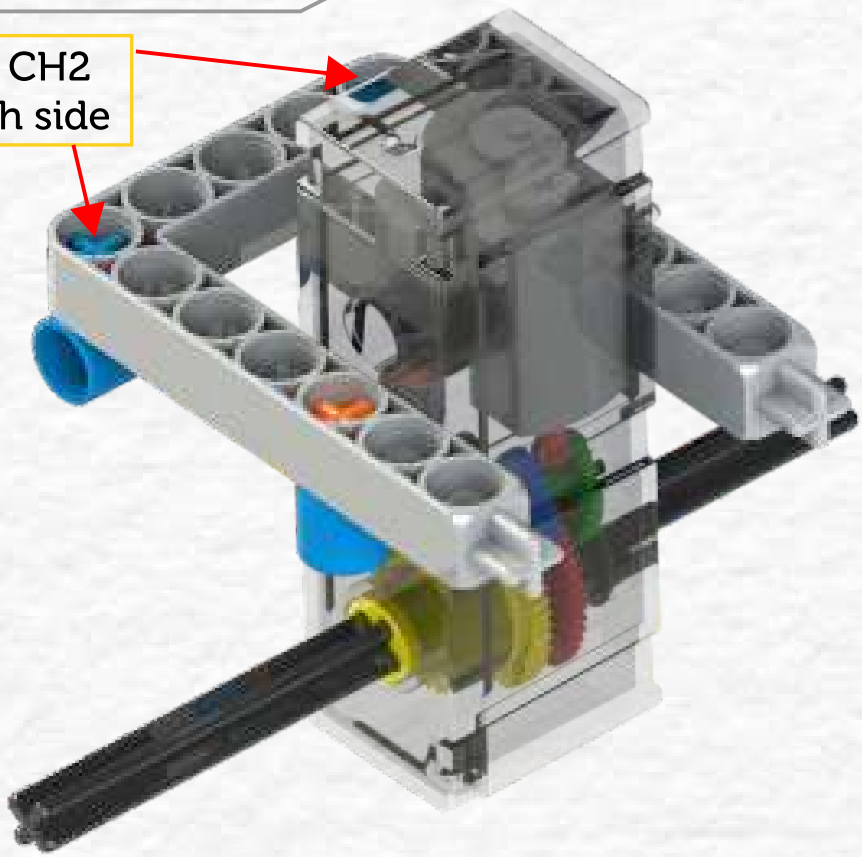
SH60	1 pc.

CH2	1 pc.



STEP - 12

Attach CH2 on both side



CH2	2 pcs.

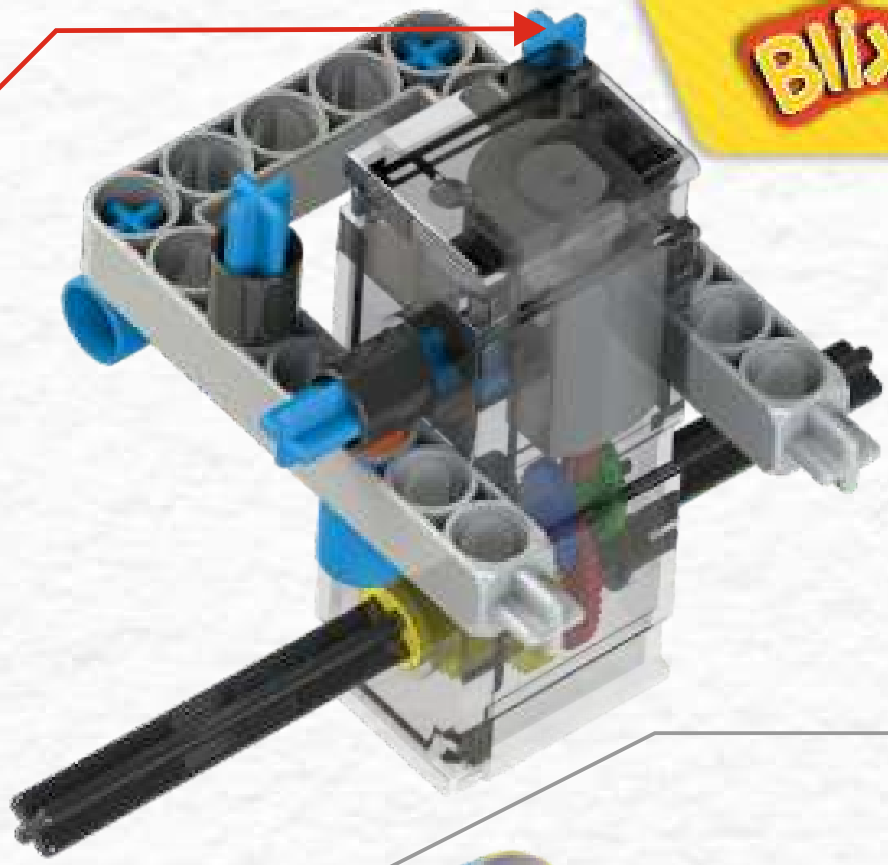
PU5 X 7	1 pc.

STEP - 13

Attach CT3 & TW1 on 3rd hole

TW1	3 pcs.

CT3	3 pcs.

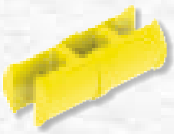


STEP - 14

TW1	2 pcs.

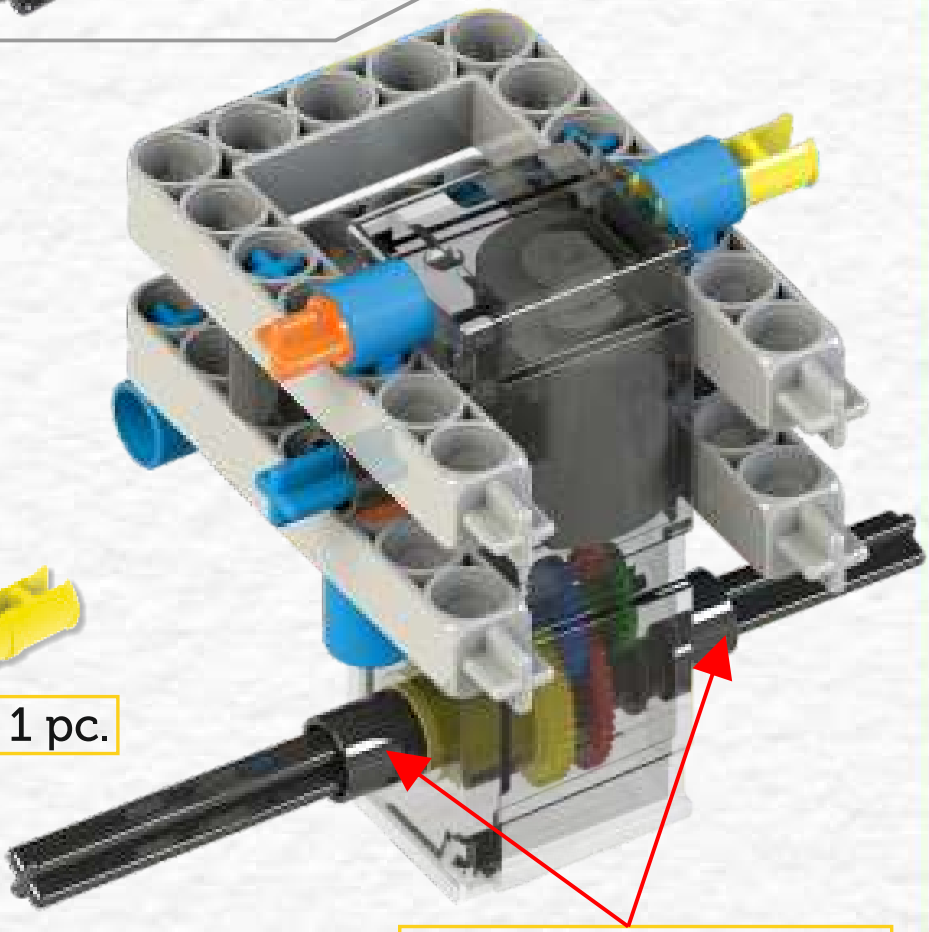
CH2	2 pcs.

CT2	1 pc.



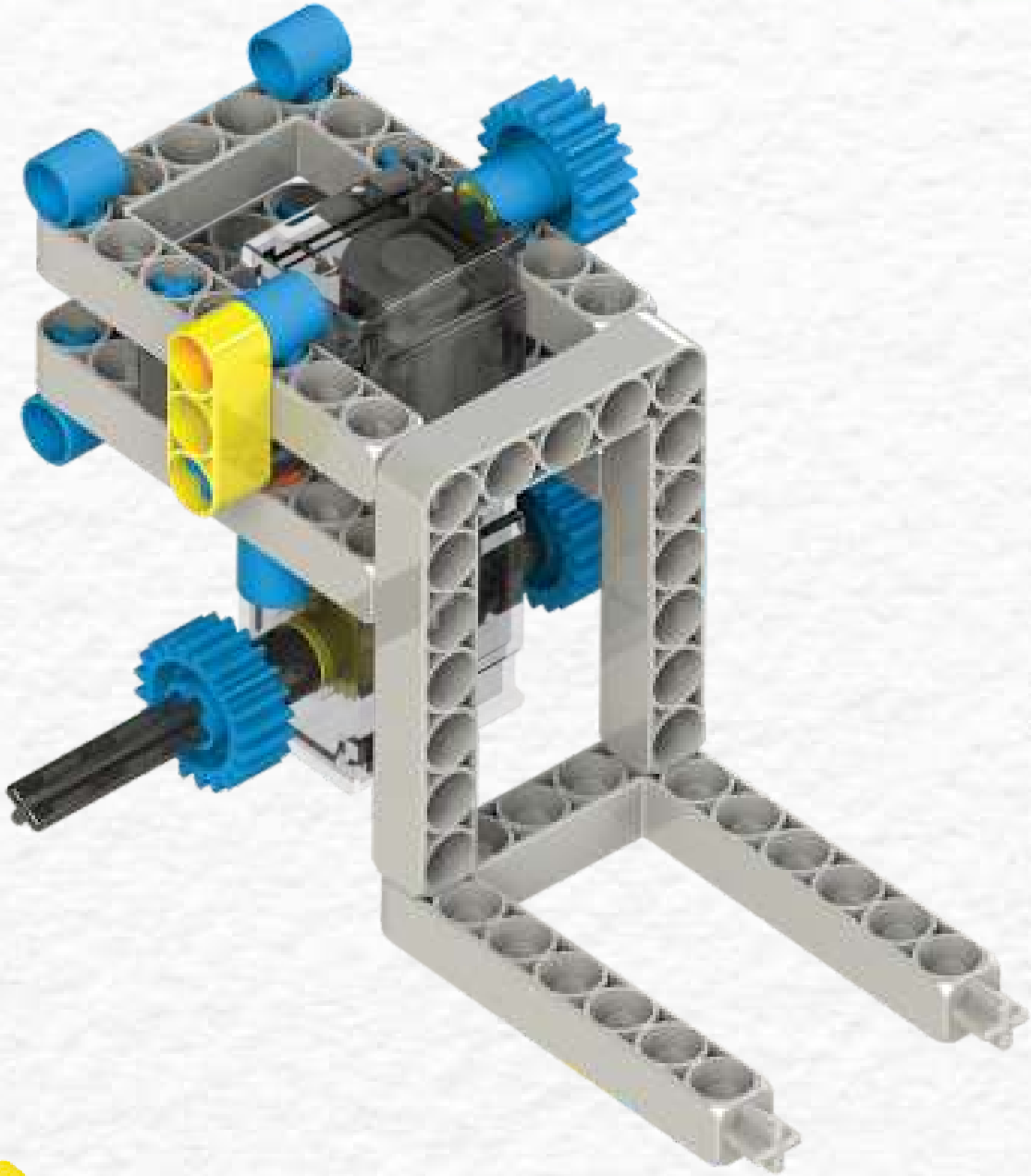
CL2	1 pc.

PU5 X 7	1 pc.



Add TW1 on both sides

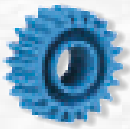
STEP - 15



P3	1 pc.

CH2	2 pcs.

G(20)	2 pcs.



G(20) Idler	1 pc.

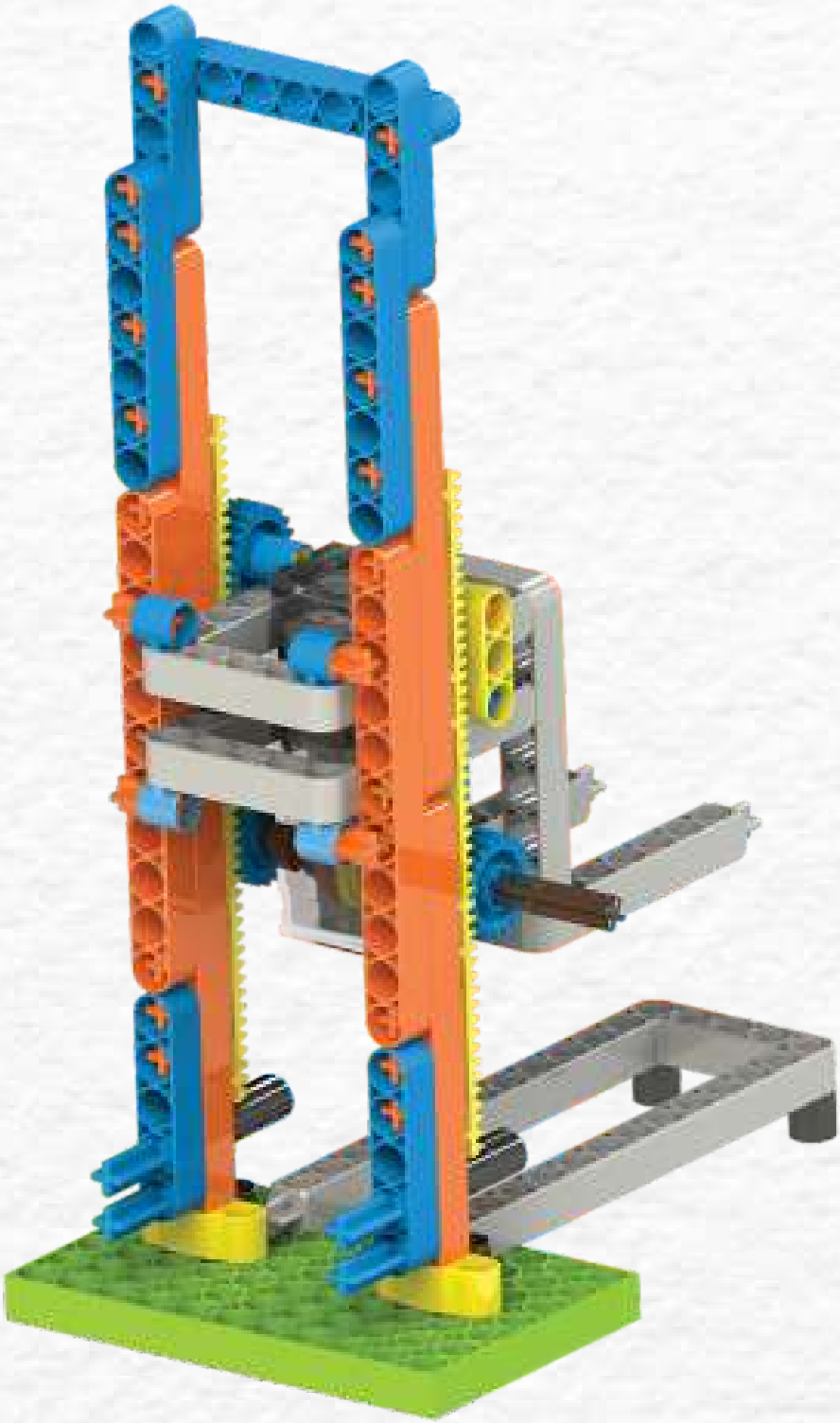
PU5 X 7	2 pcs.

STEP - 16

Assembly of step 15 & step 10

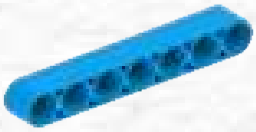
CT2	4 pcs.

CT3	4 pcs.



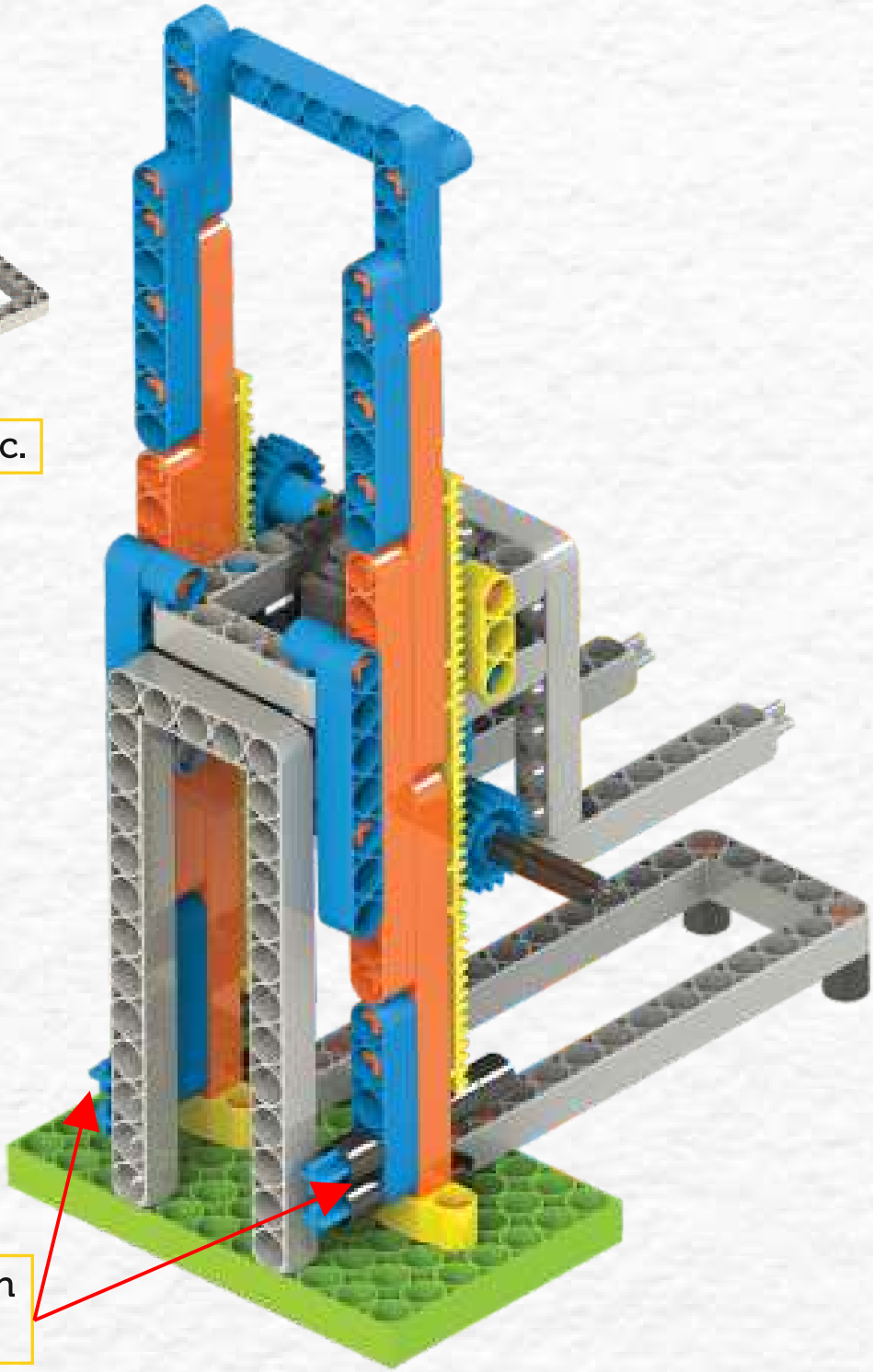
STEP - 17

TW1	4 pcs.



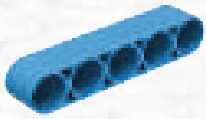
P7	2 pcs.

PU5 X 13	1 pc.

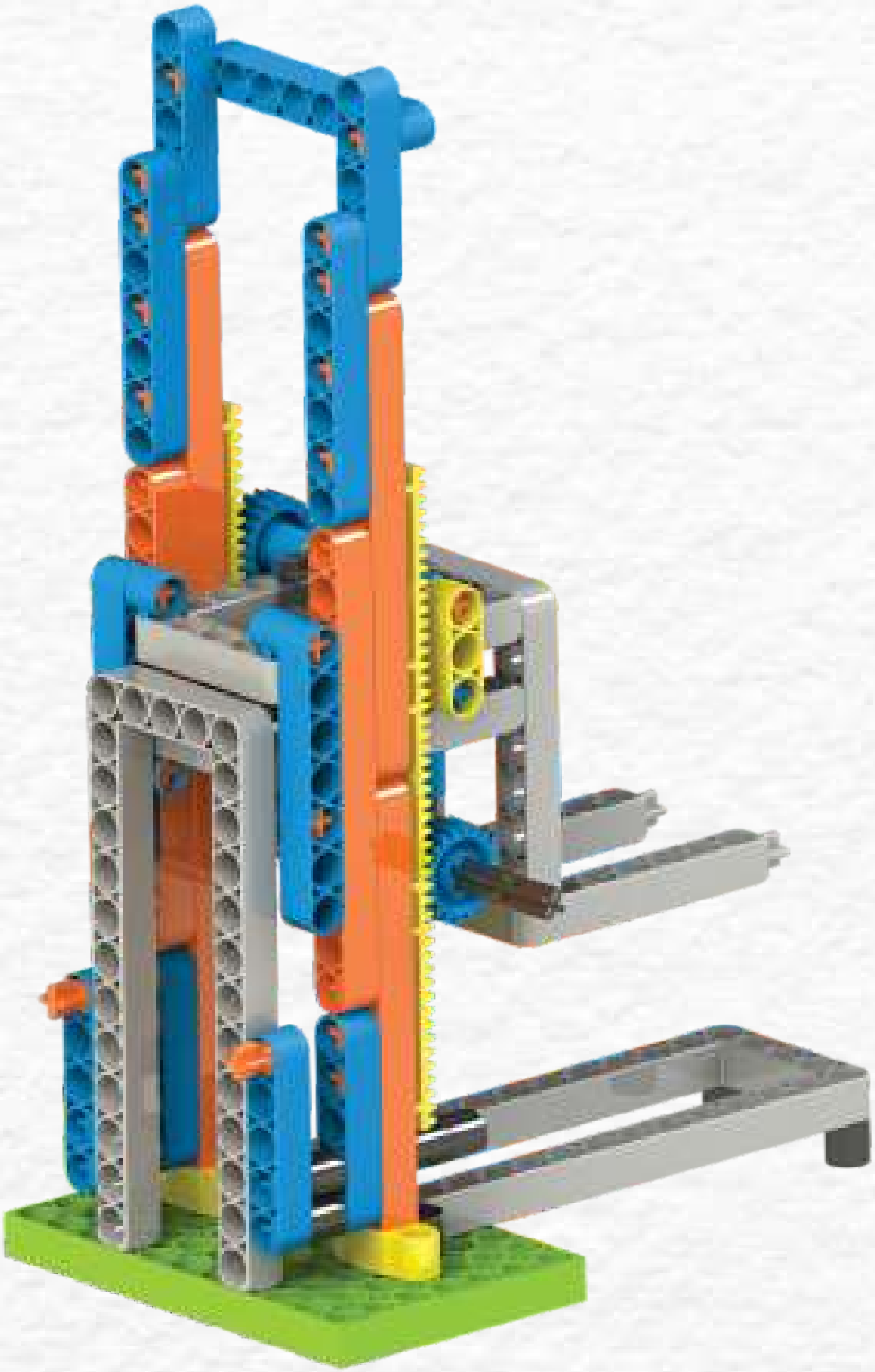


STEP - 18

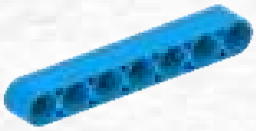
CT2	2 pcs.



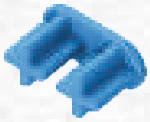
P5	2 pcs.



STEP - 19



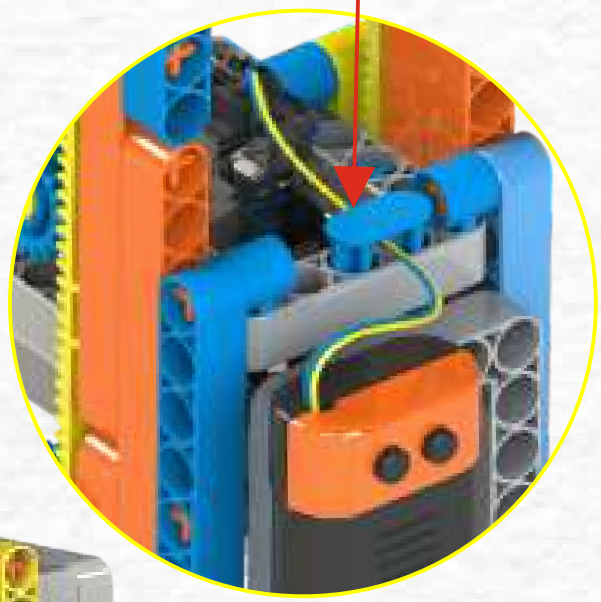
P7	1 pc.

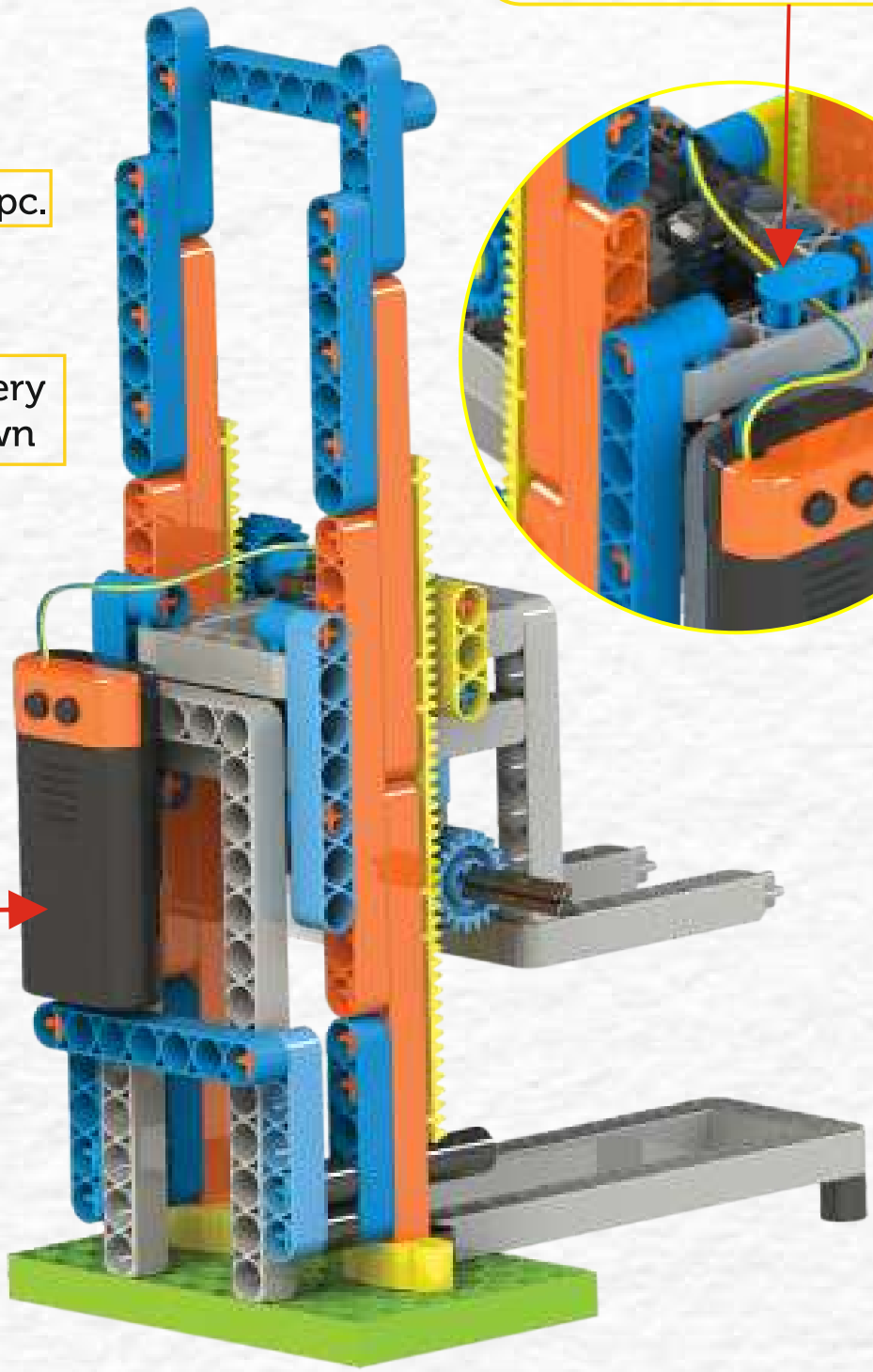


CT 1 X 2	1 pc.

Attach battery box as shown

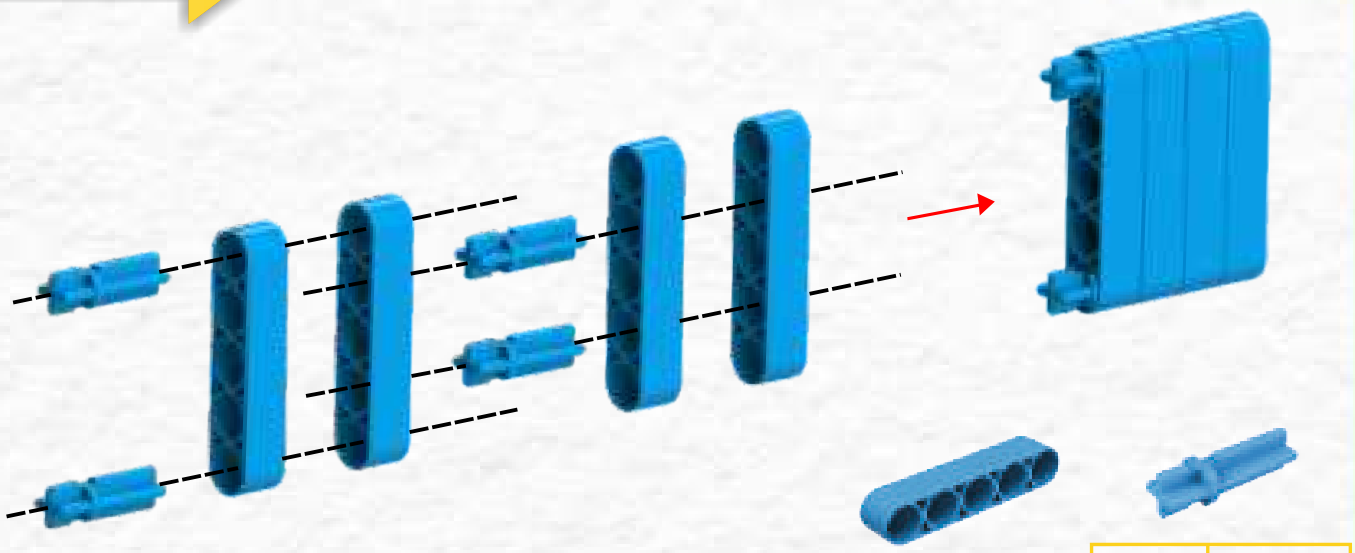
Use CT1 x 2 to prevent wire from coming in contact with moving parts





MODEL 3 - Vertical Gate

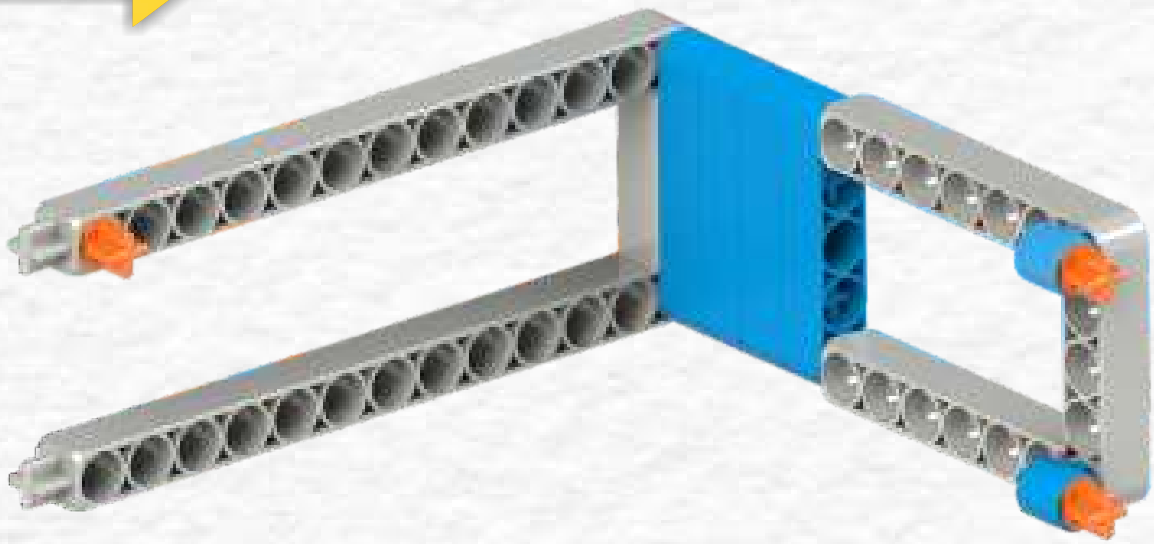
STEP - 1



P5	4 pcs.

CT3	4 pcs.

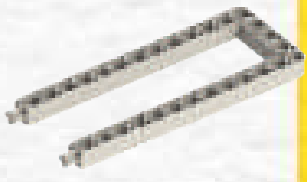
STEP - 2



CT2	3 pcs.

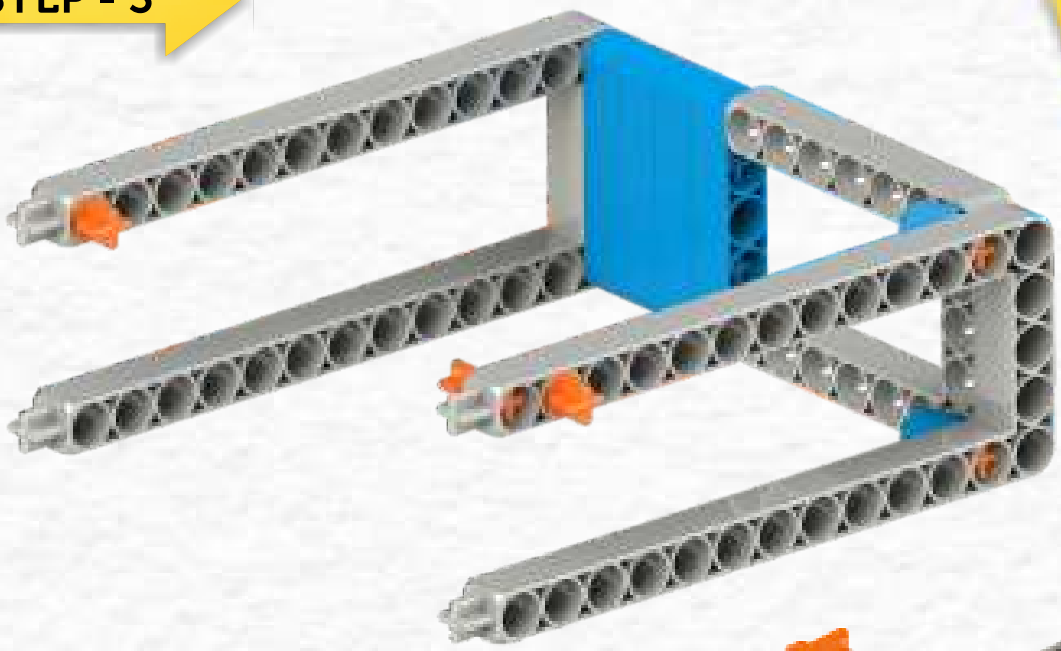
CH2	2 pcs.

PU5 X 7	1 pc.



PU5 X 13	1 pc.

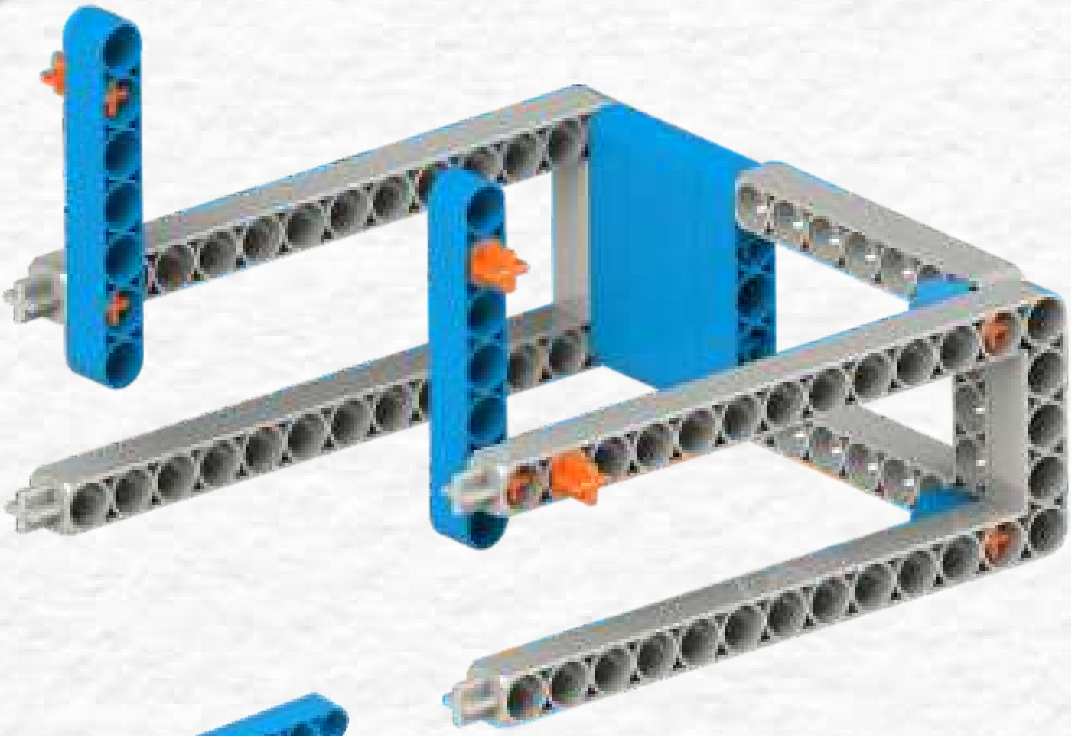
STEP - 3



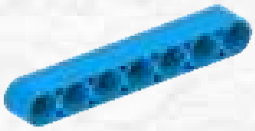
CT2	2 pcs.

PU5 X 13	1 pc.

STEP - 4



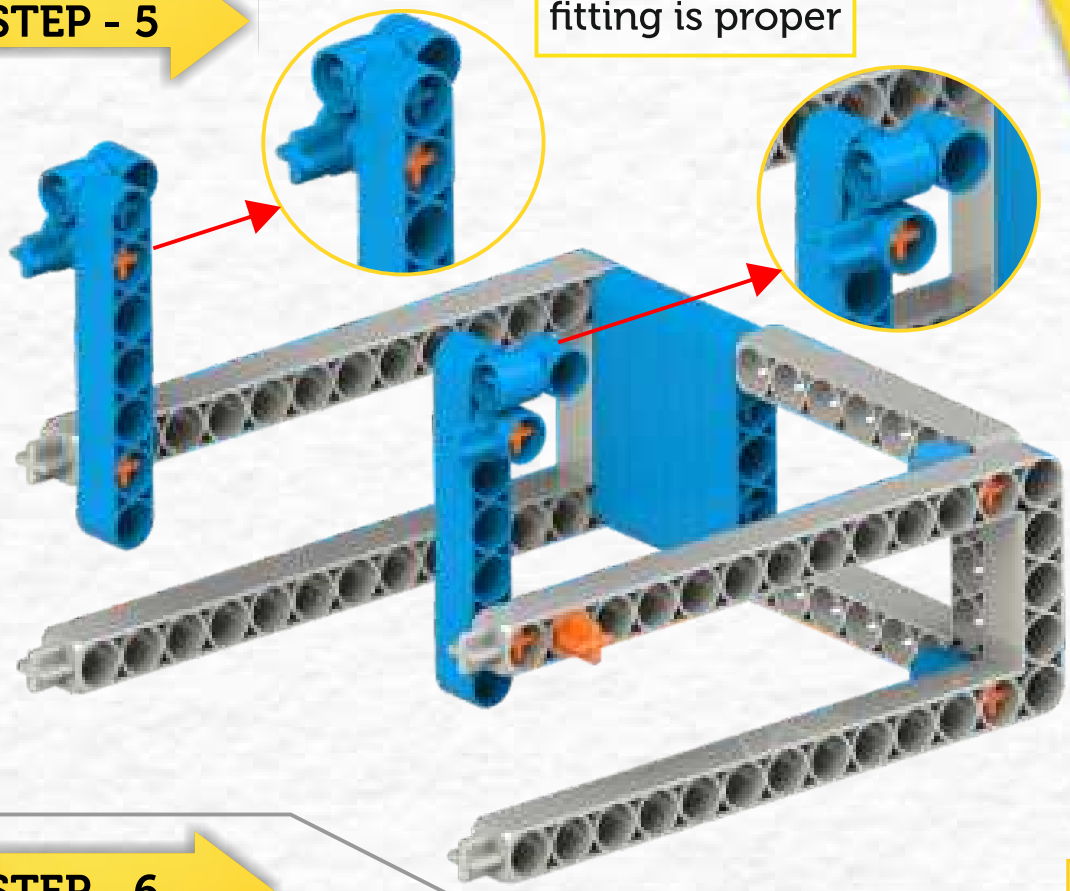
CT2	2 pcs.



P7	2 pcs.

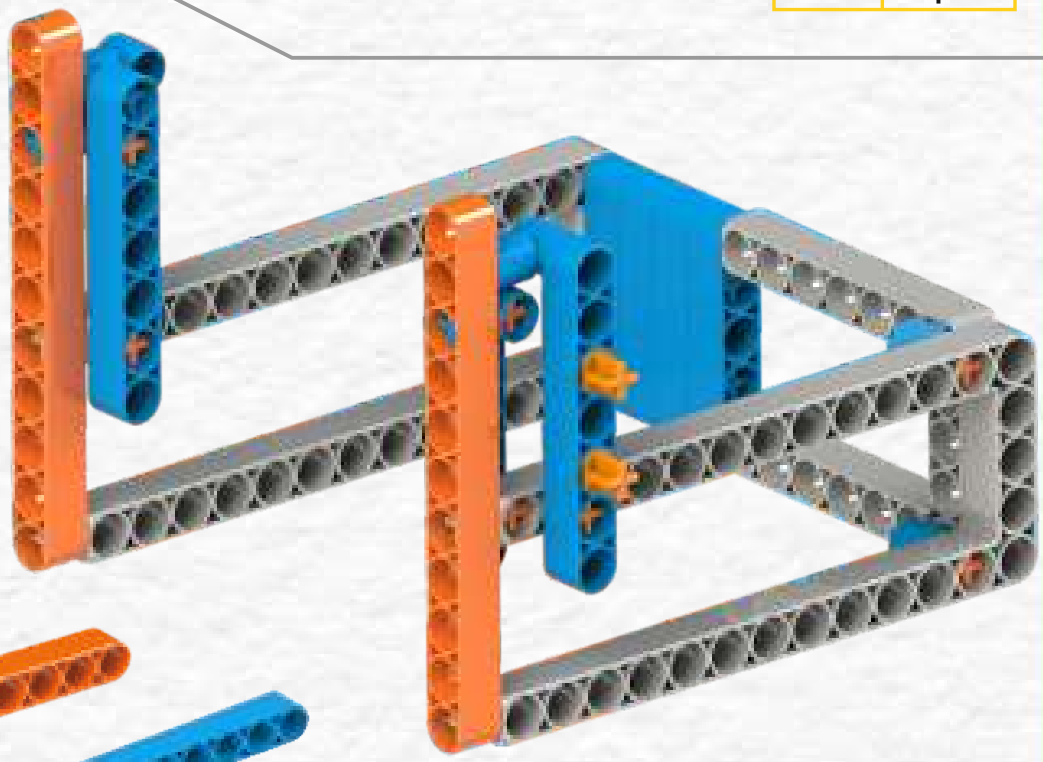
STEP - 5

Make sure this fitting is proper



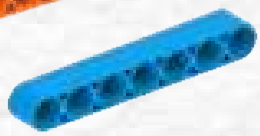
CH2	6 pcs.

STEP - 6



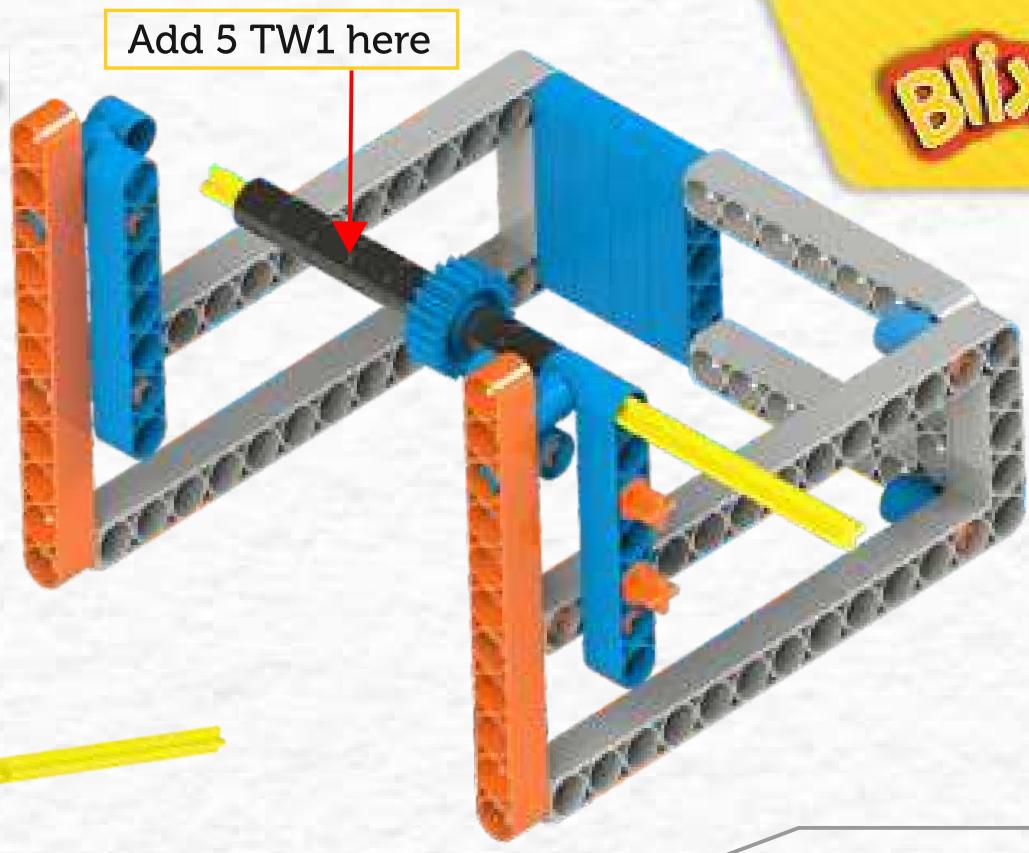
CT2	2 pcs.

P11	2 pcs.



P7	1 pc.

STEP - 7

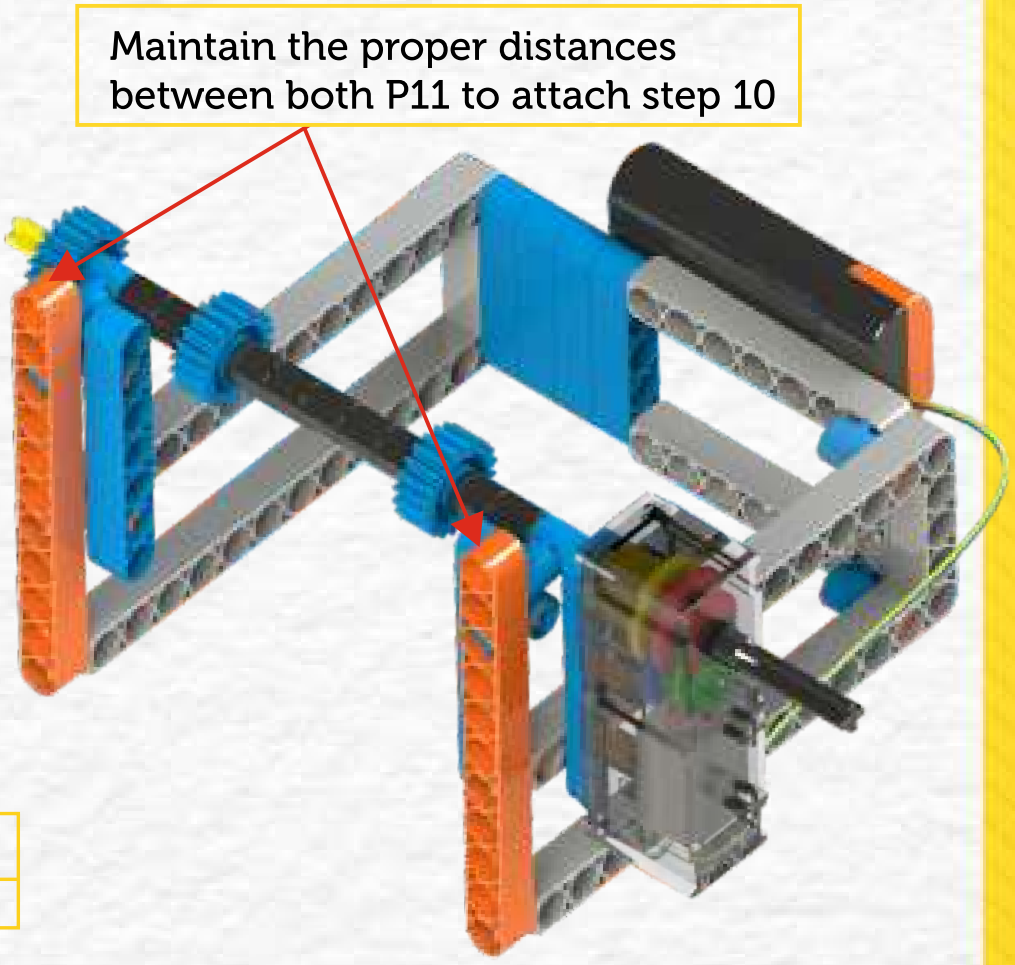


TW1	7 pcs.

G(20)	1 pc.

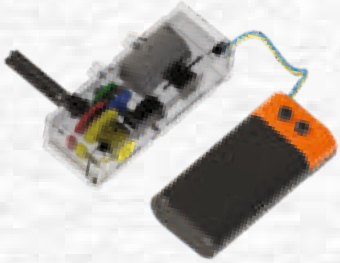
SH(170)	1 pc.

STEP - 8



TW1	2 pcs.

G(20)	2 pcs.



Motor With Battery Box 1 pc.

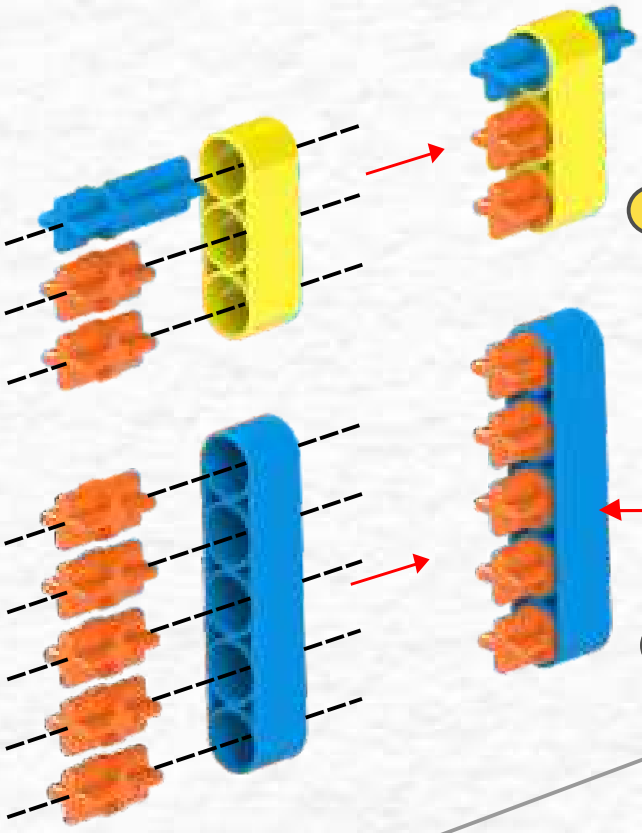
STEP - 9

P3	1 pc.

CT2	12 pcs.

CT3	1 pc.

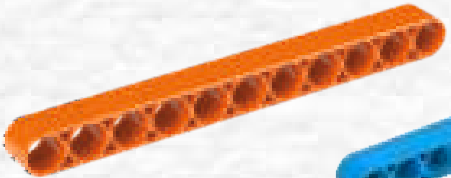
P5	2 pcs.



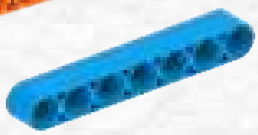
STEP - 10

P3	2 pcs.

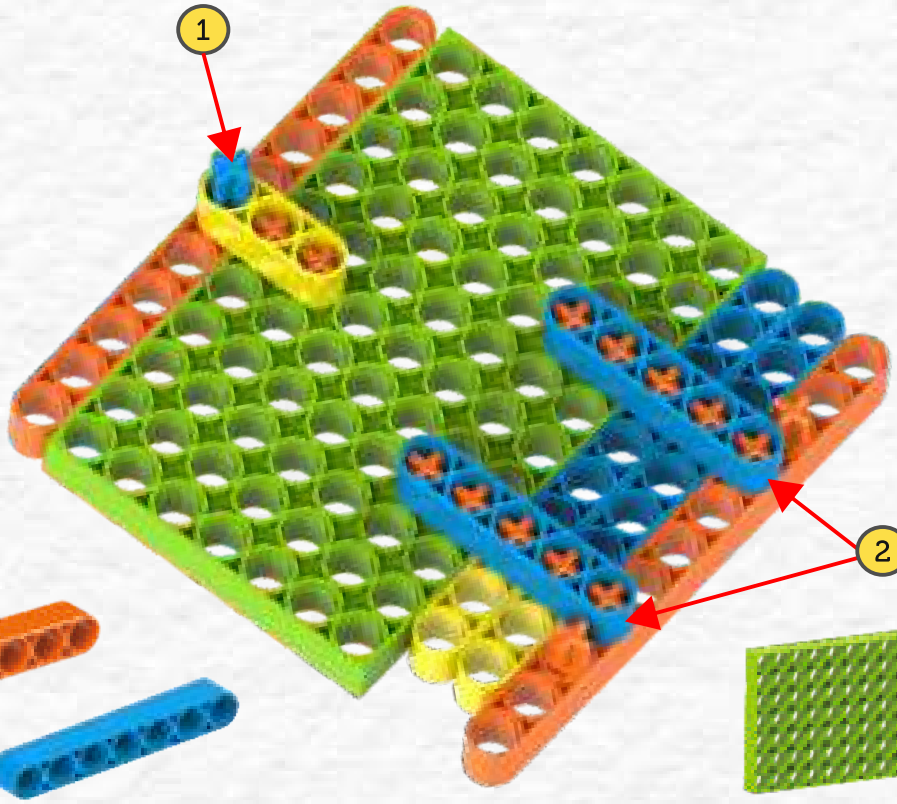
CT2	2 pcs.

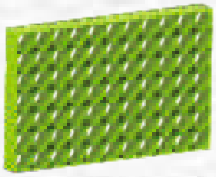


P11	2 pcs.



P7	2 pcs.

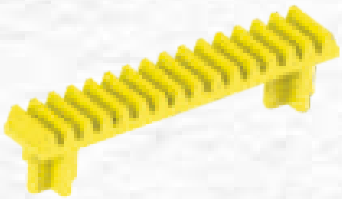




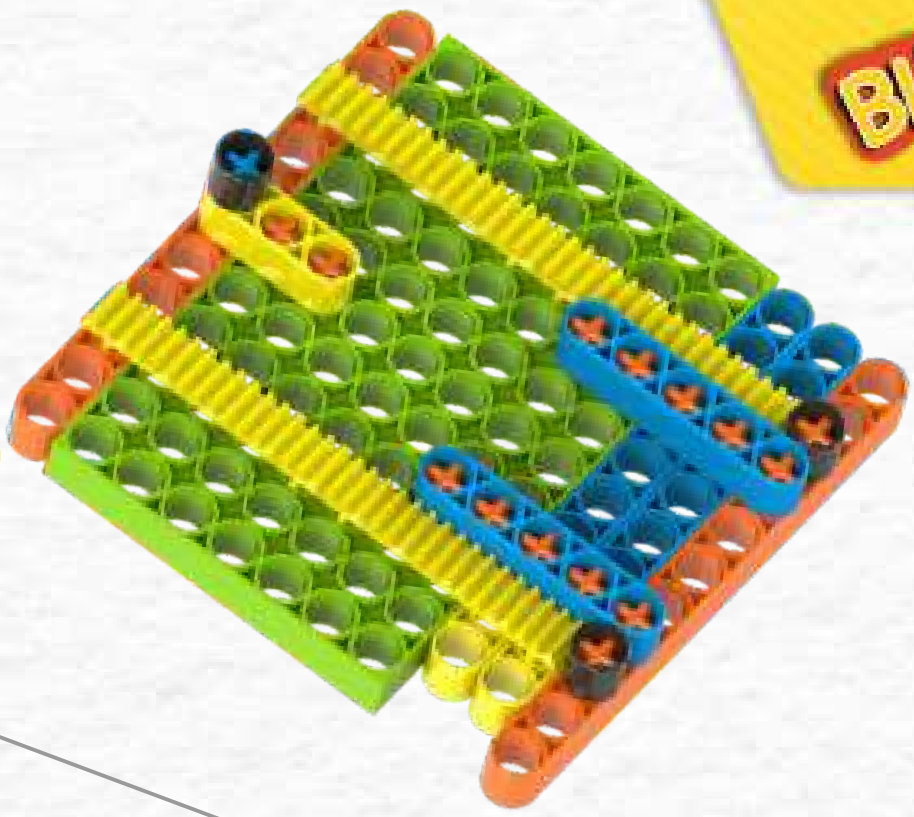
P7 X 11	1 pc.

STEP - 11

TW1	3 pcs.



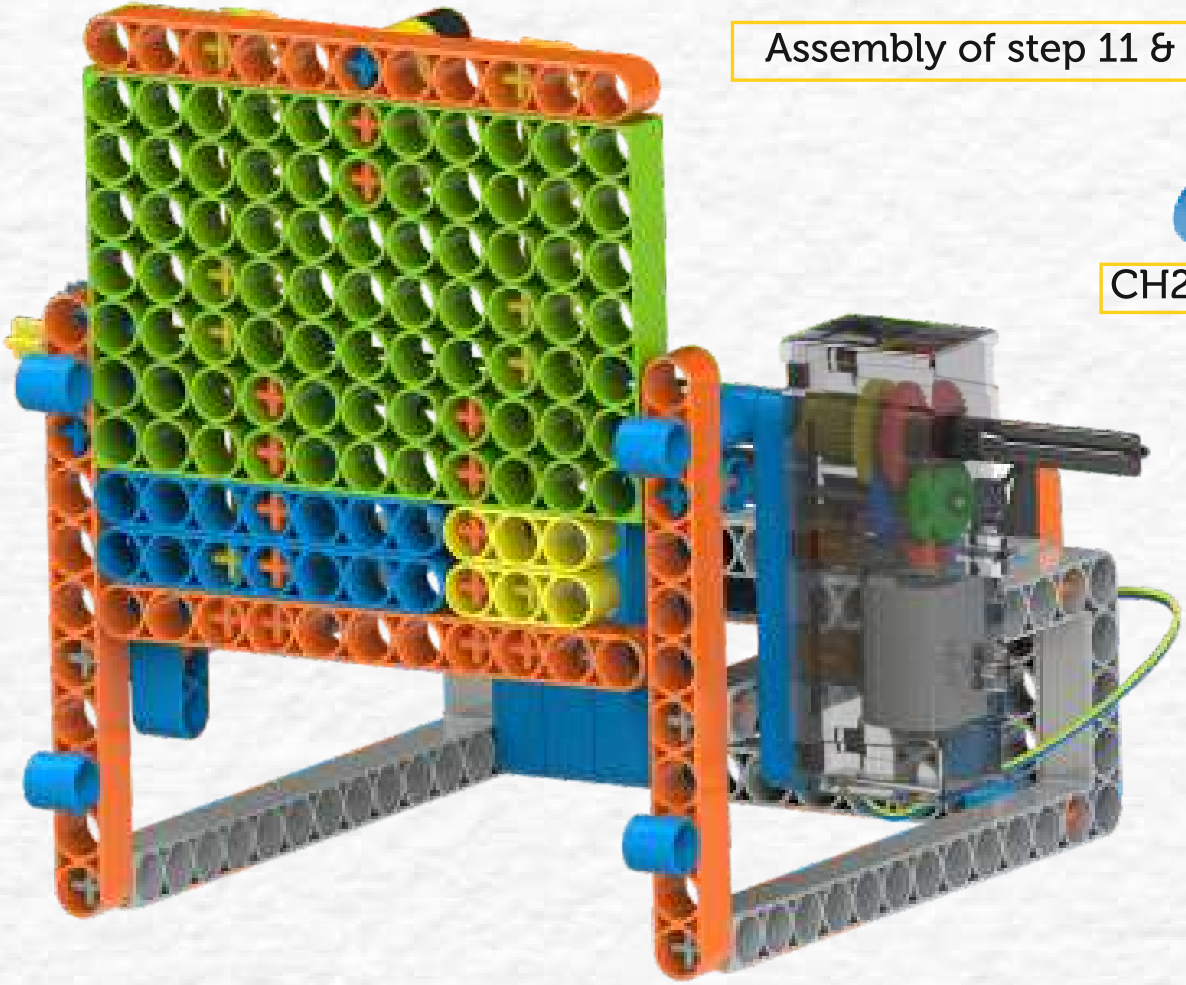
Rack	4 pcs.



STEP - 12

Assembly of step 11 & step 8

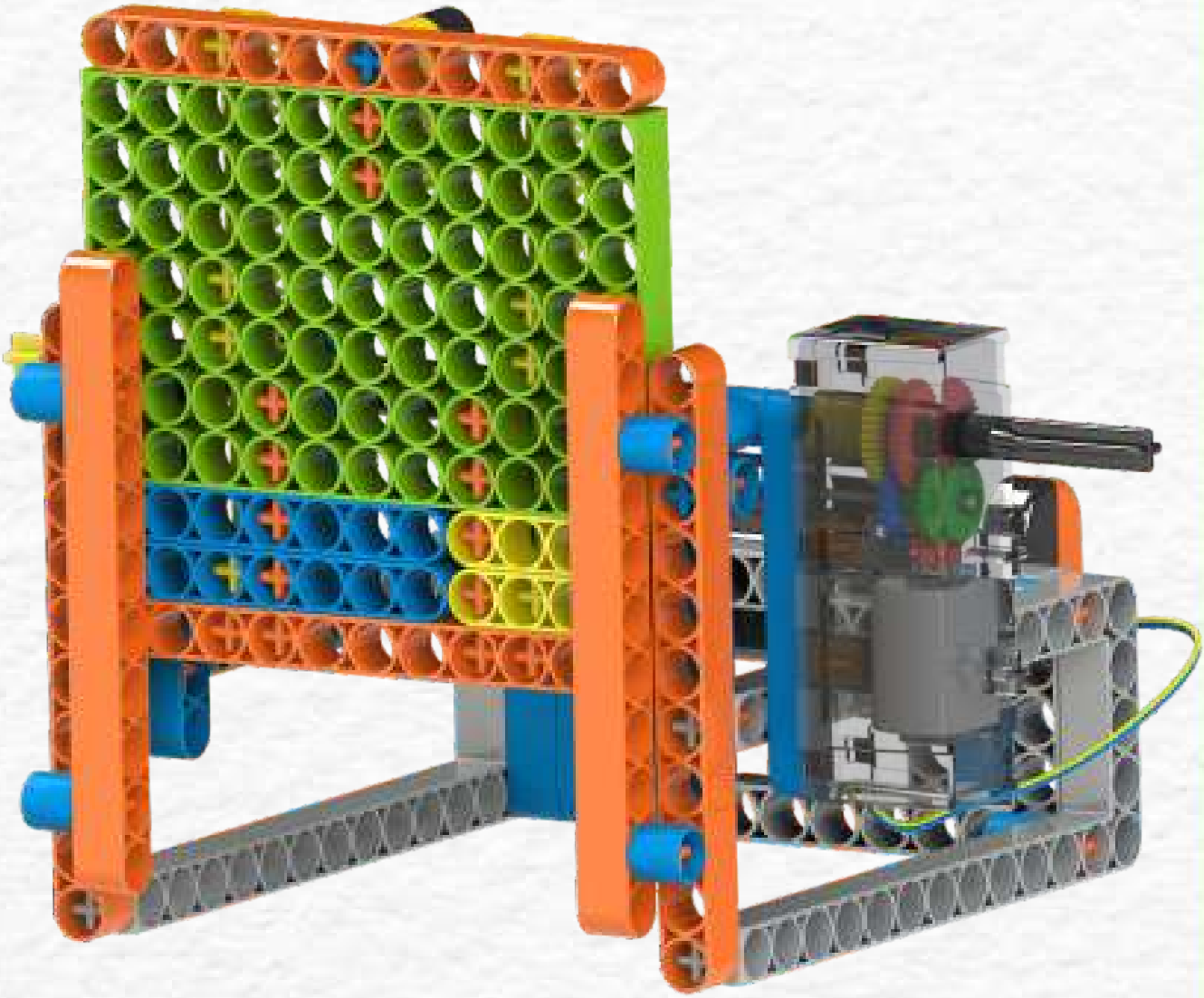
CH2	4 pcs.



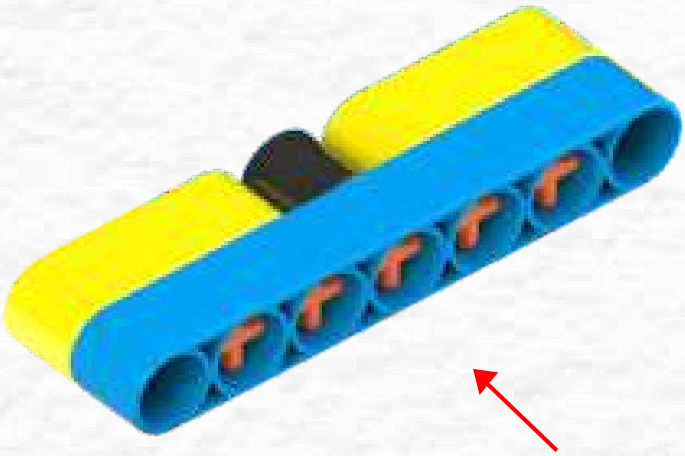
STEP - 13

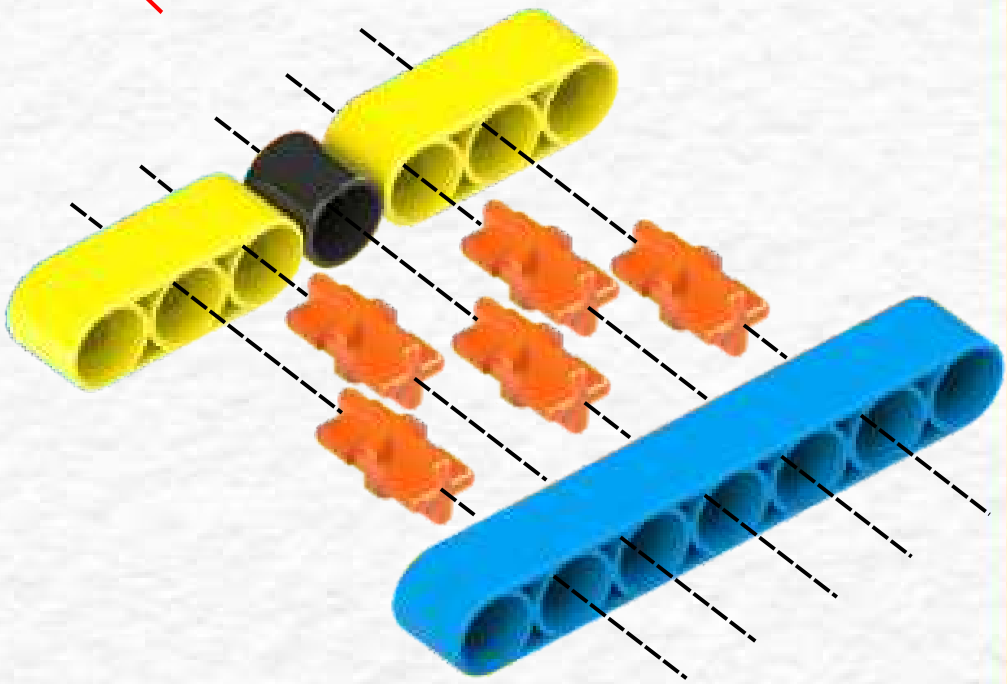
CT2	4 pcs.

P11	2 pcs.



STEP - 14

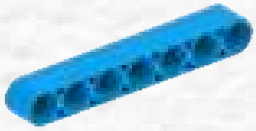




TW1	1 pc.

CT2	5 pcs.

P3	2 pcs.



P7	1 pc.

STEP - 15

TW1	4 pcs.

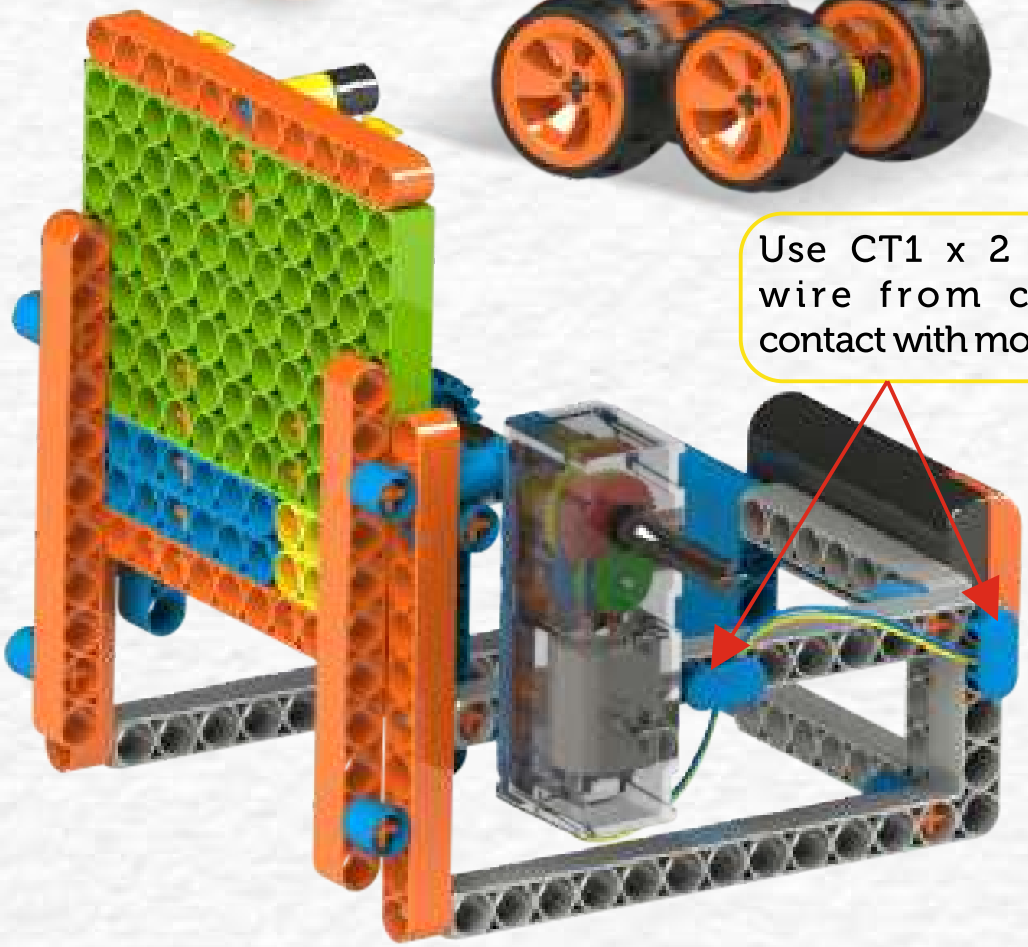
SH60	2 pcs.

STEP - 16

Wheels	4 pcs.

STEP - 17

CT 1 X 2	2 pcs.



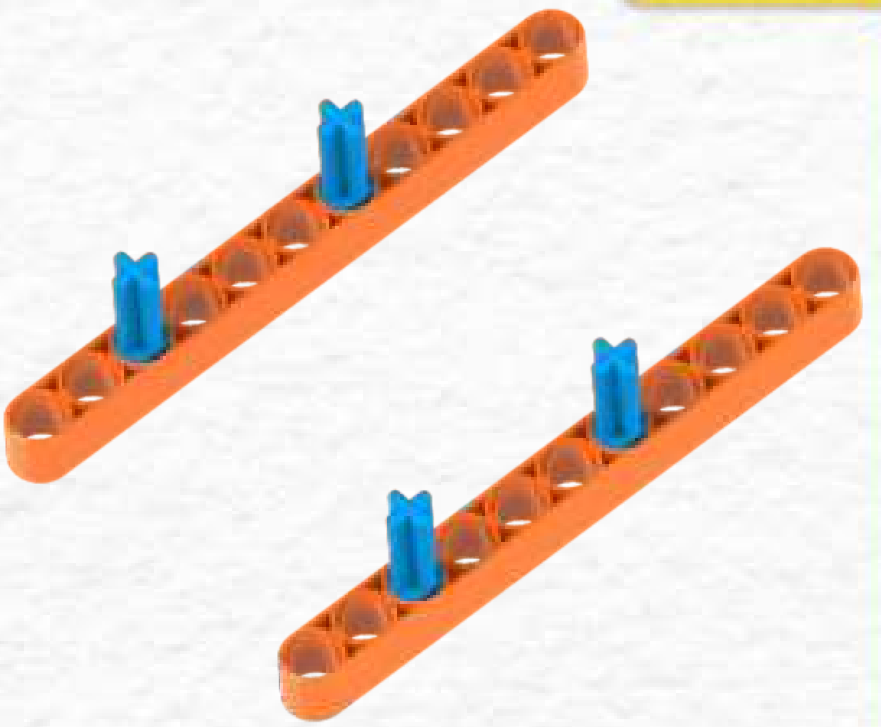
Use CT1 x 2 to prevent wire from coming in contact with moving parts

MODEL 4 - Grinding Machine

STEP - 1

CT3	4 pcs.

P11	2 pcs.

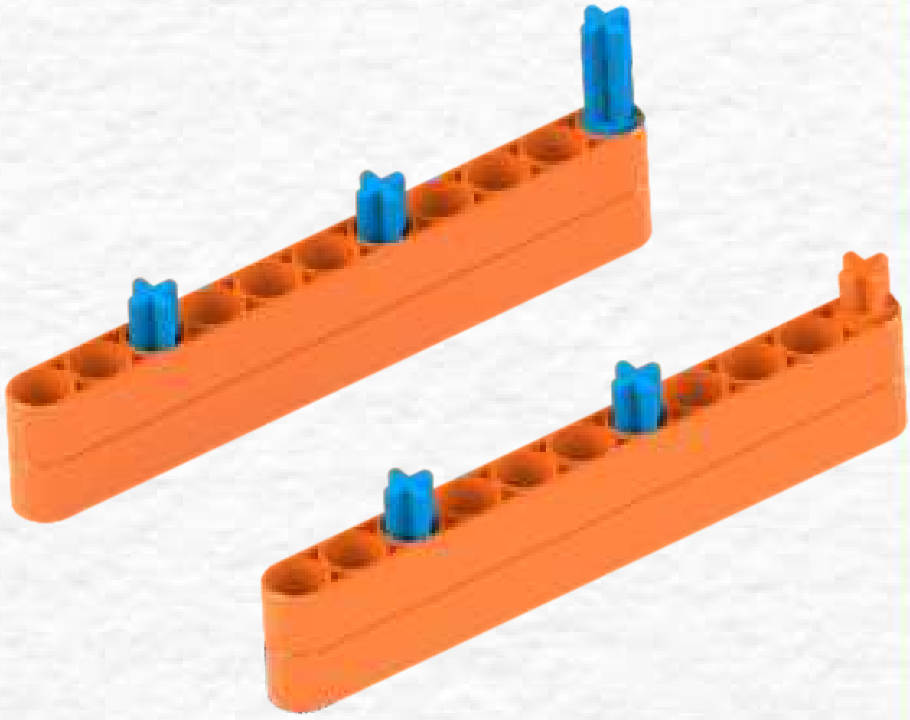


STEP - 2

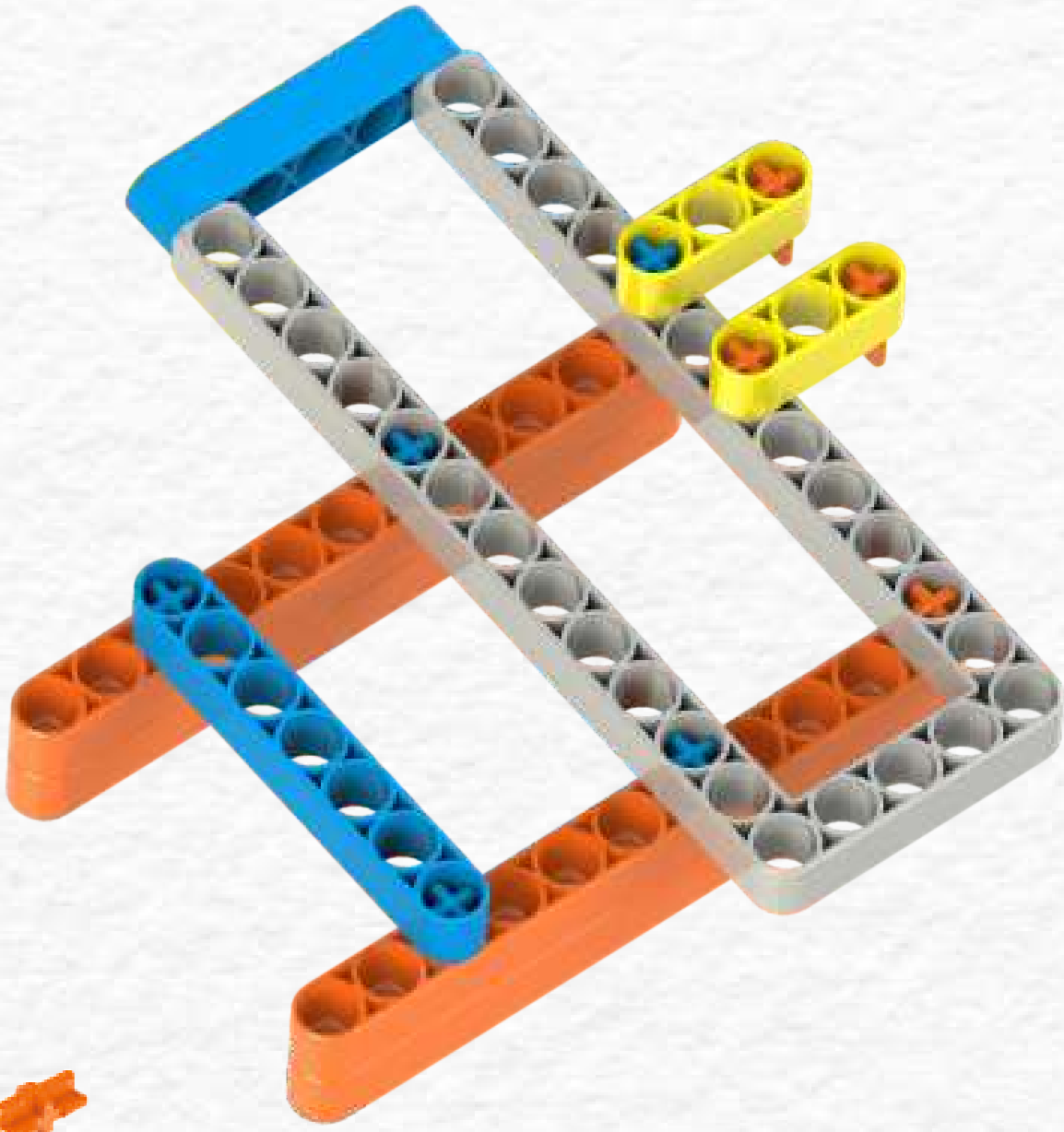
CT2	1 pc.

CT3	1 pc.

P11	2 pcs.



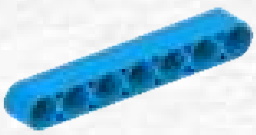
STEP - 3



CT2	3 pcs.

P3	2 pcs.

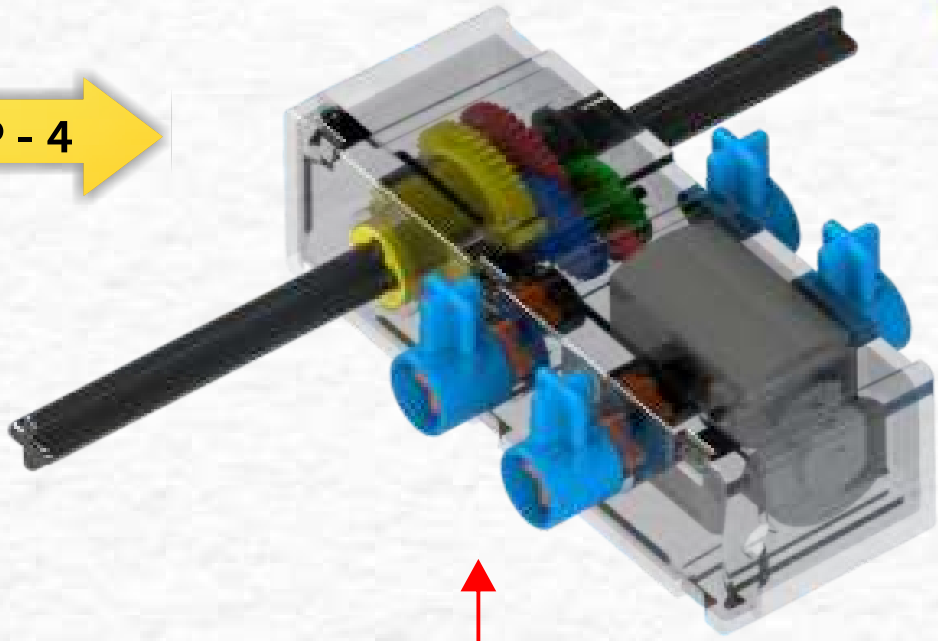
P5	1 pc.

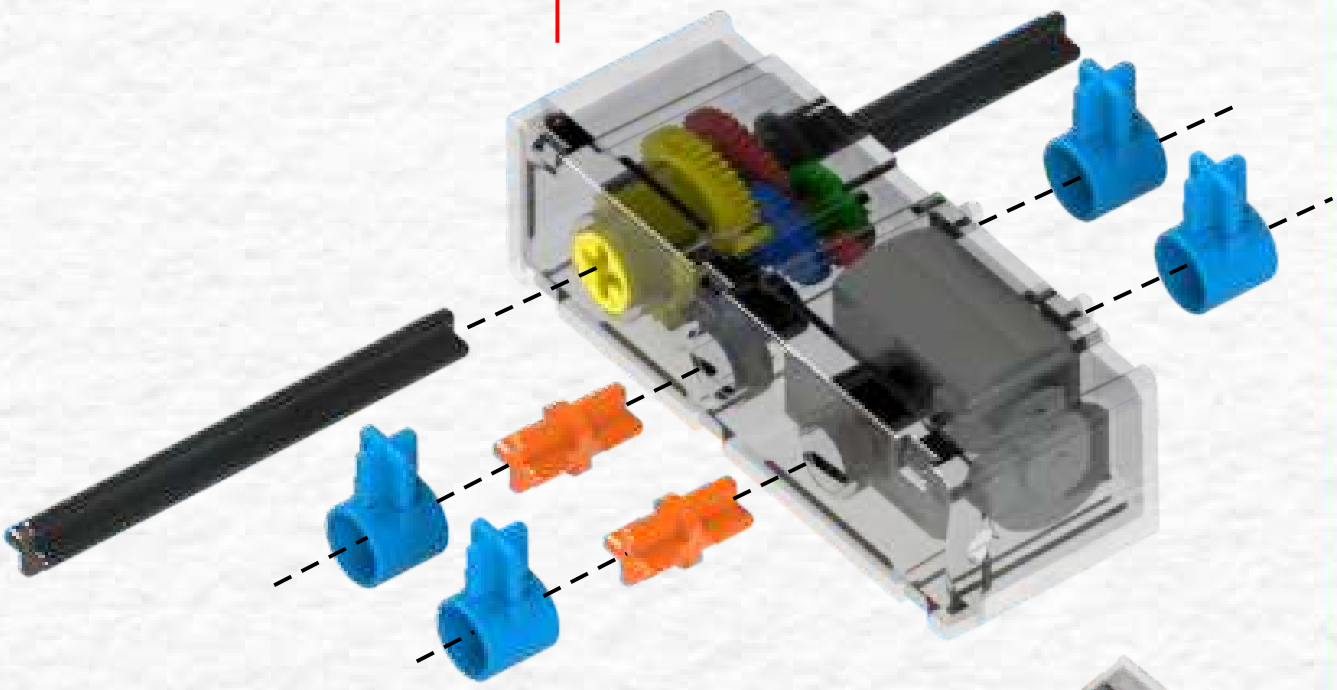


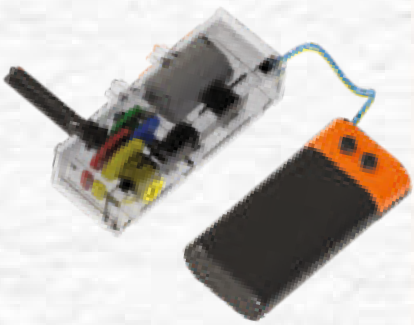
P7	1 pc.

PU5 X 13	1 pc.

STEP - 4







CH2	4 pcs.

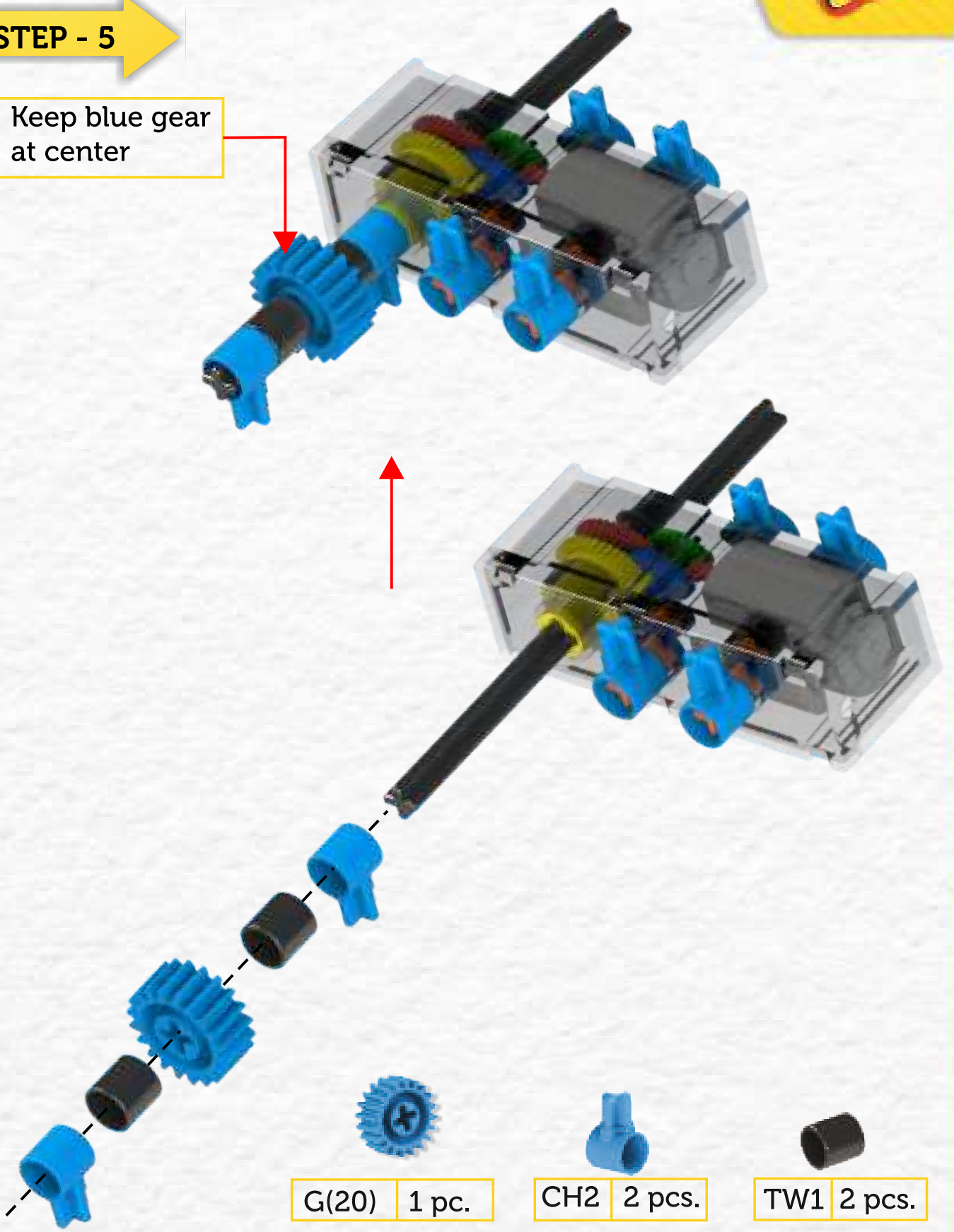
SH60	1 pc.

CT2	2 pcs.

Motor With Battery Box 1 pc.

STEP - 5

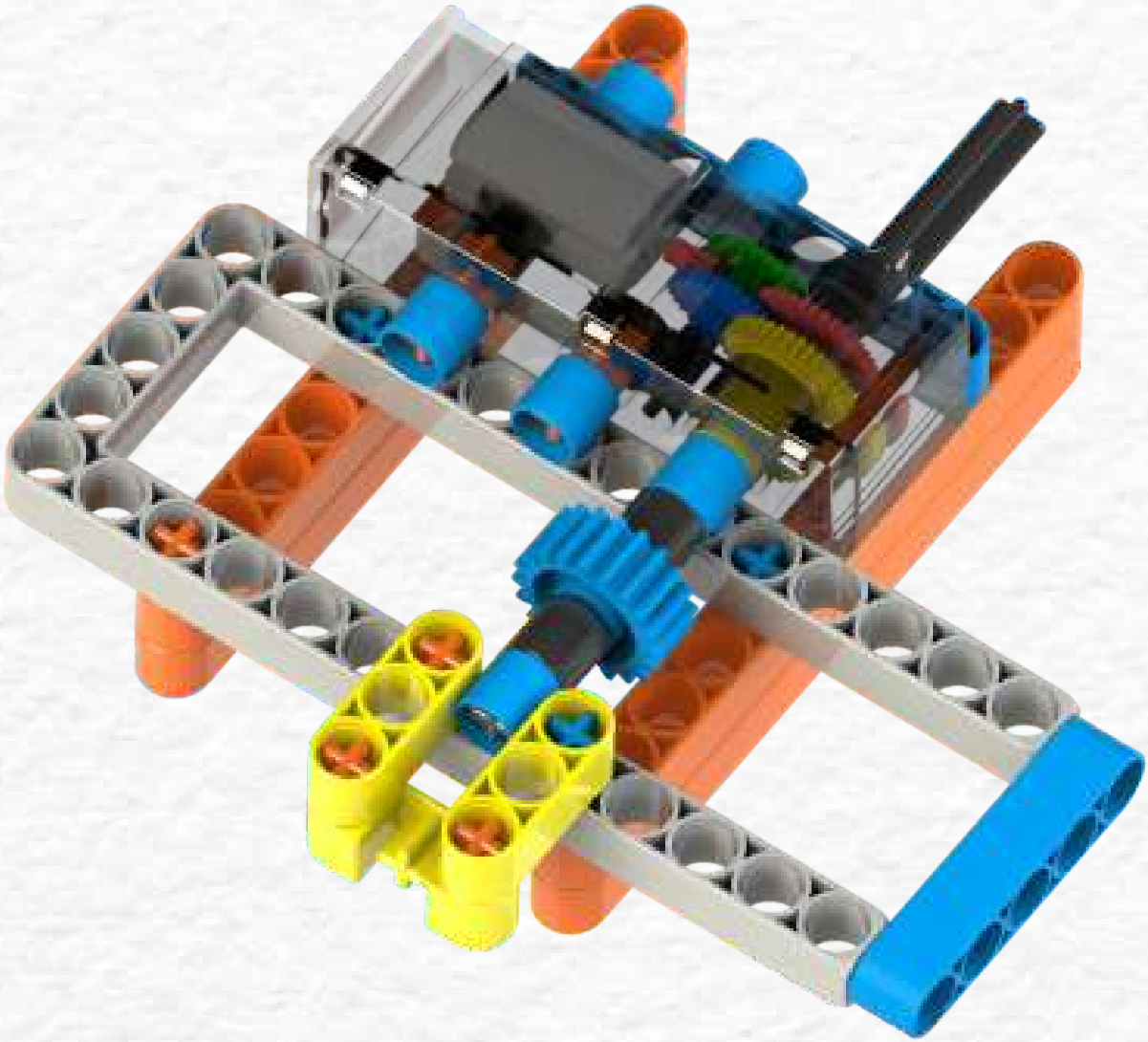
Keep blue gear at center



G(20)	1 pc.
CH2	2 pcs.
TW1	2 pcs.

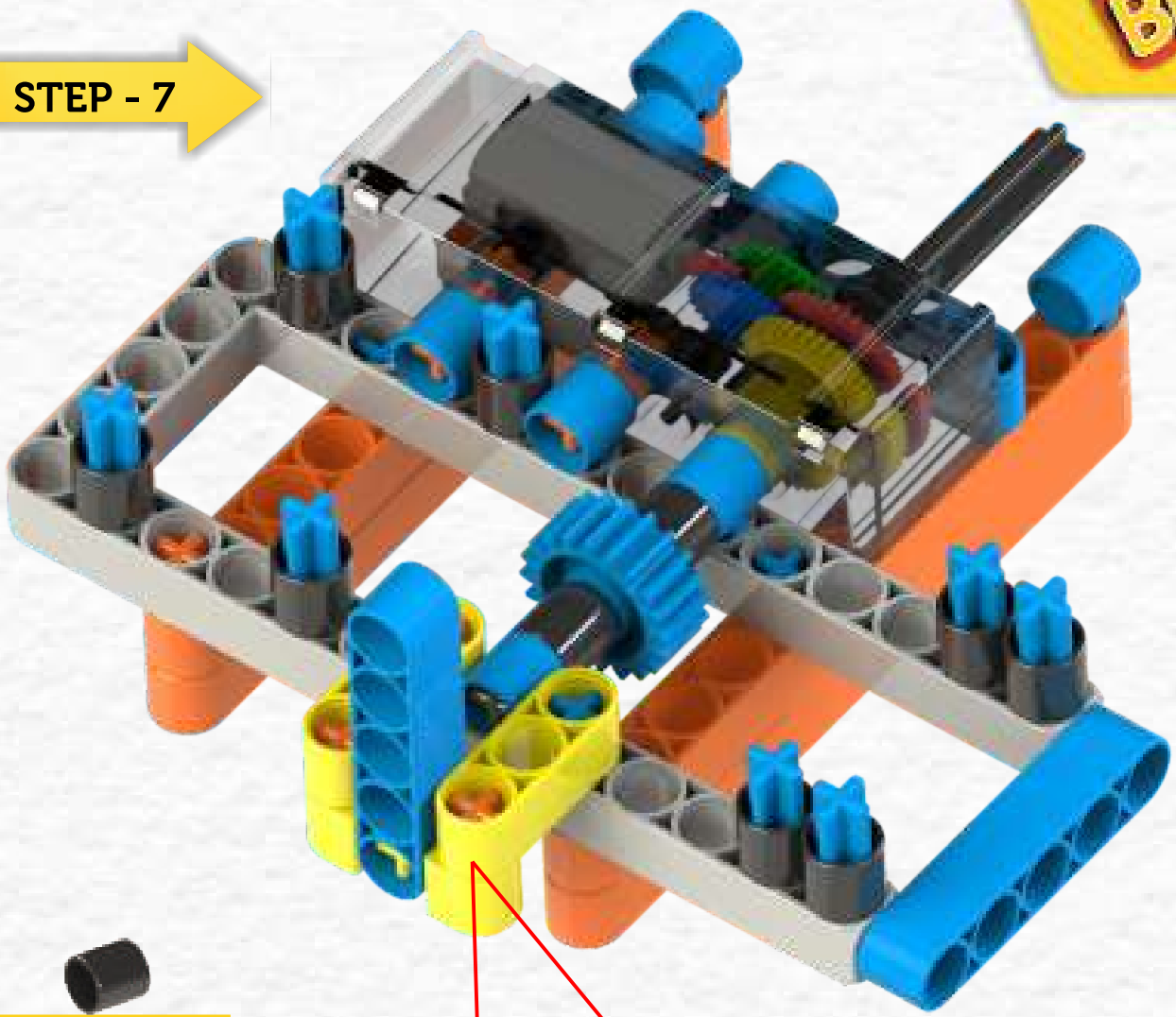
STEP - 6

Assembly of step 5 & step 3



PC 3	1 pc.

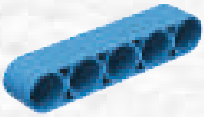
STEP - 7



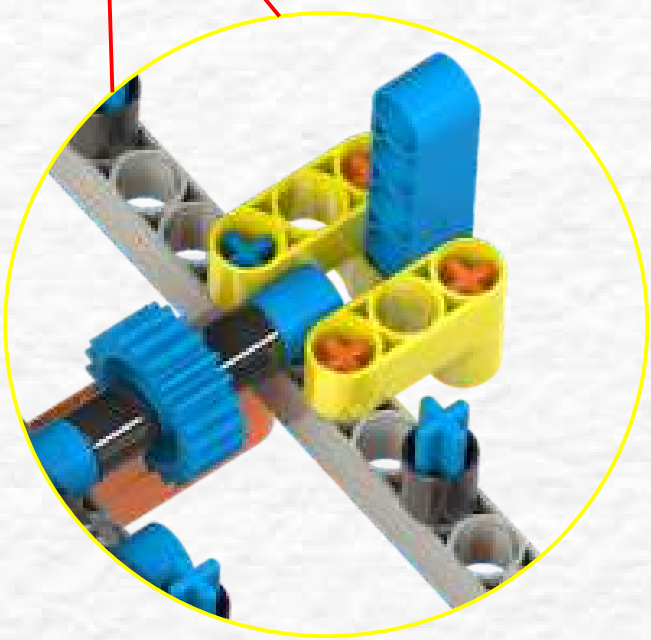
TW1	8 pcs.

CT3	8 pcs.

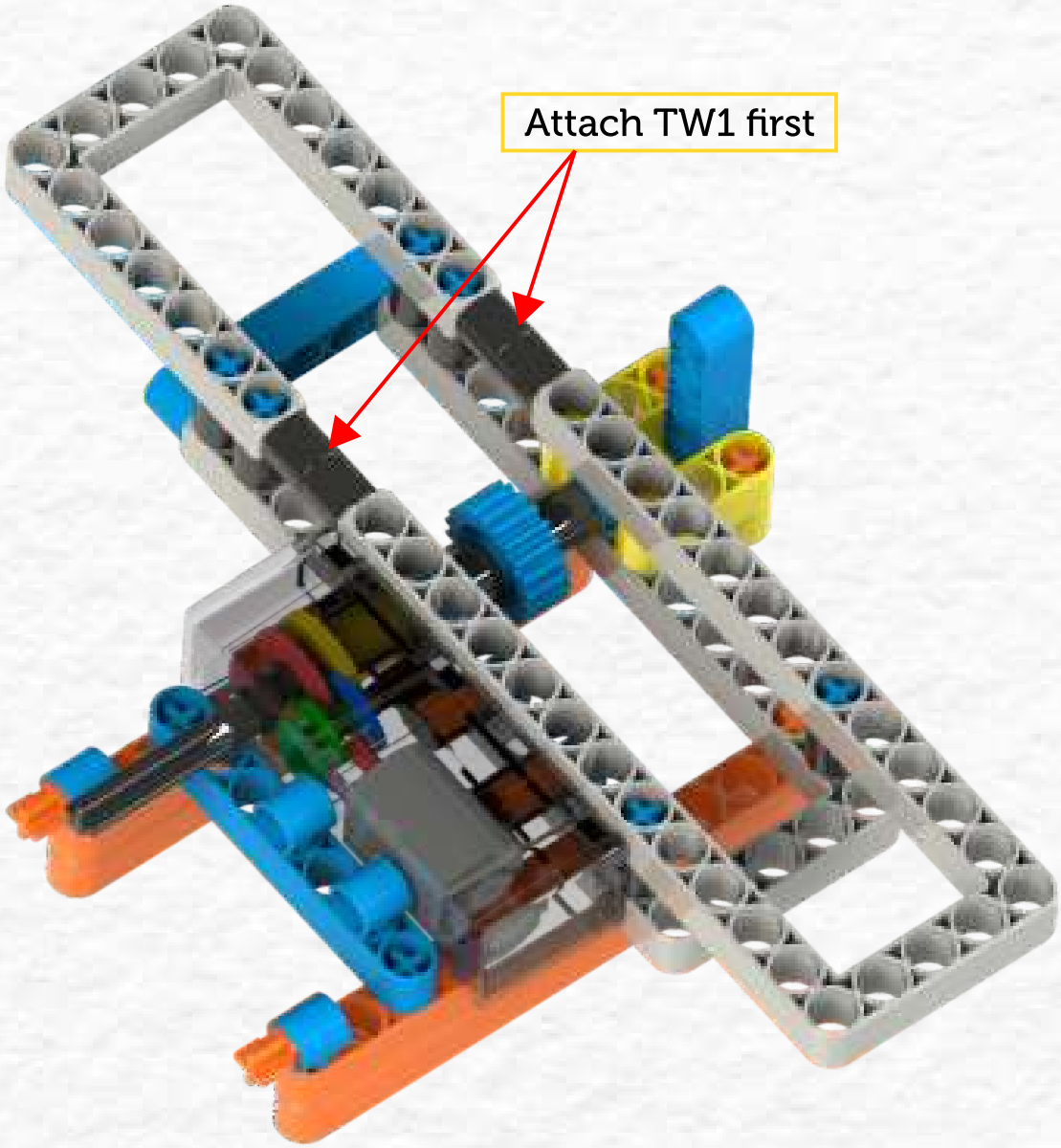
CH2	2 pcs.



P5	1 pc.



STEP - 8



TW1	4 pcs.

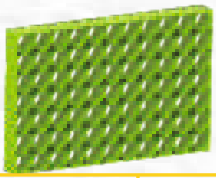
CT2	2 pcs.

PU5 X 7	1 pc.

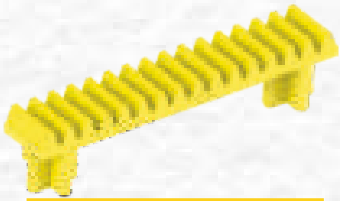
PU5 X 13	1 pc.

STEP - 9

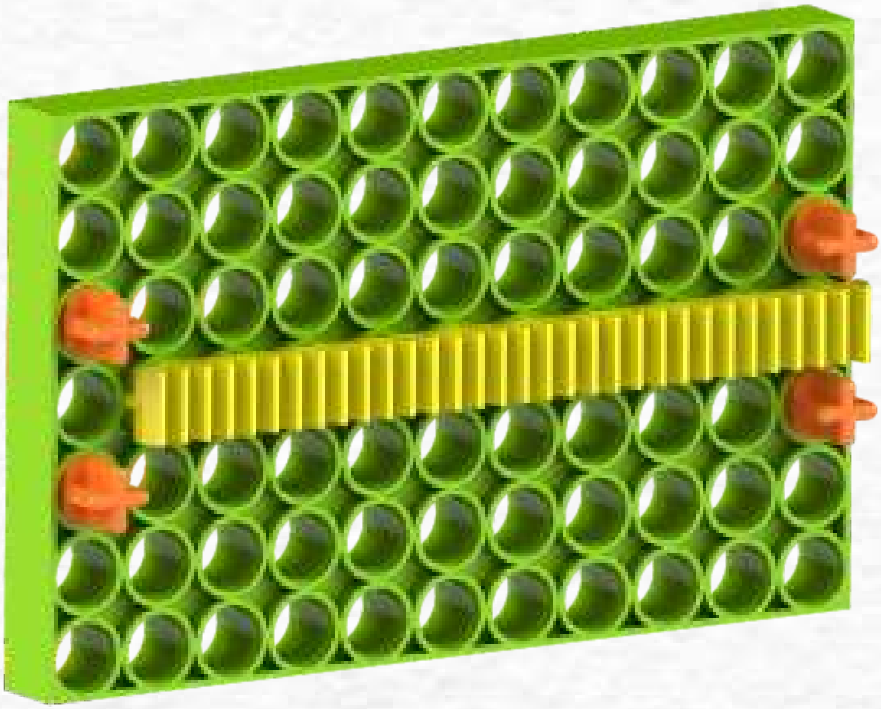
CT2	4 pcs.



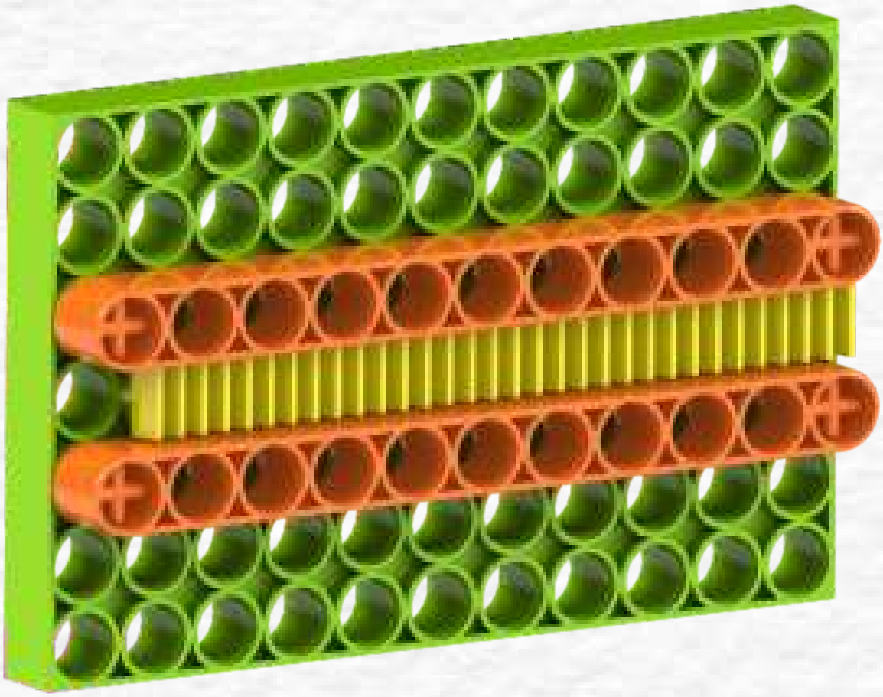
P7 X 11	1 pc.

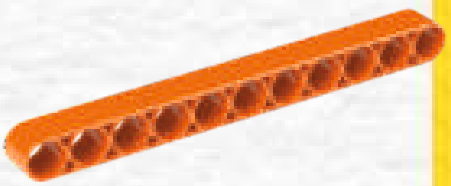


Rack	2 pcs.



STEP - 10

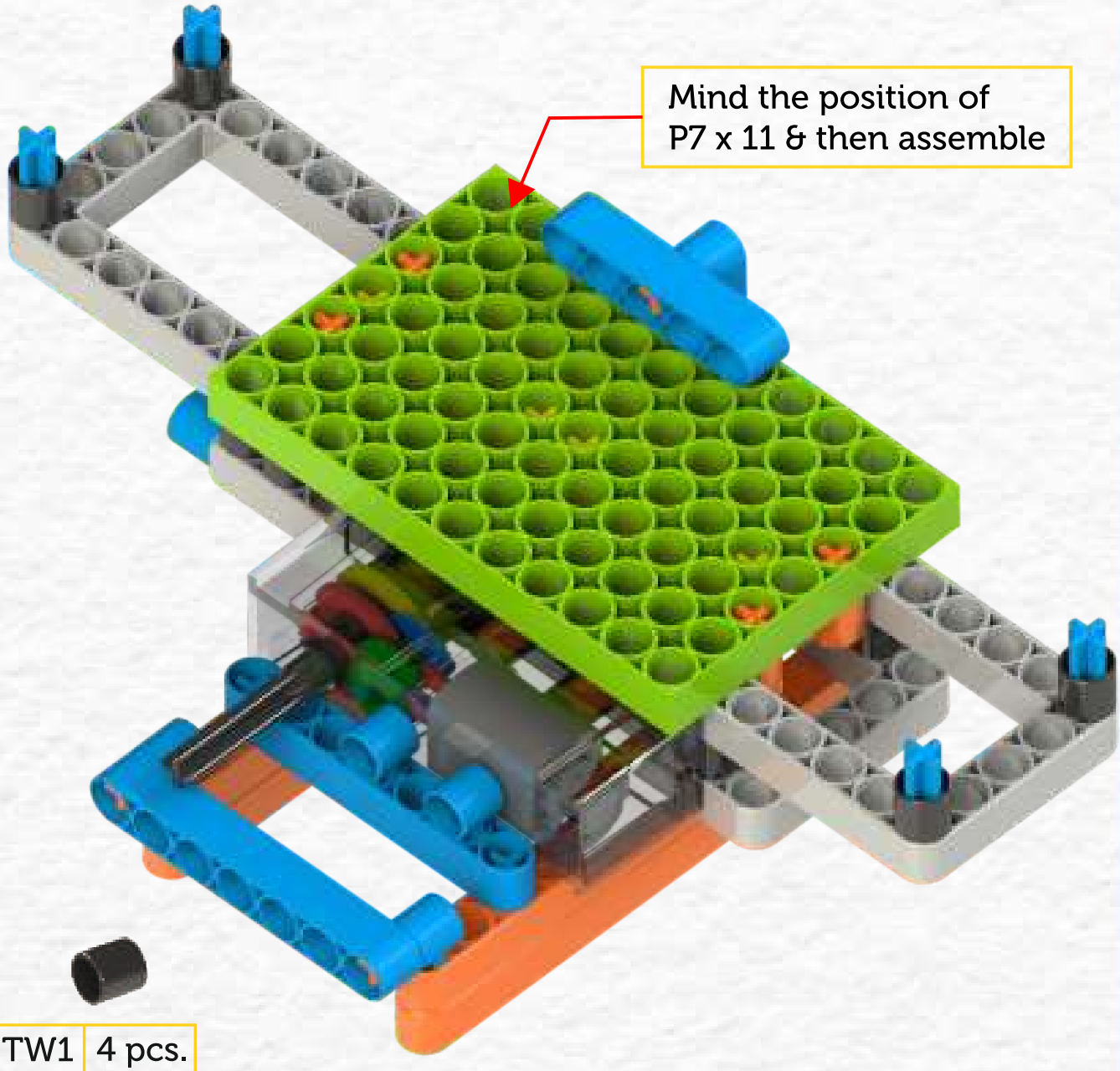




P11	2 pcs.

STEP - 11

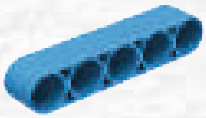
Assembly of step 10 & step 8



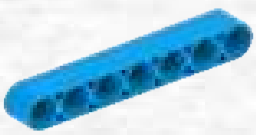
TW1	4 pcs.

CT2	1 pc.

CT3	4 pcs.

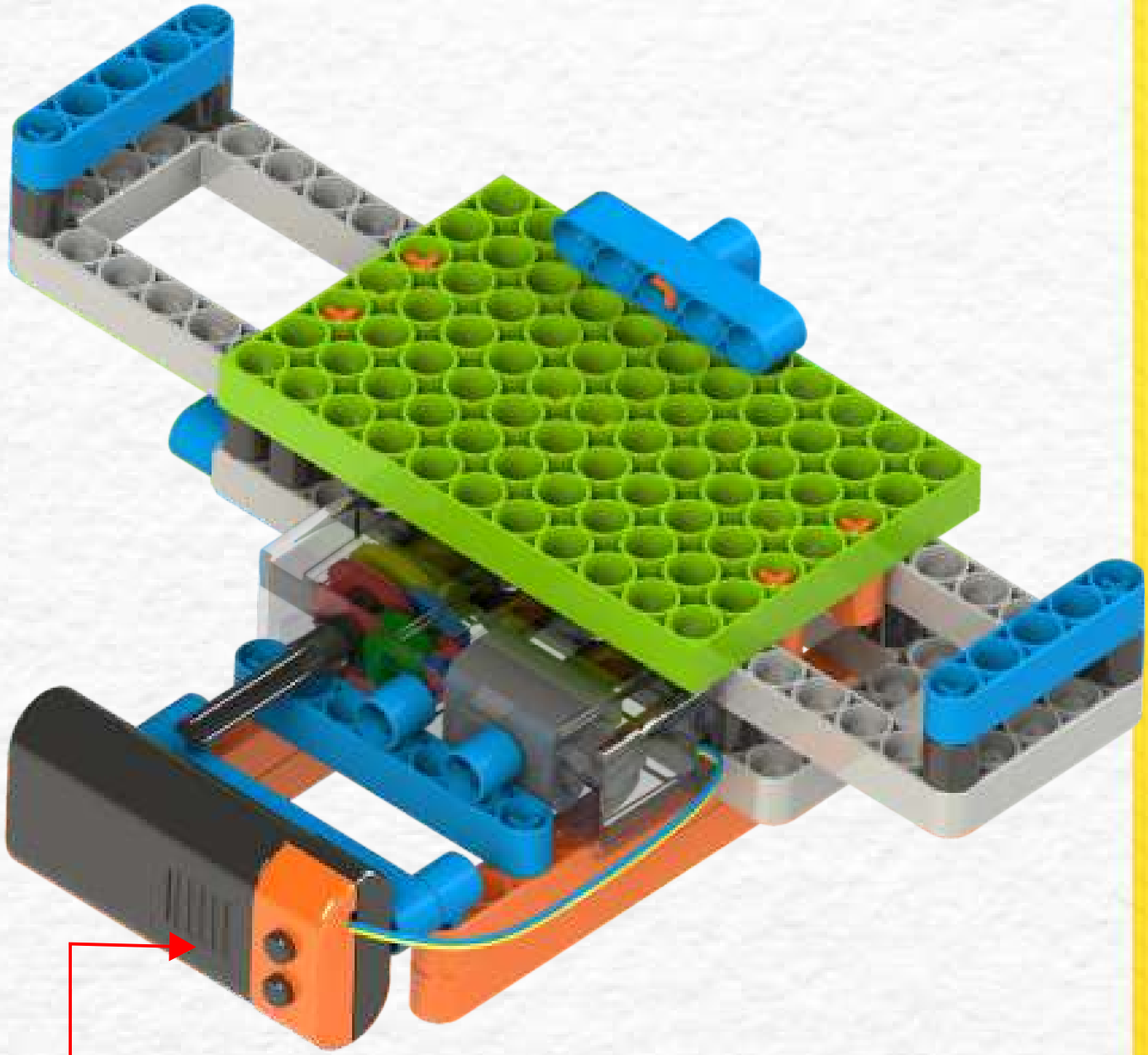


P5	1 pc.

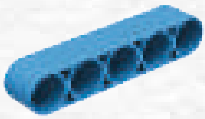


P7	1 pc.

STEP - 12

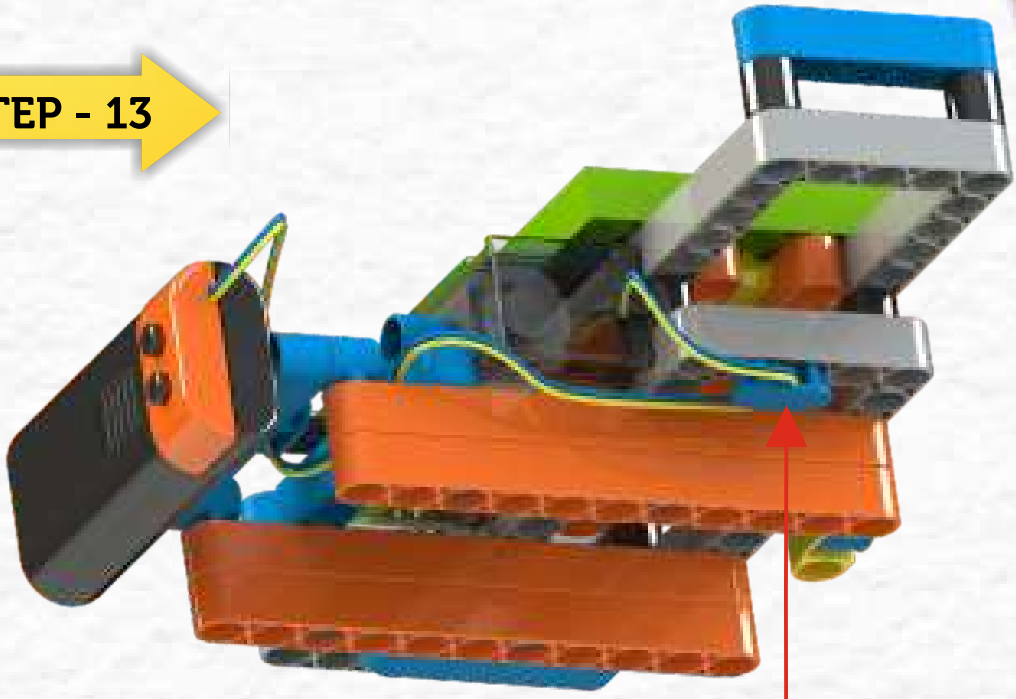


Attach battery box



P5	2 pcs.

STEP - 13

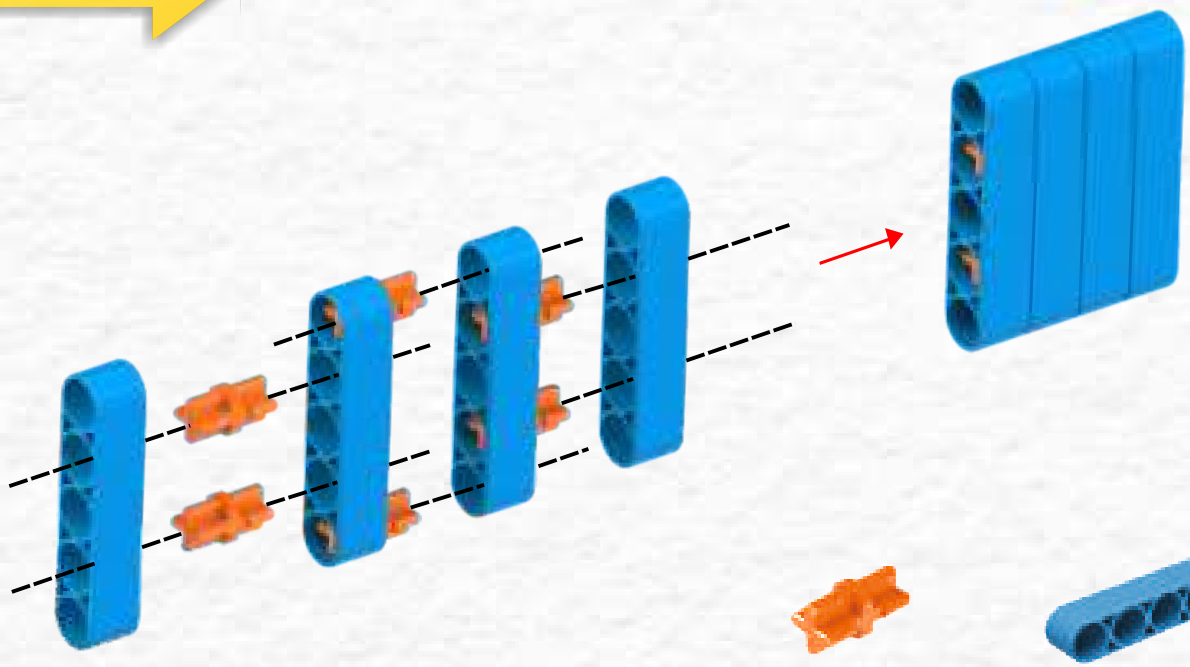


Use CT1 x 2 to prevent wire from coming in contact with moving parts

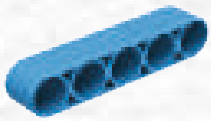
CT 1 X 2	2 pcs.

MODEL 5 - Horizontal Gate

STEP - 1

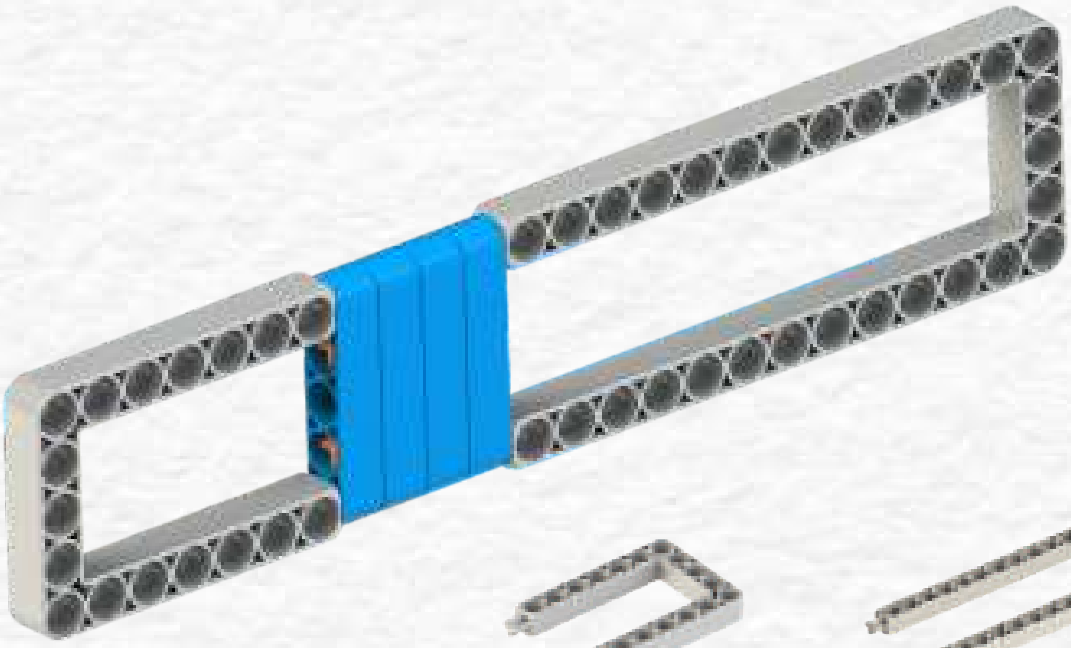


CT2	6 pcs.



P5	4 pcs.

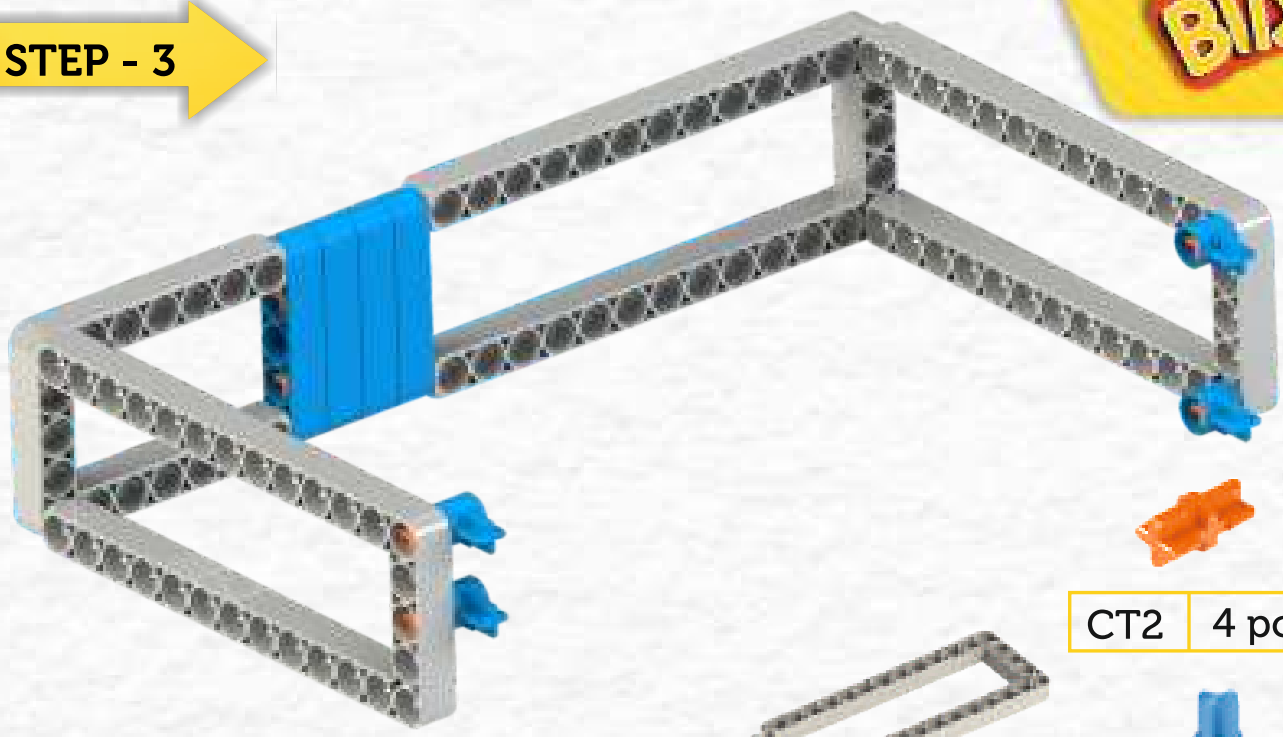
STEP - 2



PU5 X 7	1 pc.

PU5 X 13	1 pc.

STEP - 3



CT2	4 pcs.

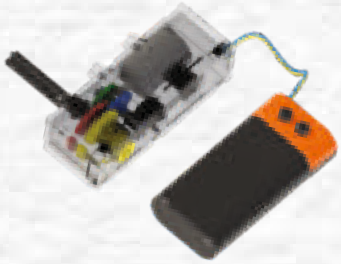
PU5 X 13	2 pcs.

CH2	4 pcs.

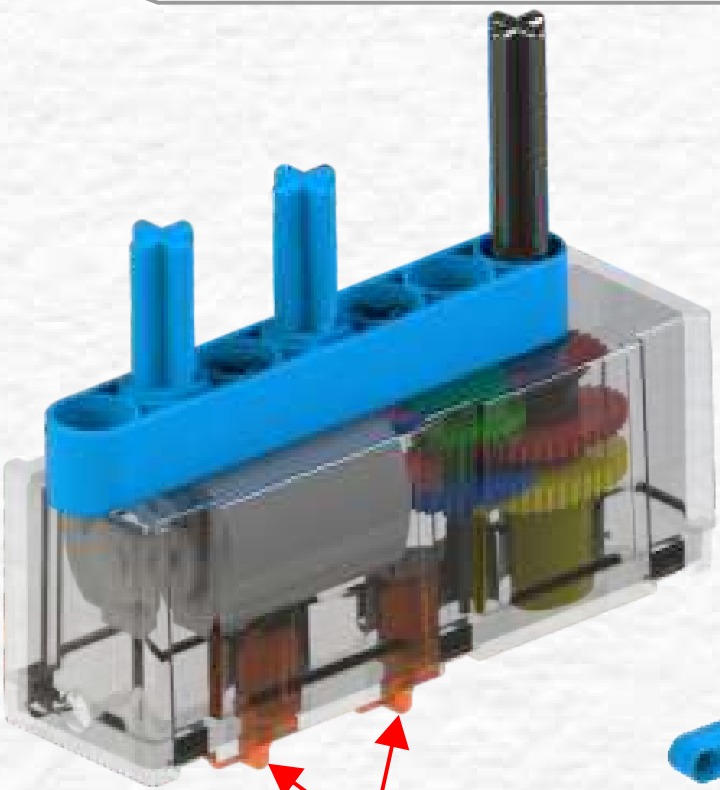
STEP -4

CT2	2 pcs.

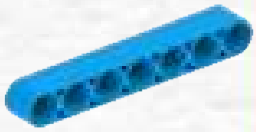
CT3	2 pcs.



Motor With Battery Box 1 pc.

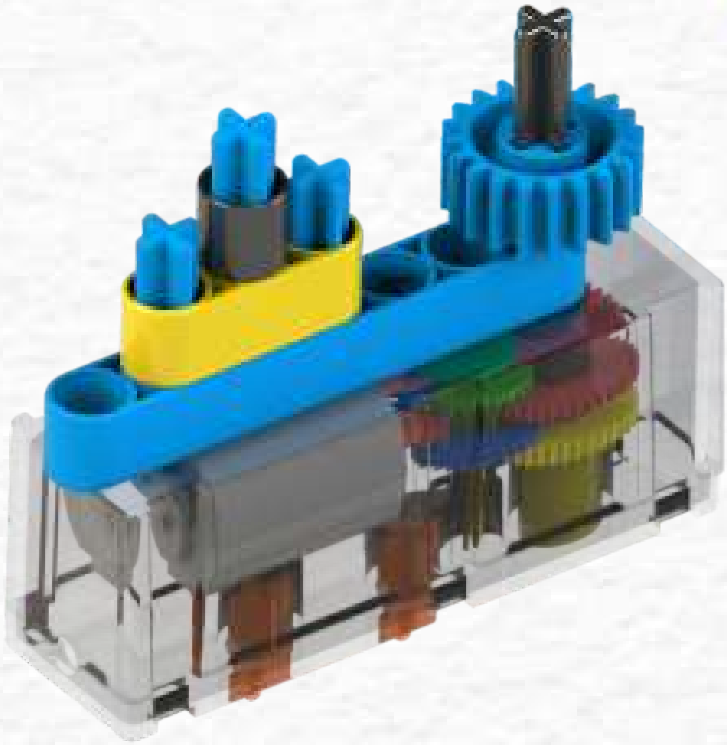


Attach CT2



P7	1 pc.

STEP -5



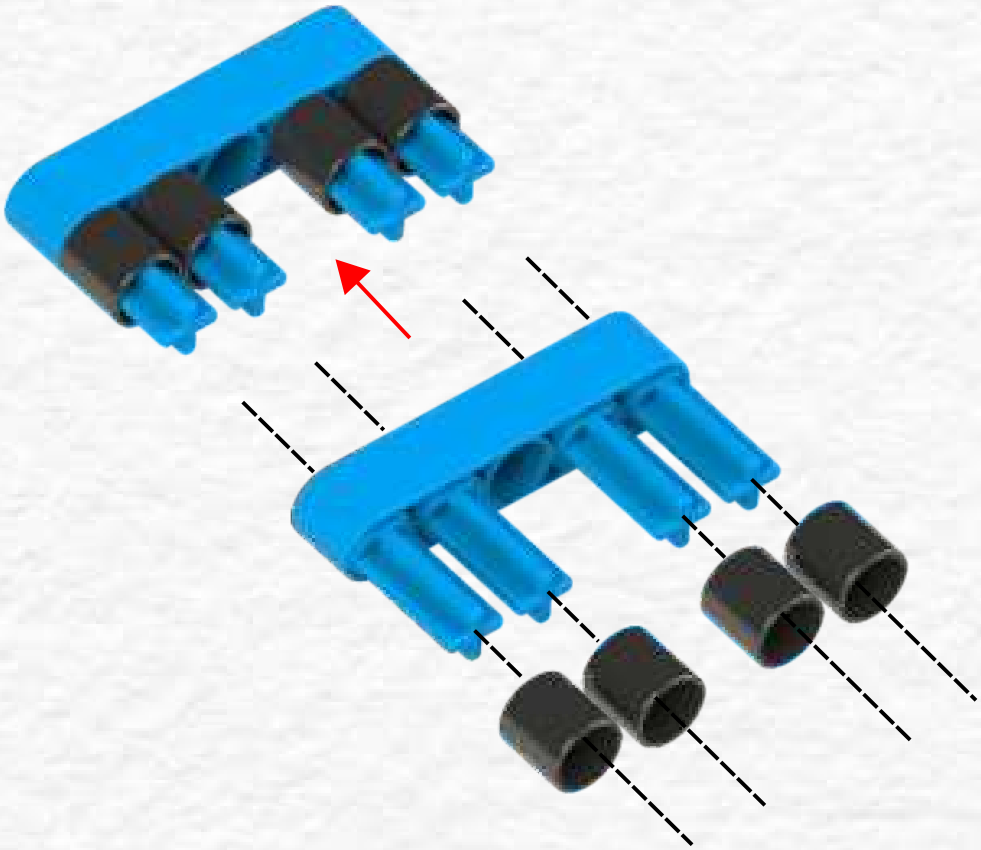
TW1	1 pc.

P3	1 pc.

CT3	1 pc.

G(20)	1 pc.

STEP - 6

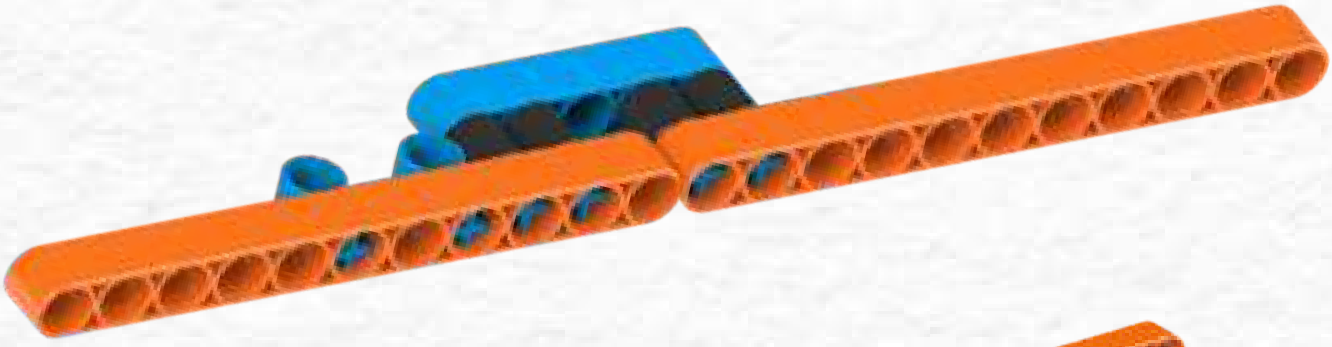


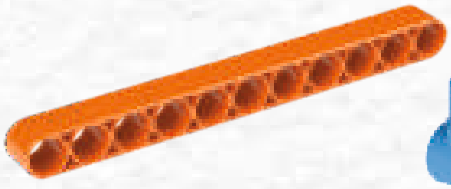
TW1	4 pcs.

CT3	4 pcs.

P5	1 pc.

STEP -7



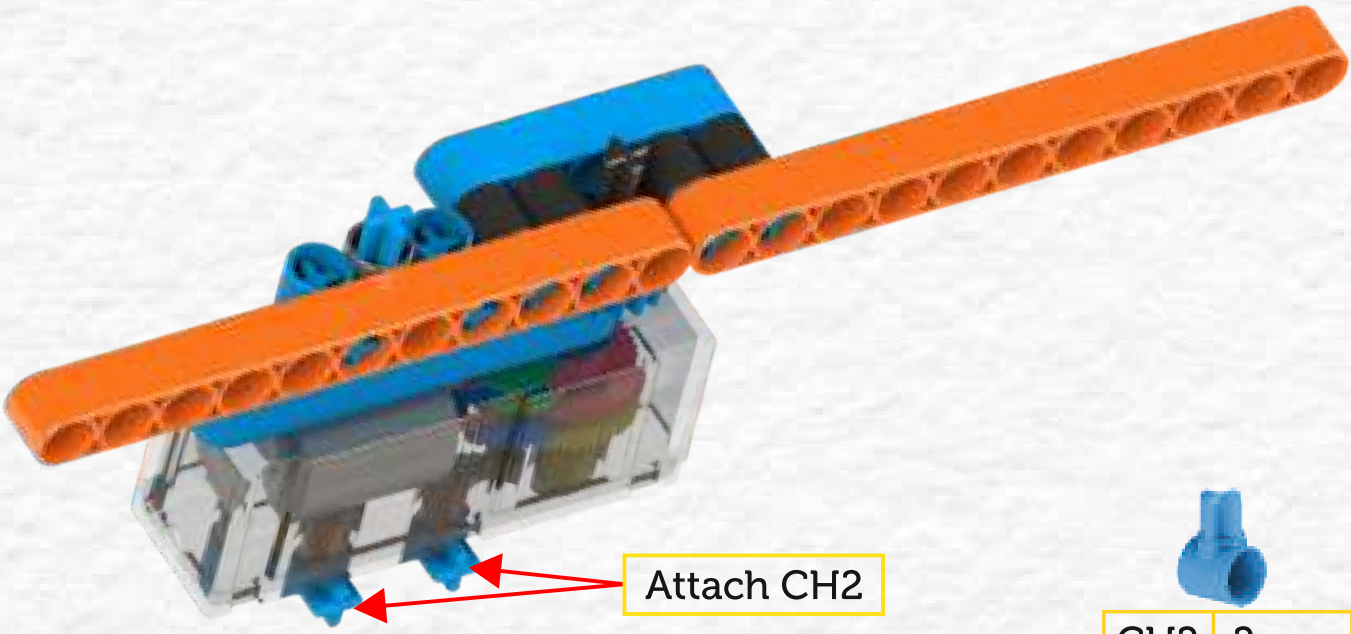


P11	2 pcs.

CH2	2 pcs.

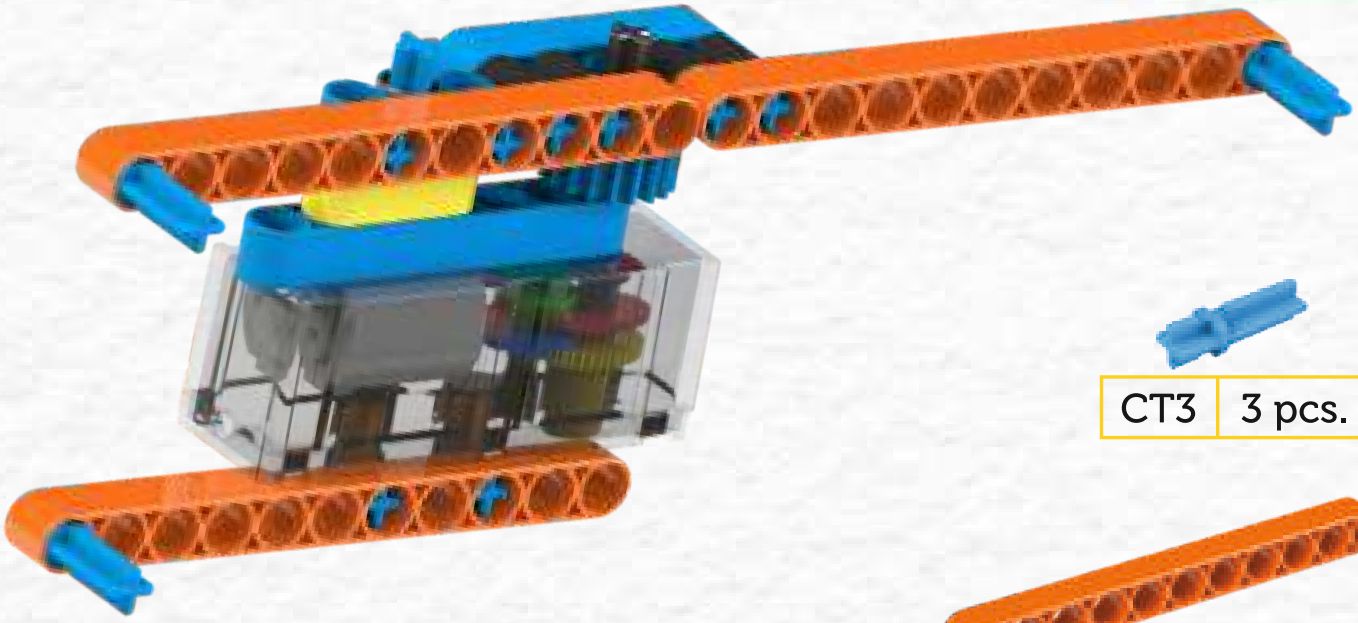
STEP - 8

Assembly of step 7 & step 5



CH2	2 pcs.

STEP -9

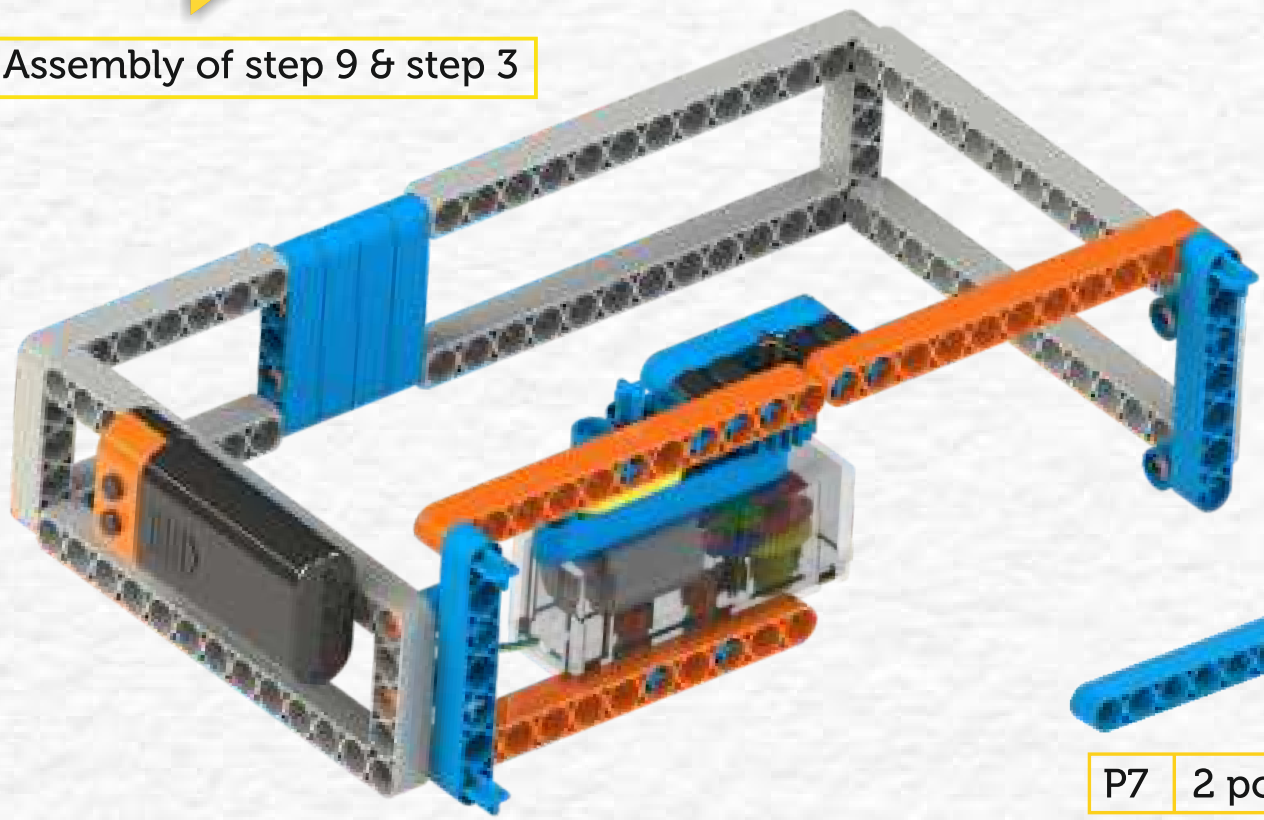


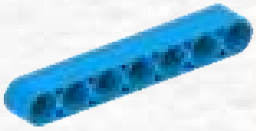
CT3	3 pcs.

P11	1 pc.

STEP - 10

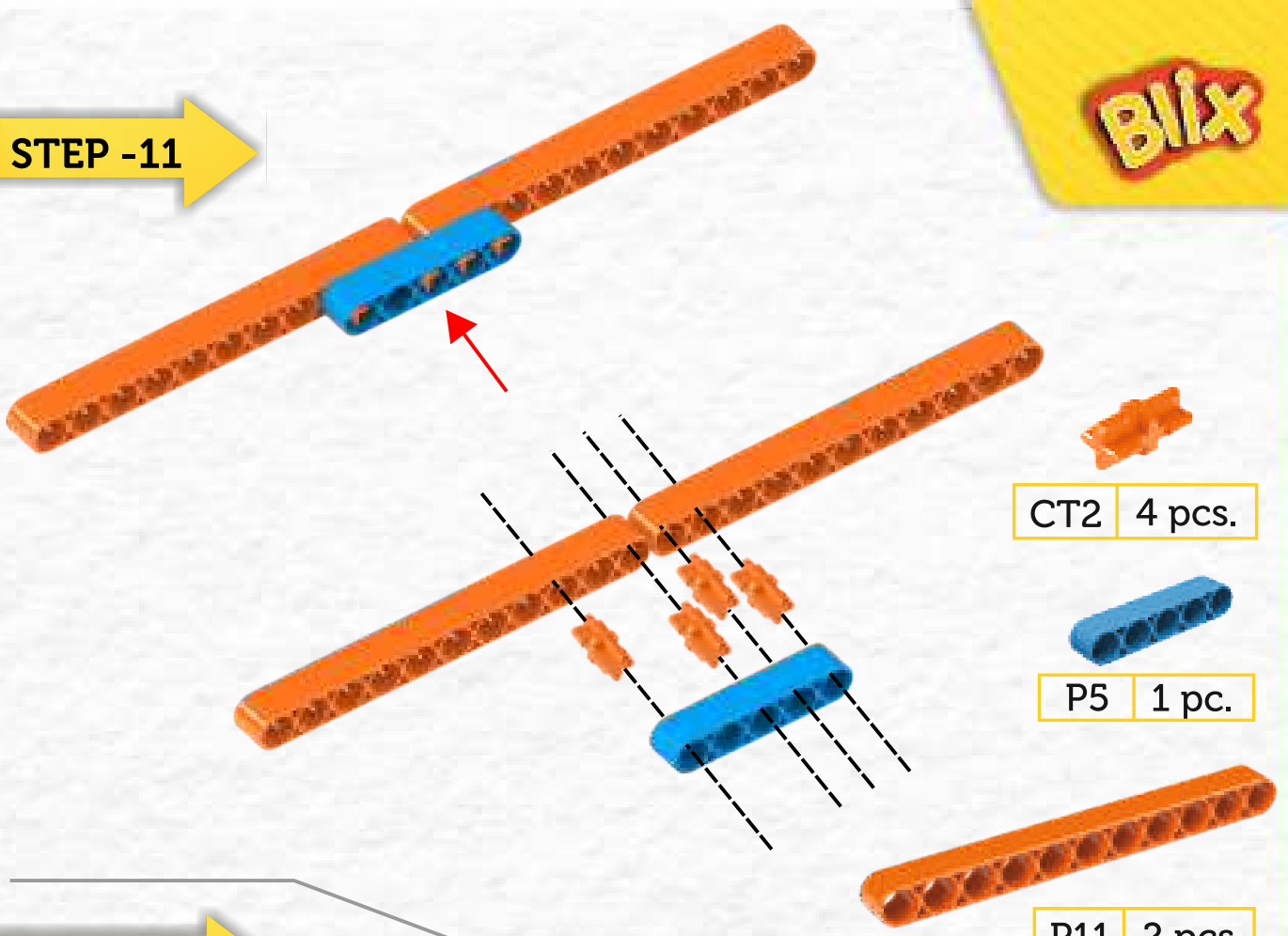
Assembly of step 9 & step 3





P7	2 pcs.

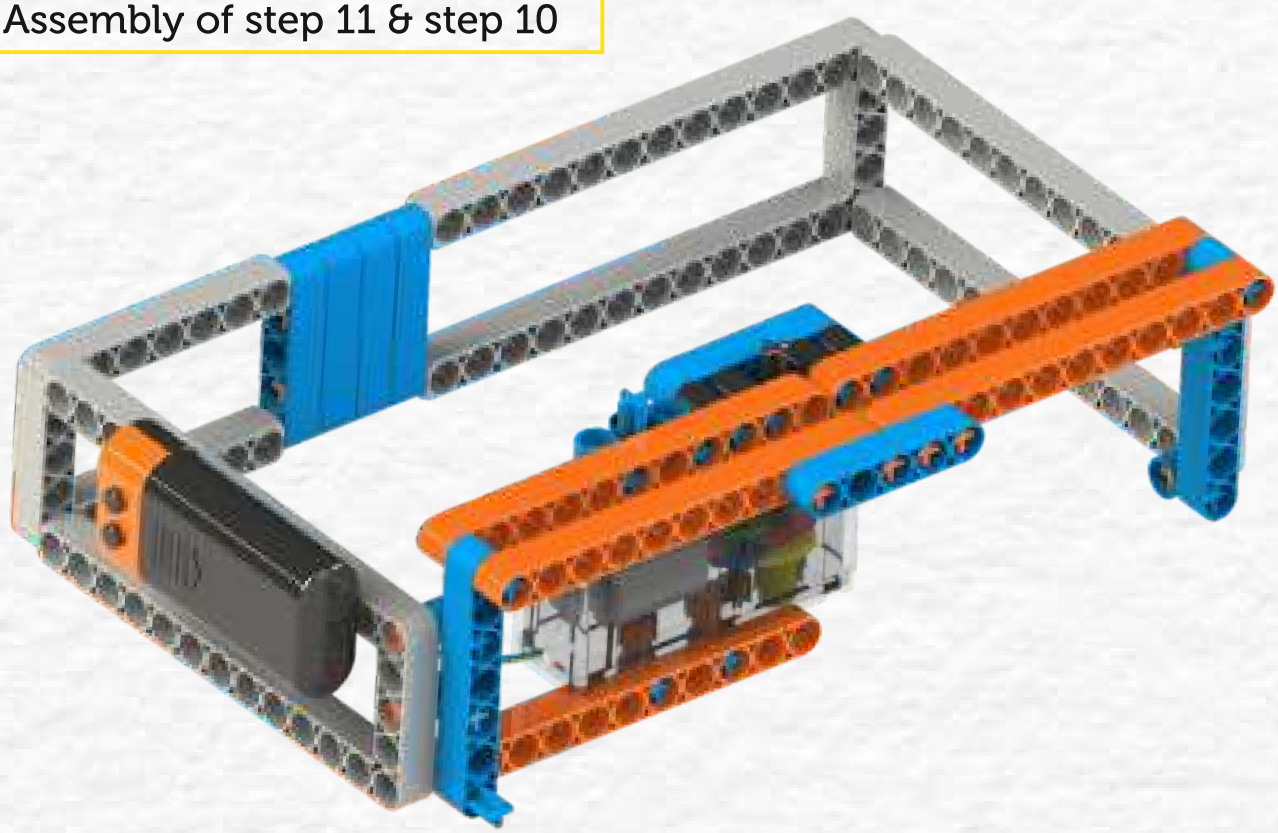
STEP -11



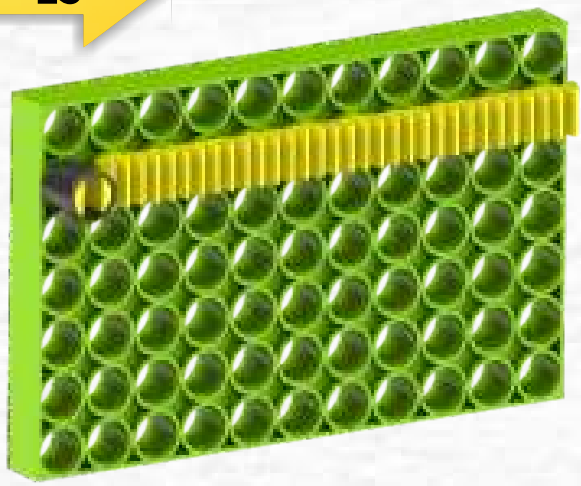
CT2	4 pcs.
P5	1 pc.
P11	2 pcs.

STEP - 12

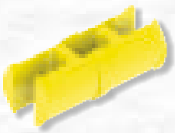
Assembly of step 11 & step 10



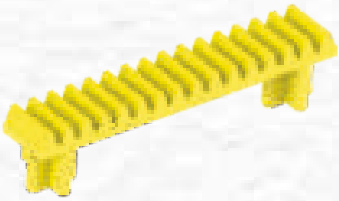
STEP -13



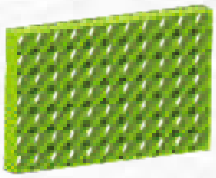
TW1	1 pc.



CL2	1 pc.



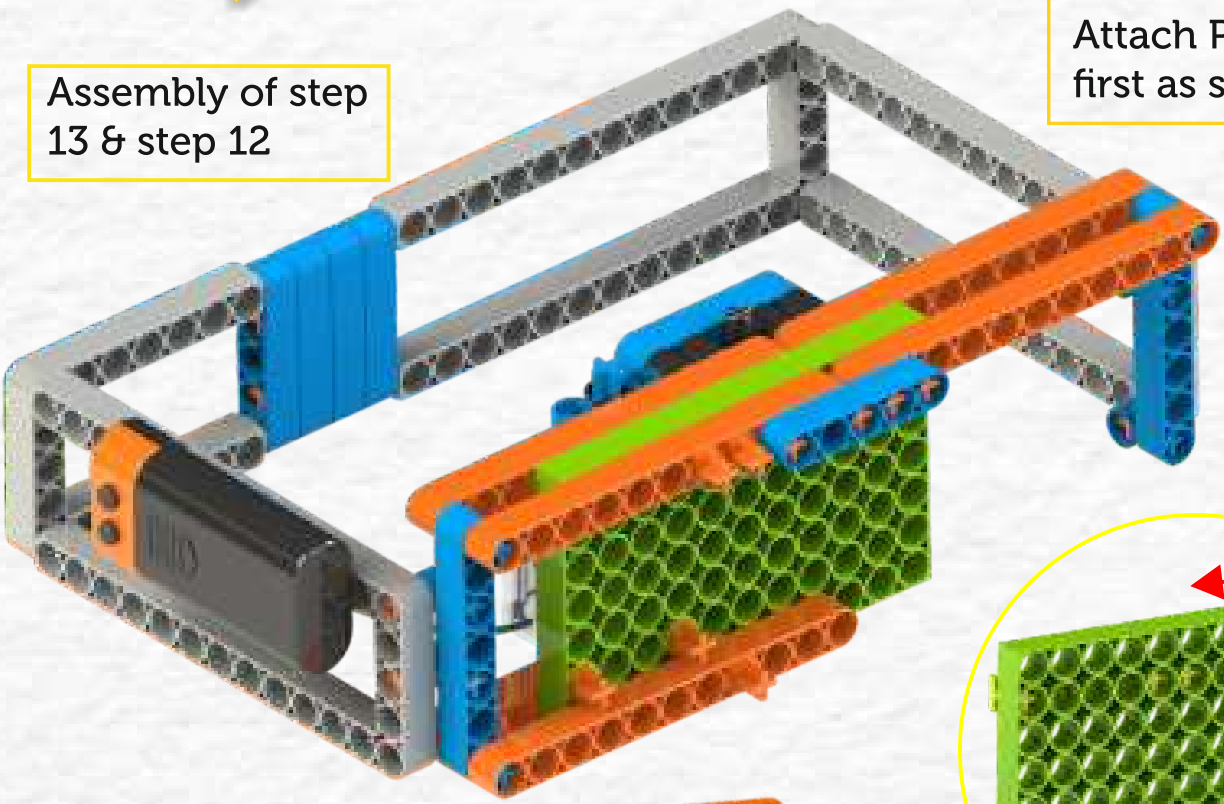
Rack	2 pcs.



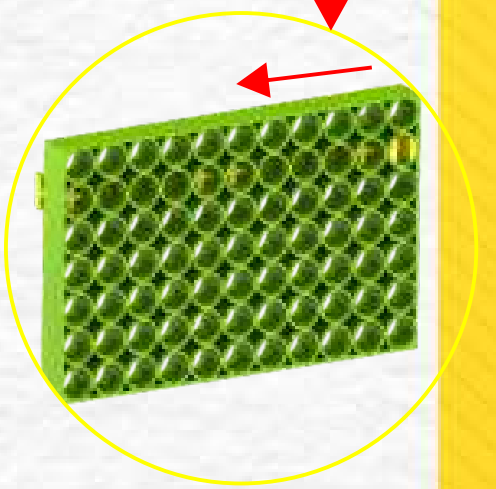
P7 X 11	1 pc.

STEP - 14

Assembly of step 13 & step 12



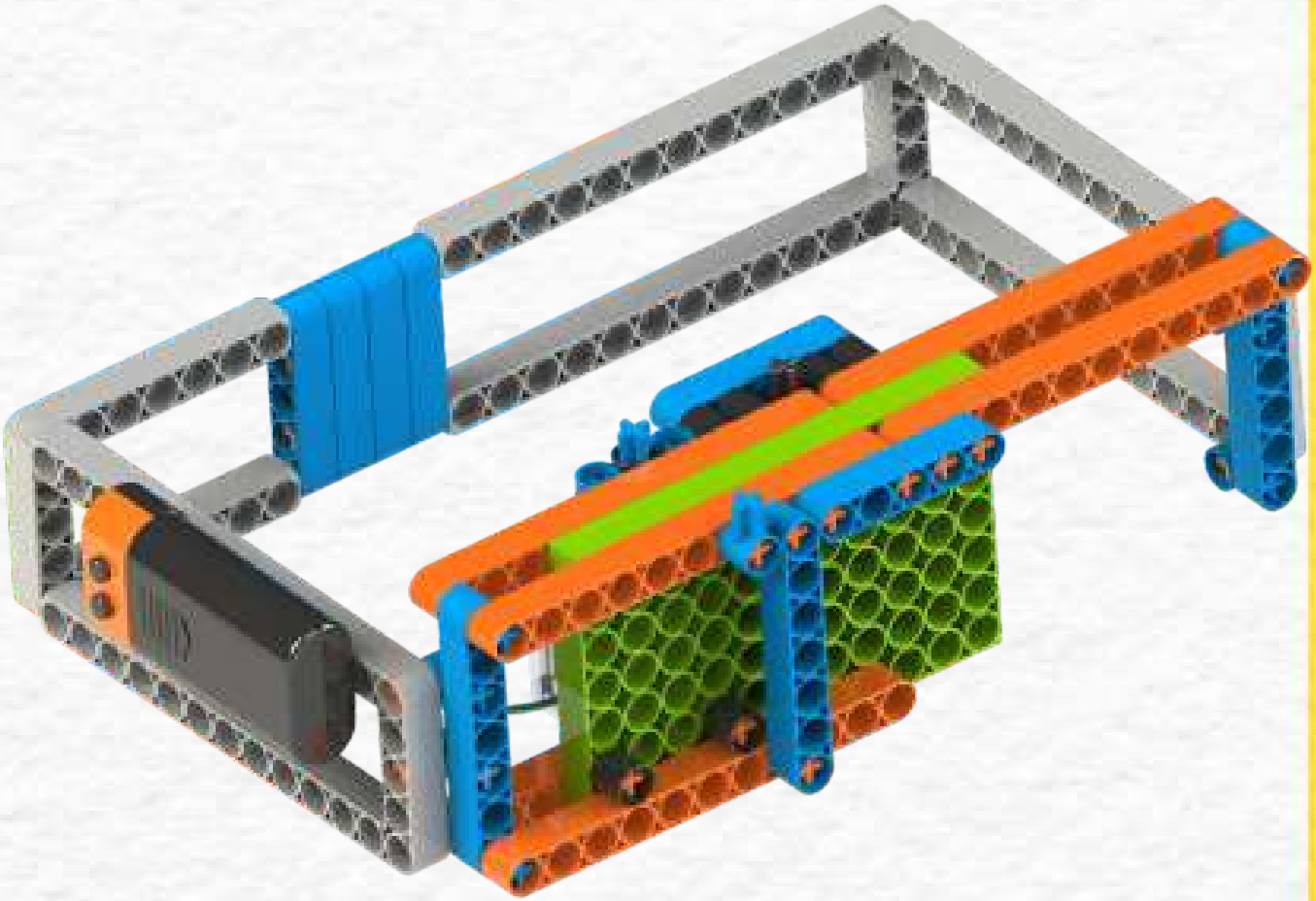
Attach P7 x 11 first as shown



CT2	5 pcs.

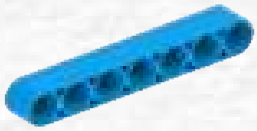
P11	1 pc.

STEP -15



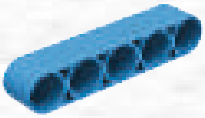
TW1	2 pcs.

CH2	1 pc.

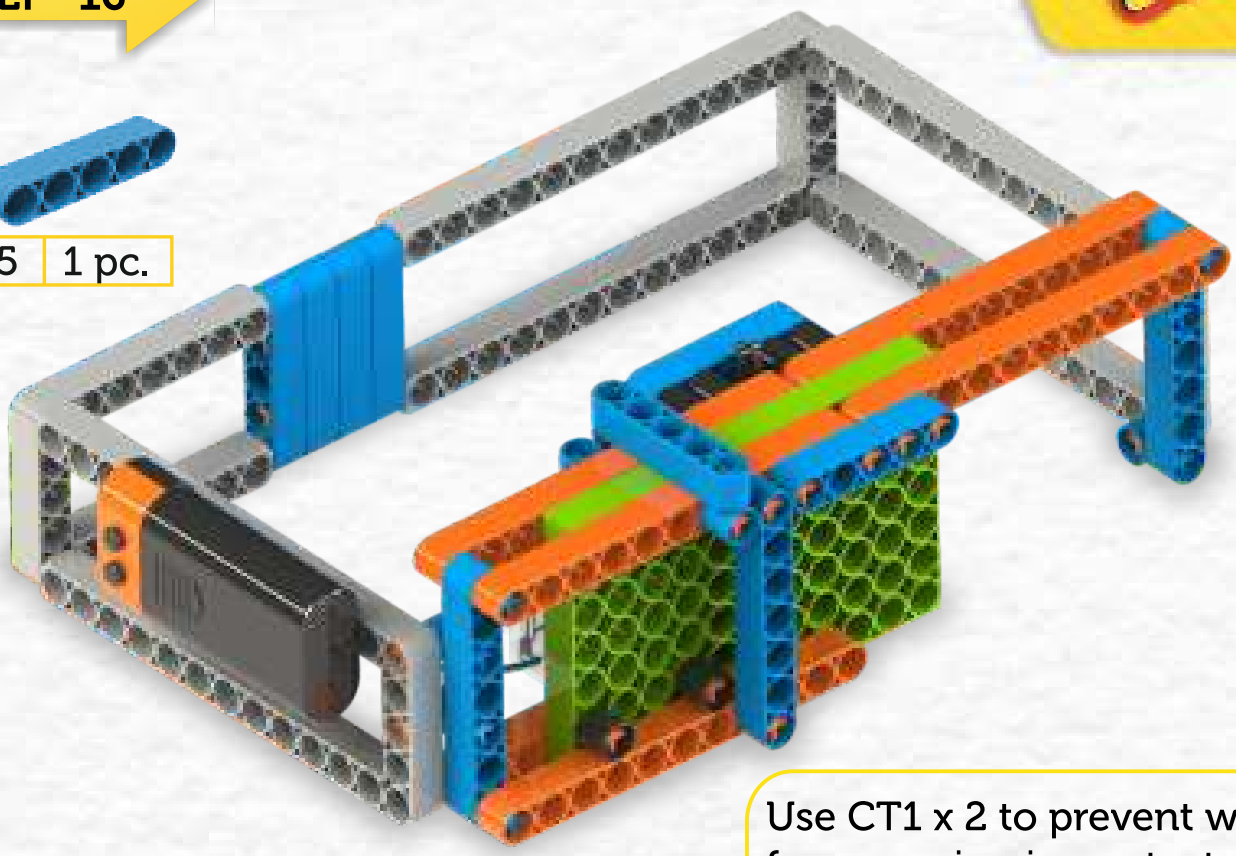


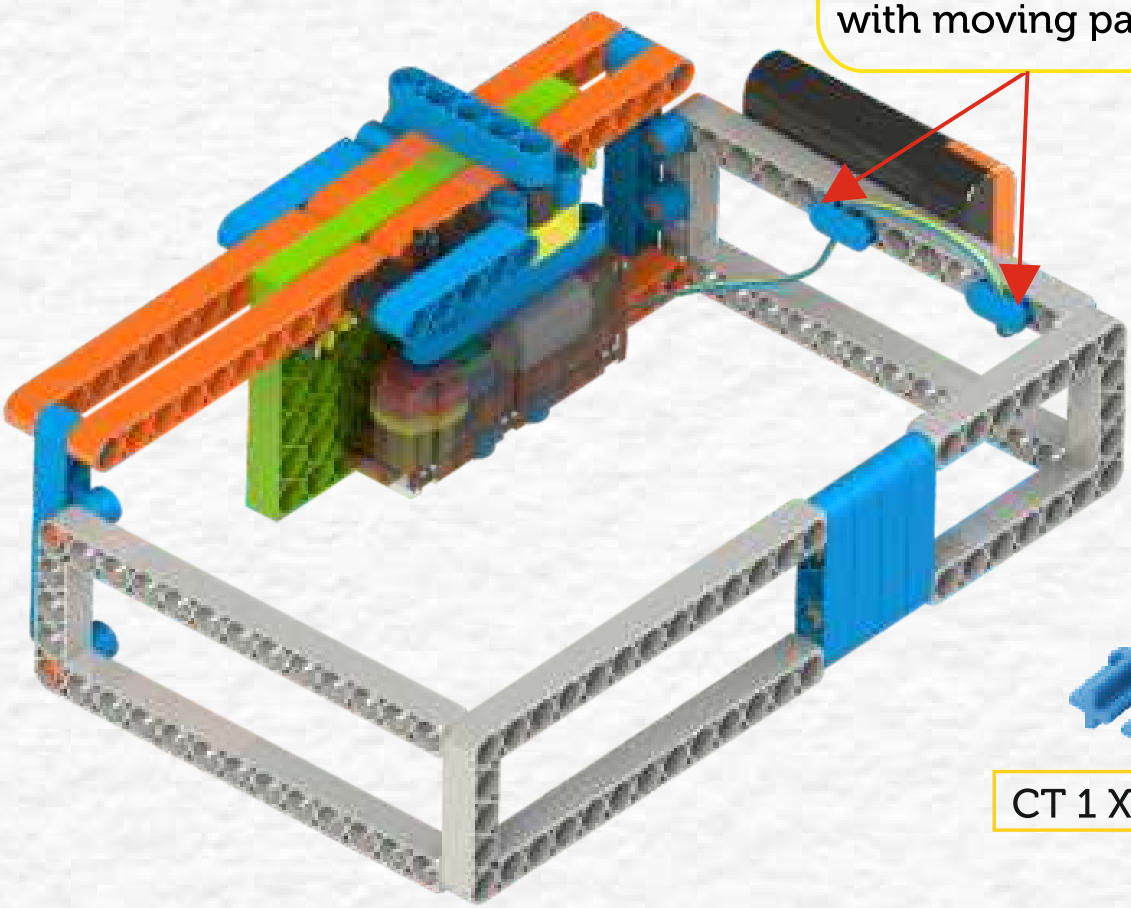
P7	1 pc.

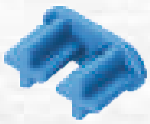
STEP -16



P5	1 pc.







CT 1 X 2	2 pcs.

MODEL 6 - Steering Car

STEP - 1

TW1	4 pcs.

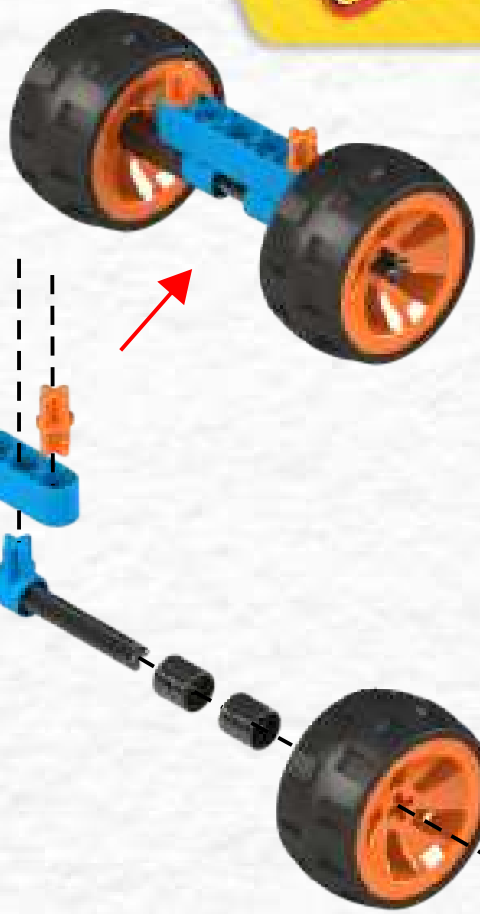
CH2	2 pcs.

CT2	2 pcs.

P5	1 pc.

SH 100	1 pc.

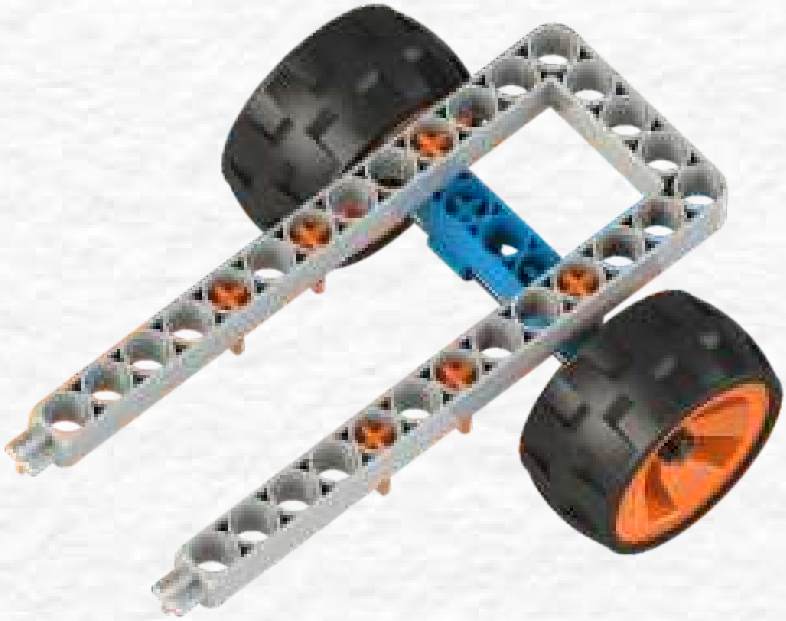
Wheels	2 pcs.



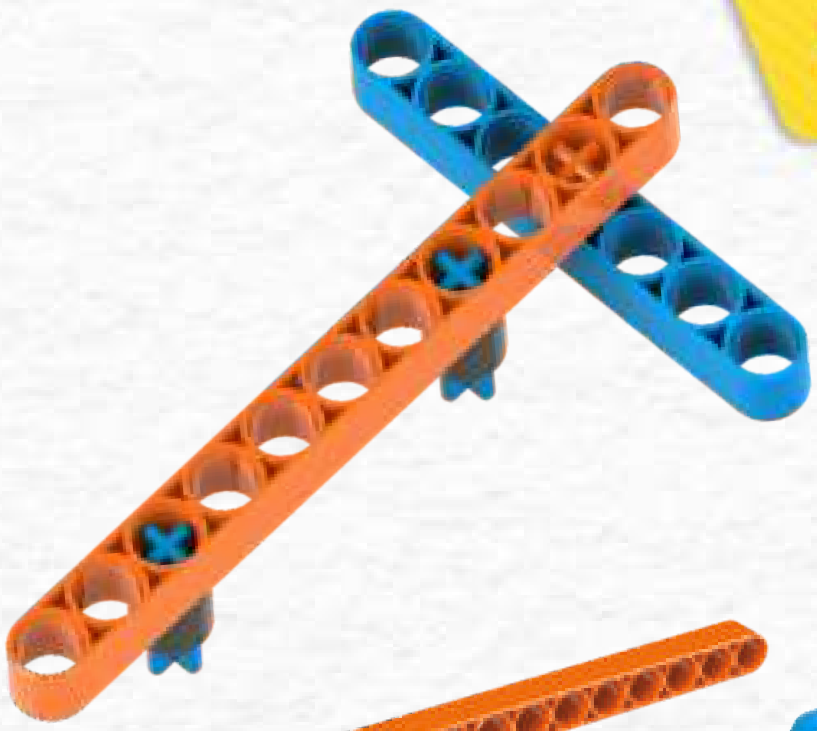
STEP - 2

CT2	4 pcs.

PU5 X 13	1 pc.



STEP - 3

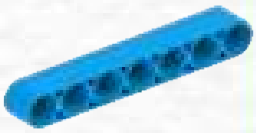


TW1	2 pcs.

CT2	1 pc.

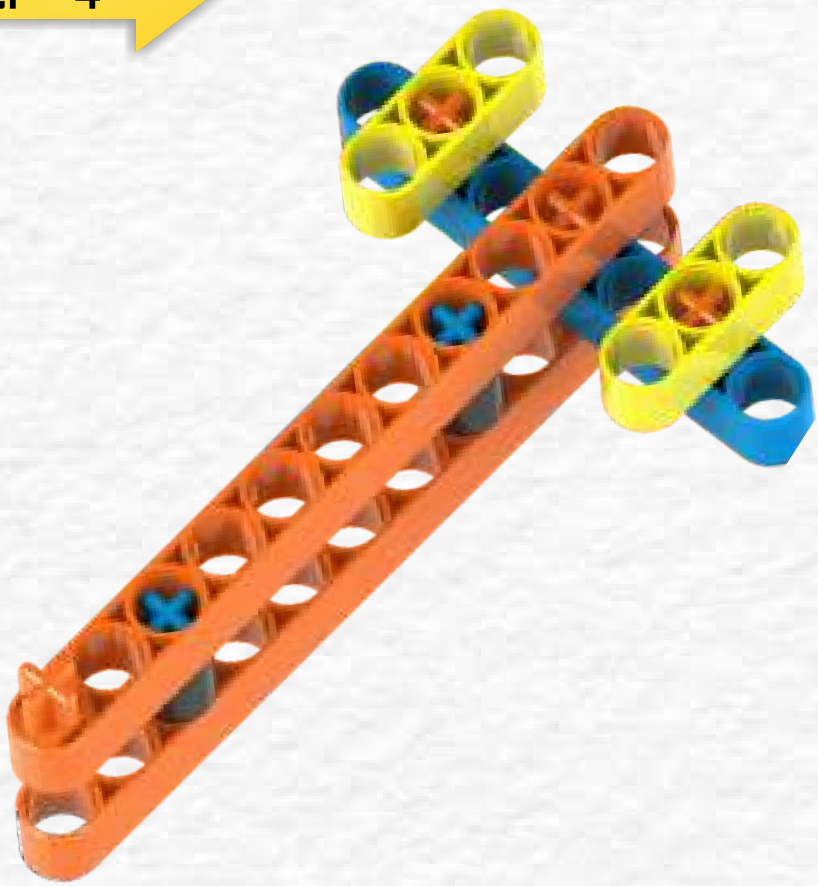
CT3	2 pcs.

P11	1 pc.



P7	1 pc.

STEP - 4



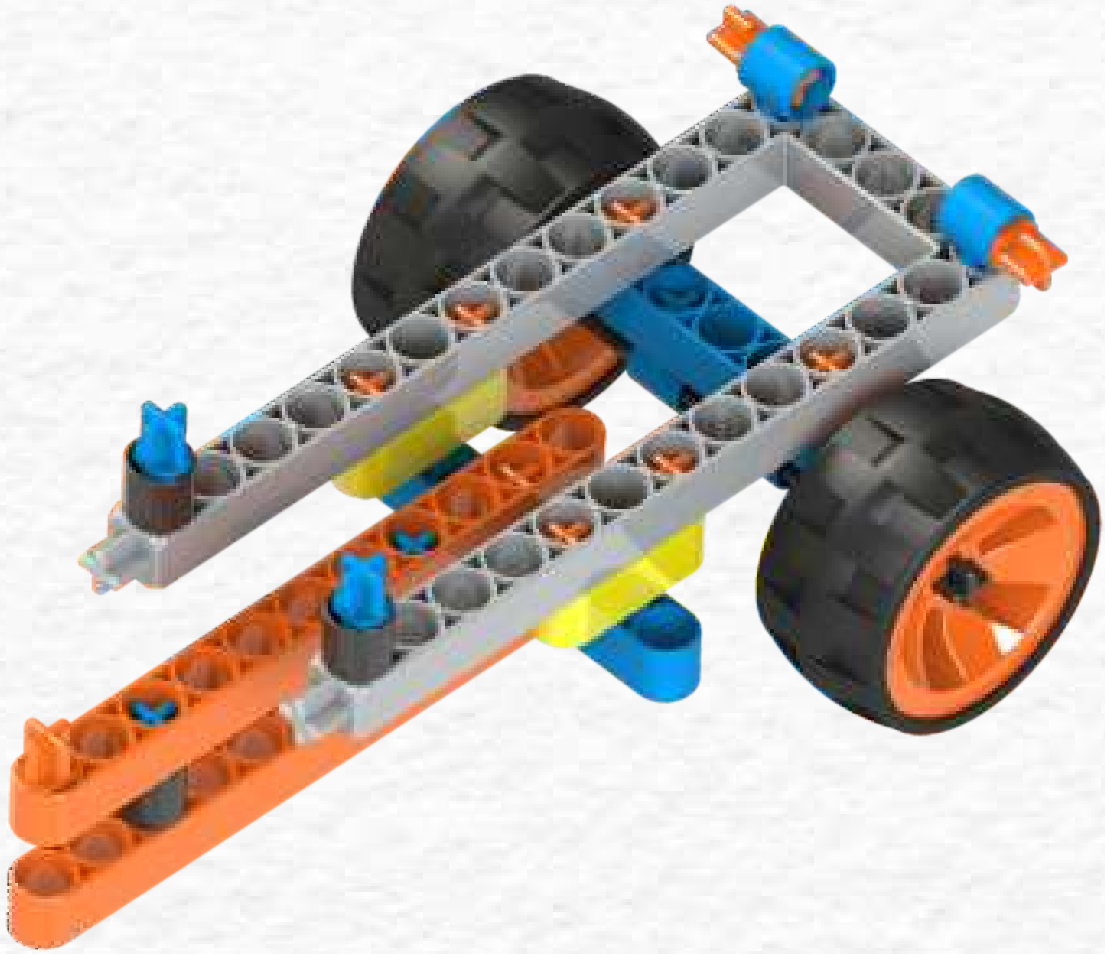
P3	2 pcs.

CT2	3 pcs.

P11	1 pc.

STEP - 5

Assembly of step 4 & step 2



TW1	2 pcs.

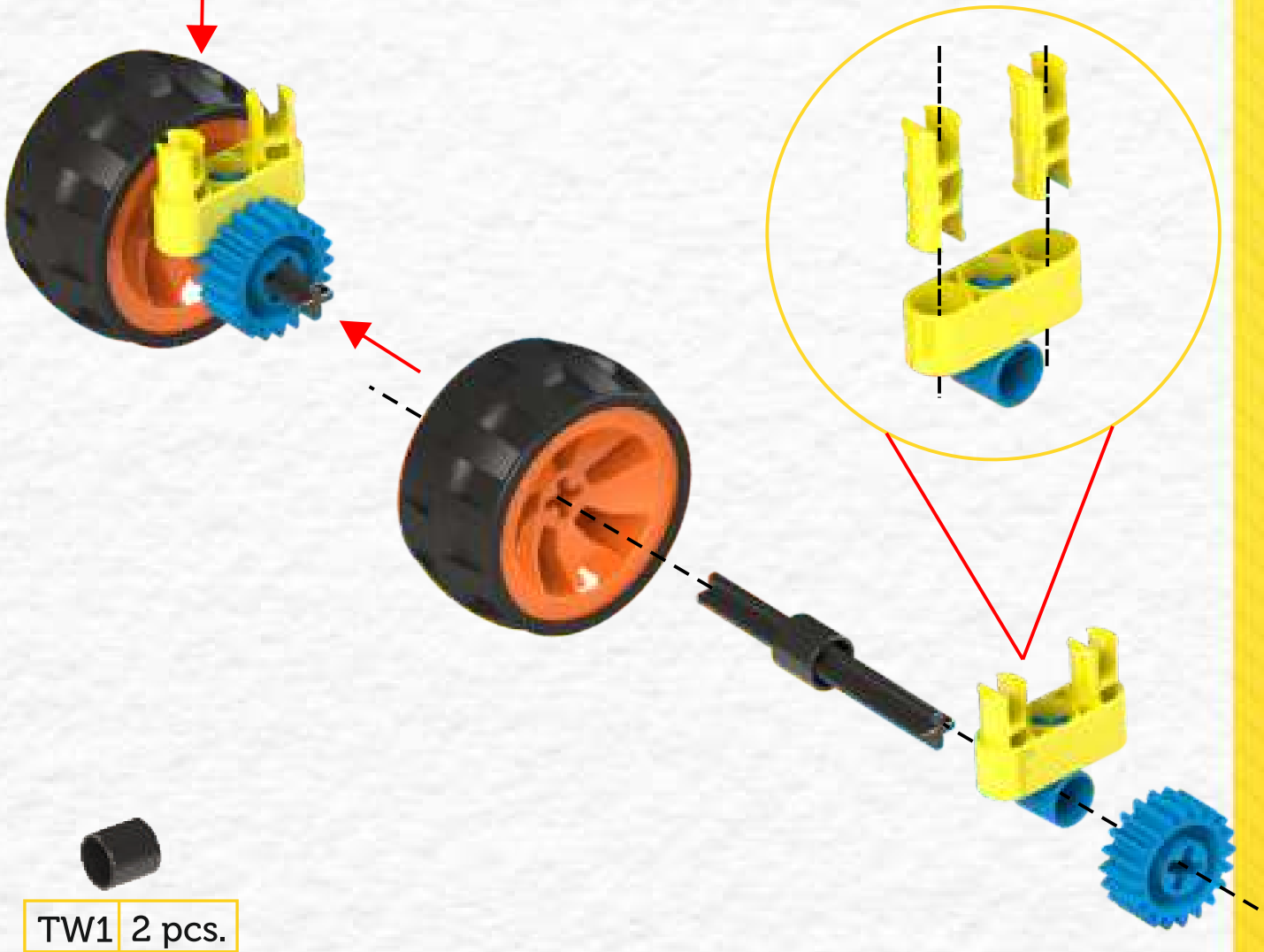
CT2	2 pcs.

CH2	2 pcs.

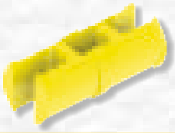
CT3	2 pcs.

STEP - 6

Make two assemblies



TW1	2 pcs.



CL2	4 pcs.

P3	2 pcs.

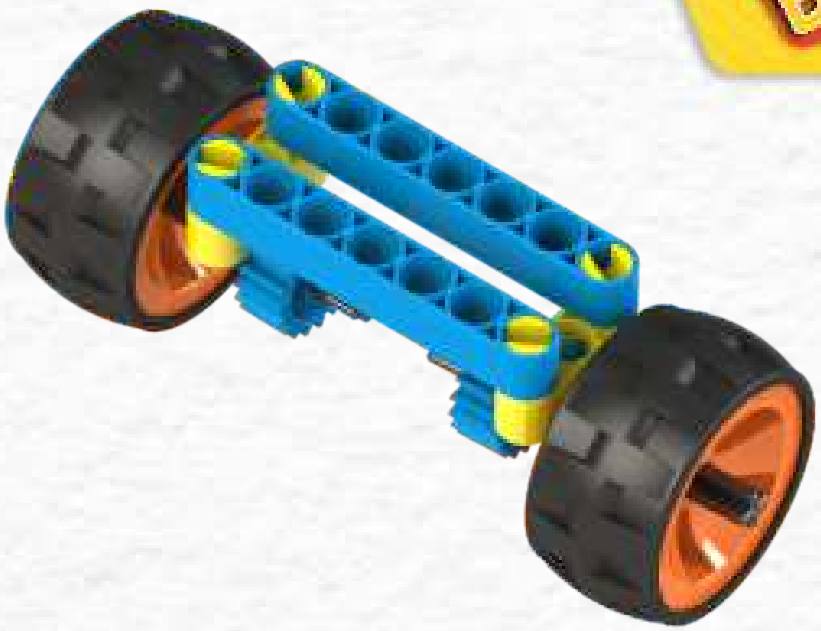
Wheels	2 pcs.

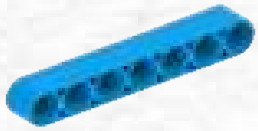
G(20)	2 pcs.

SH60	2 pcs.

CH2	2 pcs.

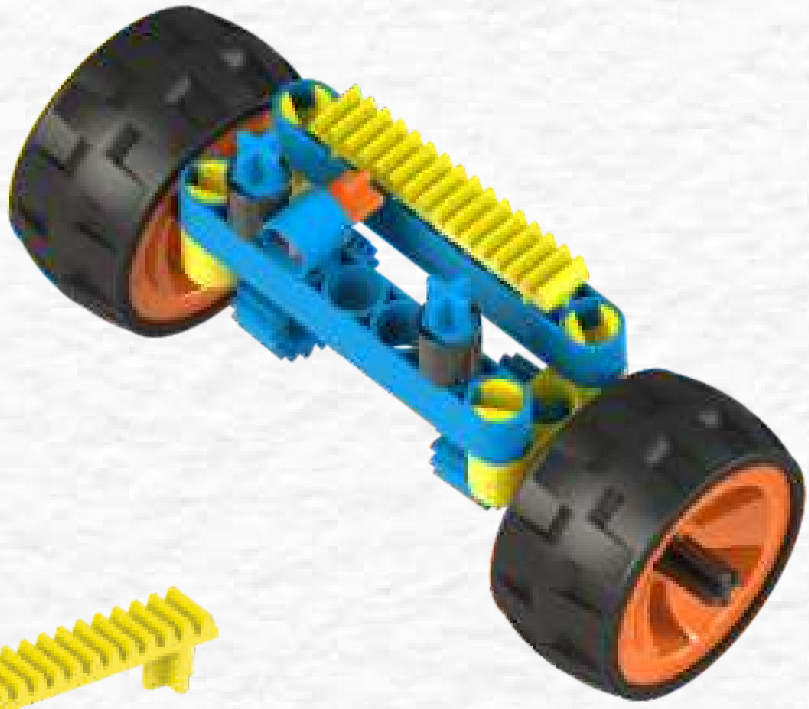
STEP - 7





P7	2 pcs.

STEP - 8

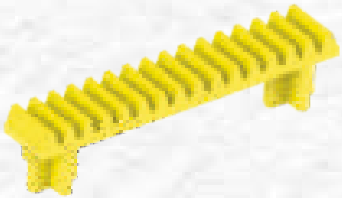


TW1	2 pcs.

CT2	1 pc.

CH2	1 pc.

CT3	2 pcs.



Rack	1 pc.

STEP - 9

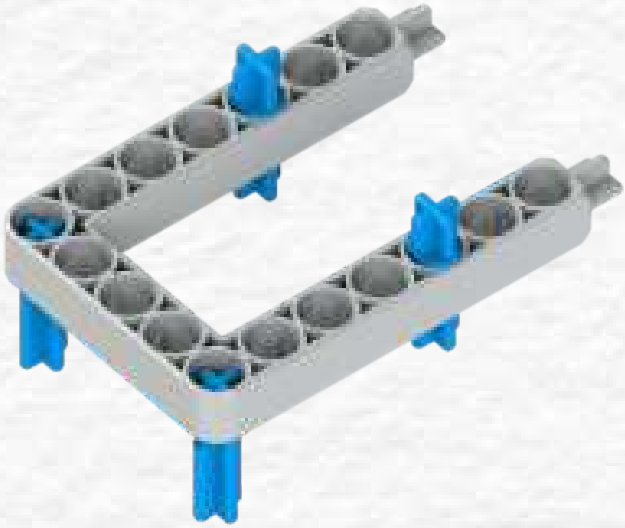
Assembly of step 8 & step 5

Attach here

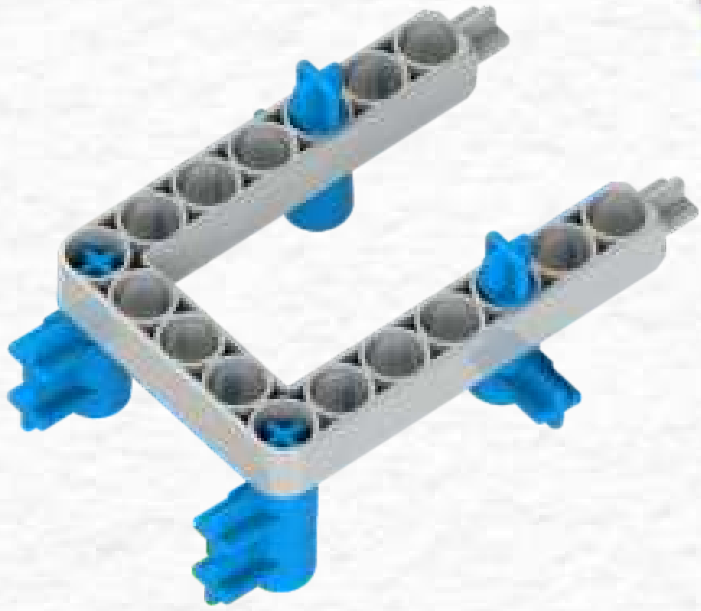
STEP - 10

CT3	4 pcs.

PU5 X 7	1 pc.

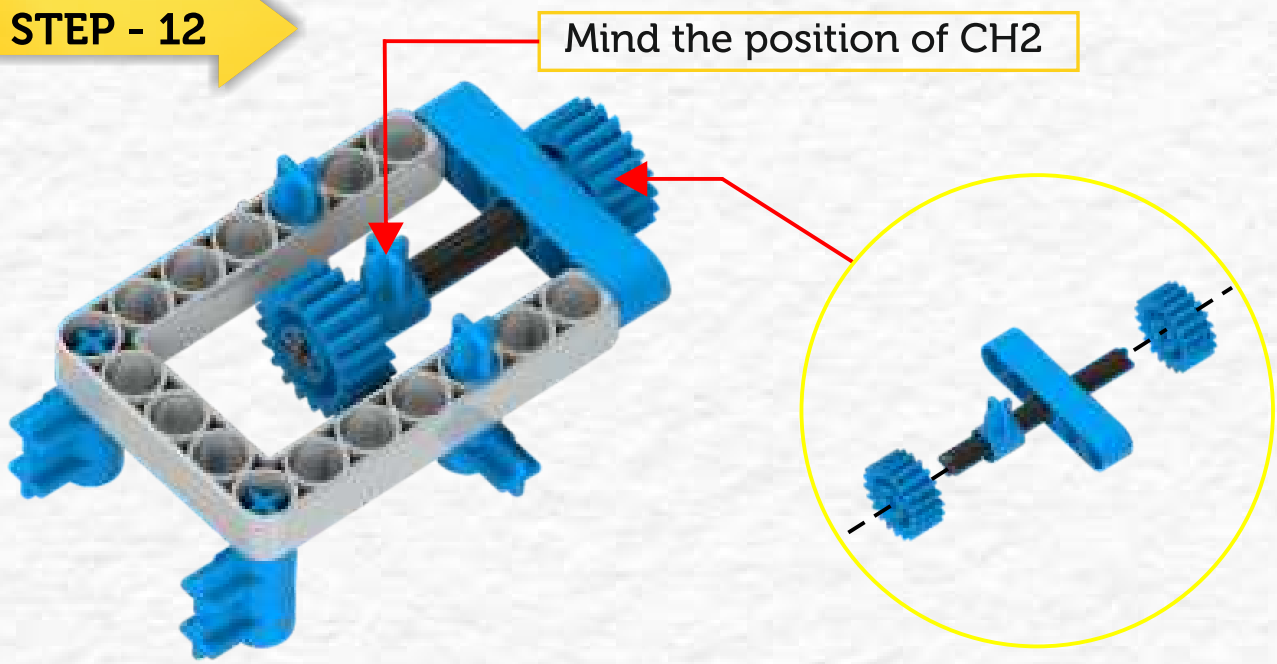


STEP - 11



CH2	6 pcs.

STEP - 12



P5	1 pc.

CH2	1 pc.

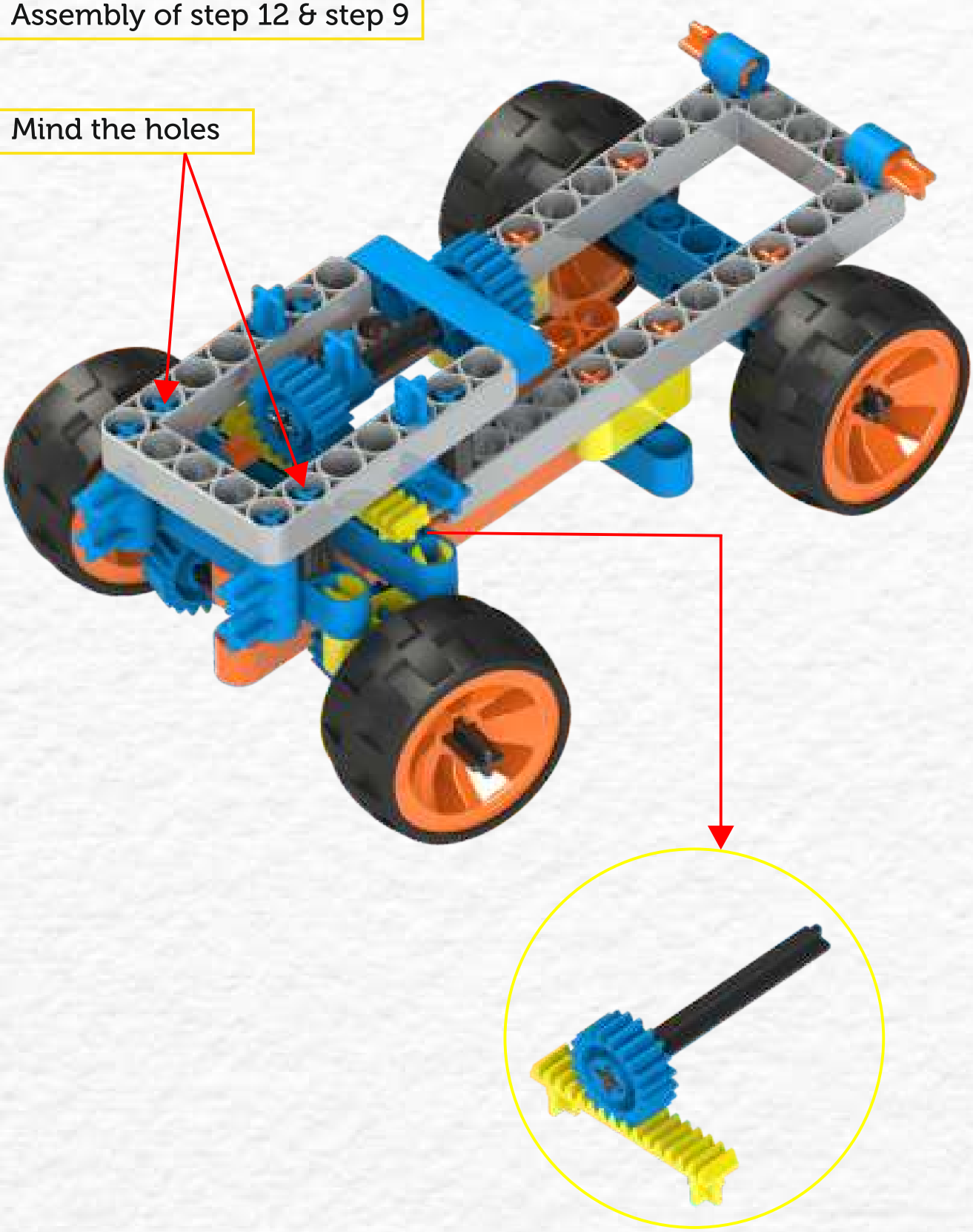
G(20)	2 pcs.

SH60	1 pc.

STEP - 13

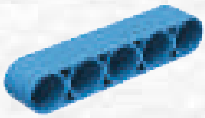
Assembly of step 12 & step 9

Mind the holes

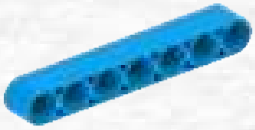


STEP - 14

TW1	2 pcs.

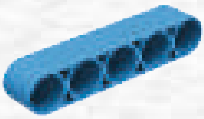


P5	1 pc.



P7	1 pc.

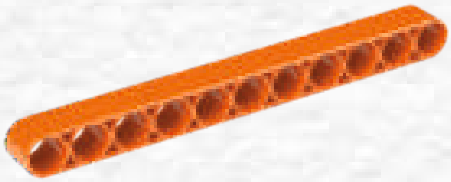
STEP - 15



P5	4 pcs.

STEP - 16

CT 1 X 2	4 pcs.



P11	2 pcs.

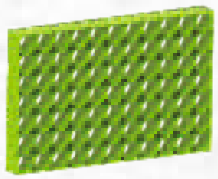
Back side View

STEP - 17

CT2	4 pcs.

CH2	4 pcs.

STEP - 18



P7 X 11	1 pc.

ZEPHYR
More Than Just Play

Blix
RACK & PINION

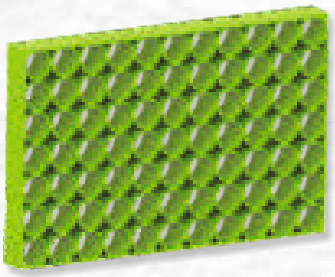
CT2	45 pcs.

CT3	20 pcs.

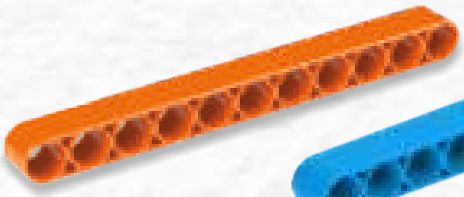
CL2	6 pcs.

CH2	20 pcs.

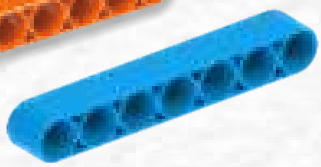
TW1	20 pcs.



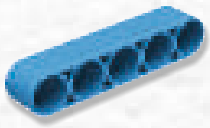
P7 X 11	1 pc.



P11	6 pcs.



P7	6 pcs.



P5	7 pcs.

P3	6 pcs.

PC 3	2 pcs.

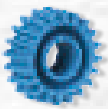
SH60	3 pcs.

SH 100	3 pcs.

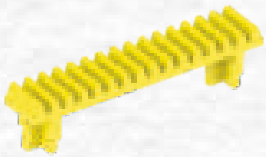
SH 170	1 pc.

CT 1 X 2	4 pcs.

G(20)	5 pcs.



G(20) Idler	2 pcs.

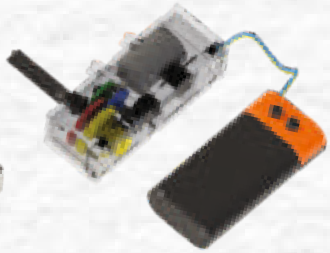


Rack	6 pcs.

Wheels	4 pcs.

PU5 X 7	4 pcs.

PU5 X 13	4 pcs.



Motor With Battery Box	1 pc.

Remover Tool	1 pc.